

Dynamics of Conserved-Charge Fluctuations in Heavy-Ion Collisions



Oleh Savchuk, University of Houston,
January 14th 2026

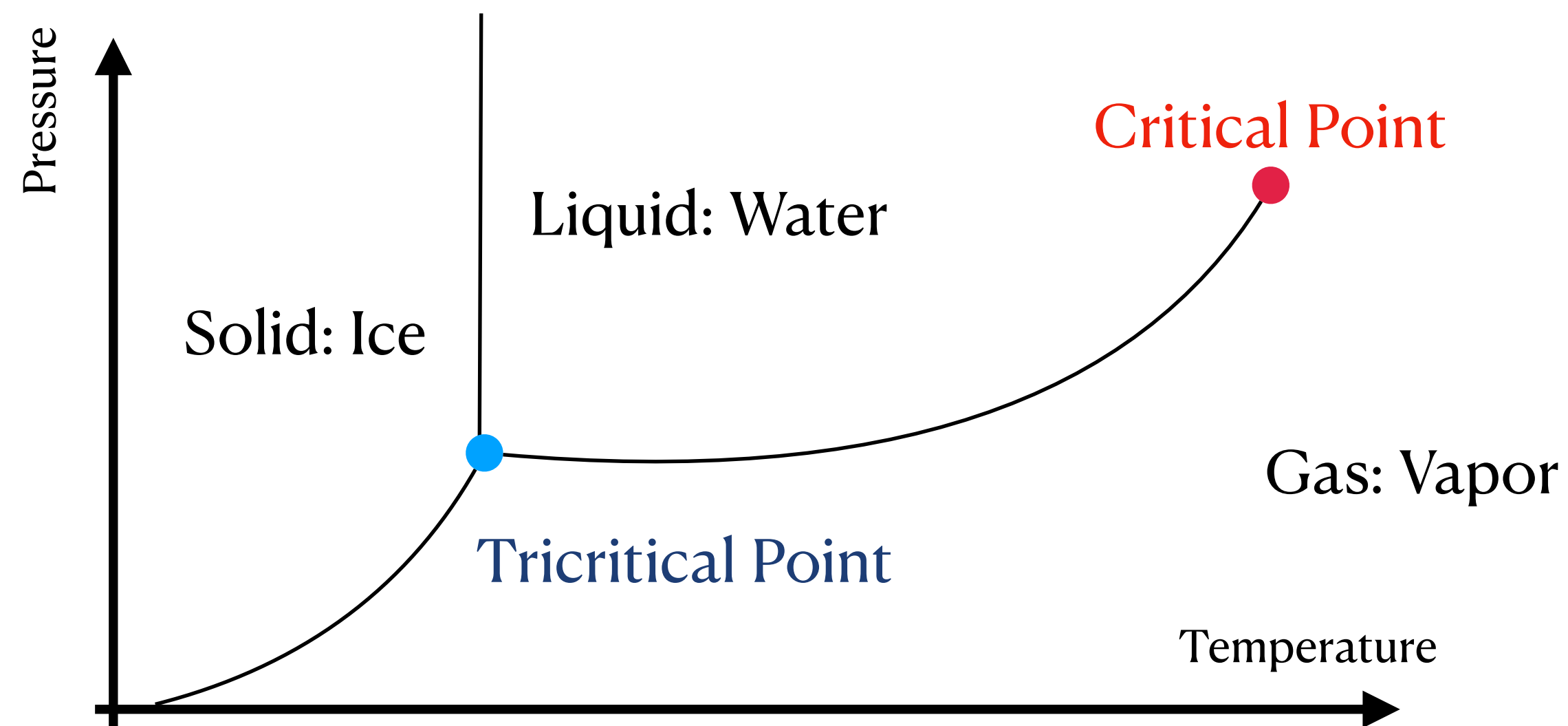
Facility for
Rare Isotope Beams
at Michigan State University

Introduction: Thermodynamic properties

Thermodynamics describes relation between the pressure, bulk volume, density/chemical potential and temperature.

$$\Omega(\mu, T, V) = -VP(\mu, T)$$

$$F(T, V, N) = U - TS$$



Transition from liquid to gaseous phase happens with a jump in density.

This is a case of the first order phase transition with density being first derivative of pressure:

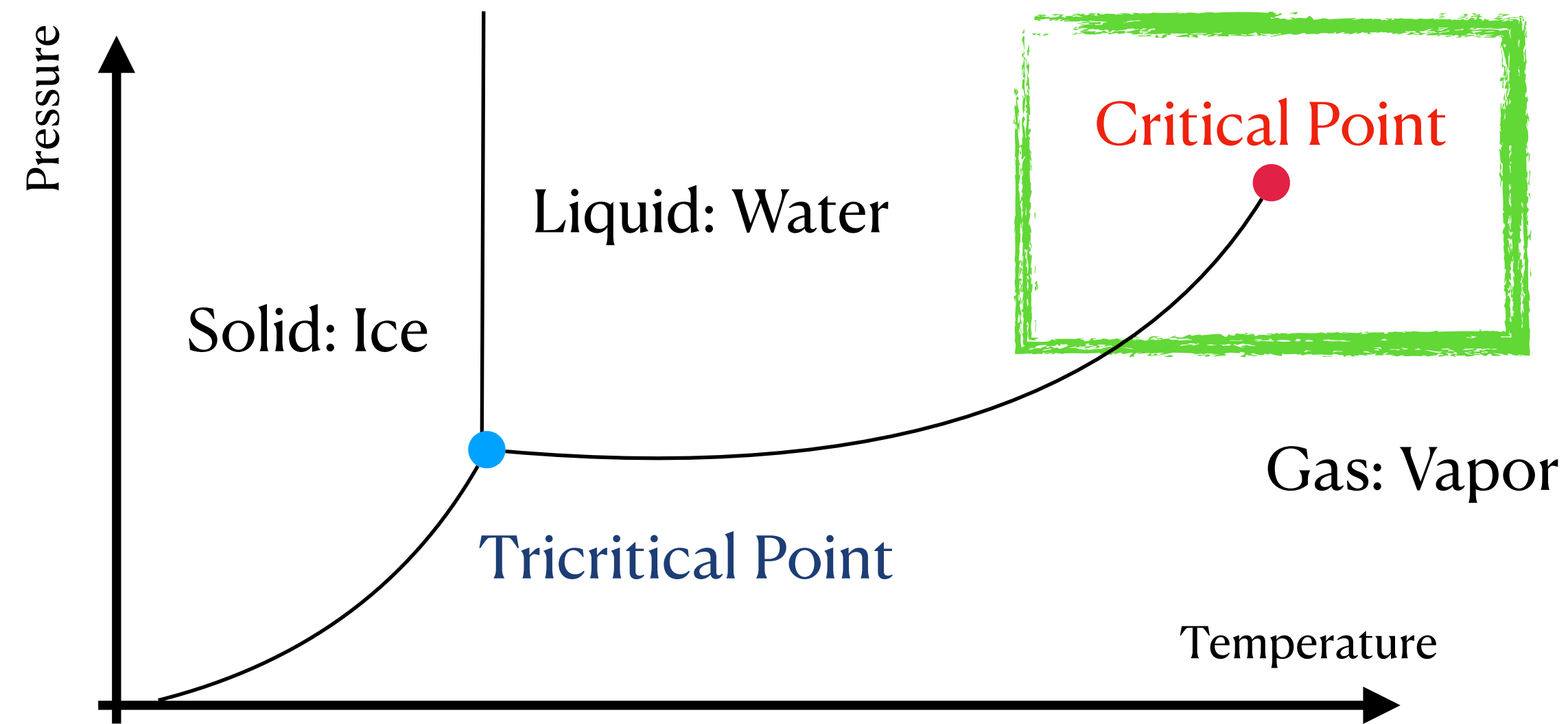
$$n = \frac{\partial P(\mu, T)}{\partial \mu}$$

The most basic knowledge we have about water is when it boils or when it freezes

There is also a second order phase transition when the jump happens in second derivative:

$$\frac{\langle n^2 \rangle - \langle n \rangle^2}{\langle n \rangle} = \frac{\mu}{T} \frac{\partial^2 P}{\partial \mu^2}$$

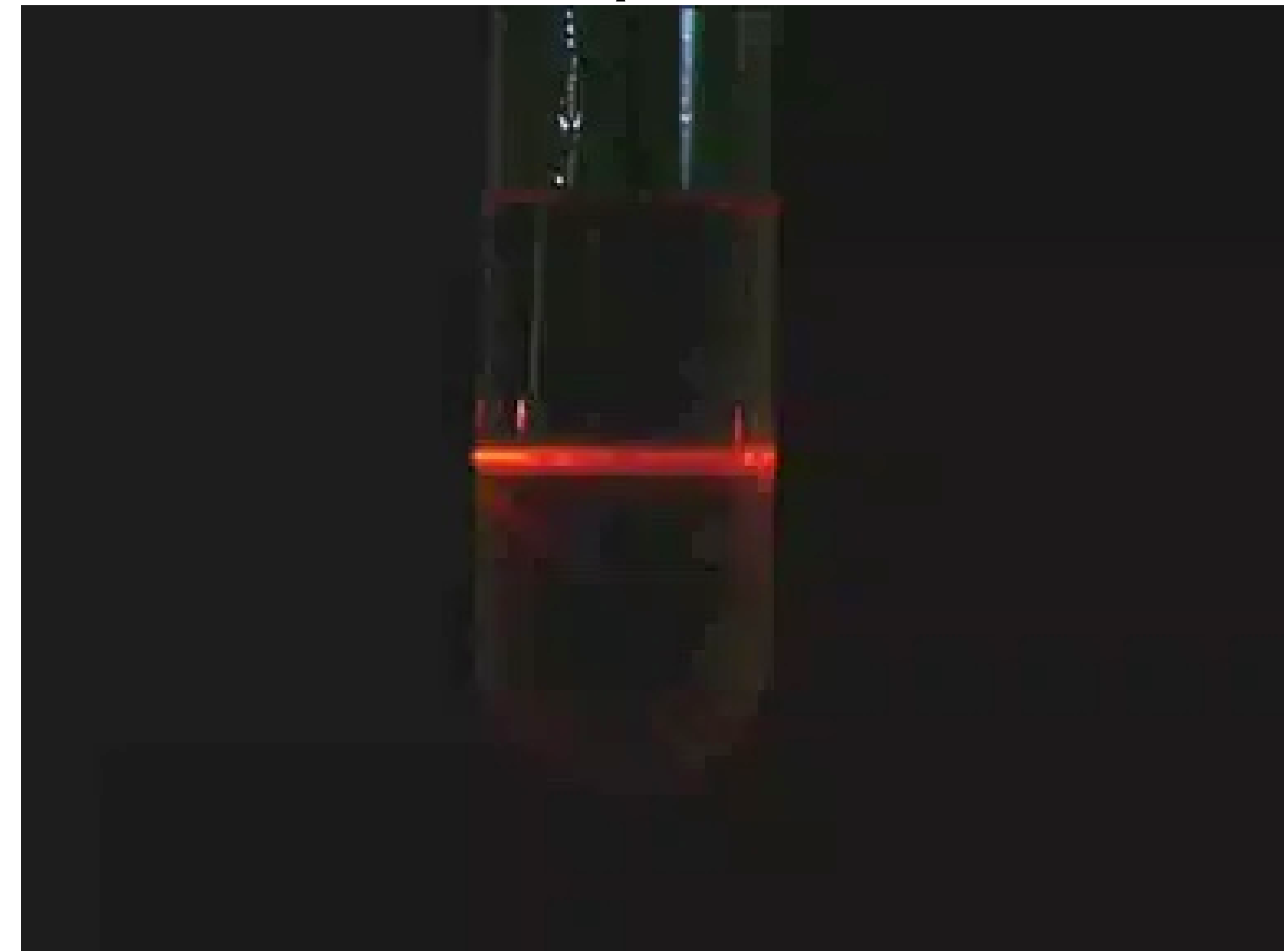
Introduction: Thermodynamic properties



Above critical point the transition between liquid and gas is smooth/continuous. At the critical point interesting physics occurs.

Large fluctuations and infinite correlation length!

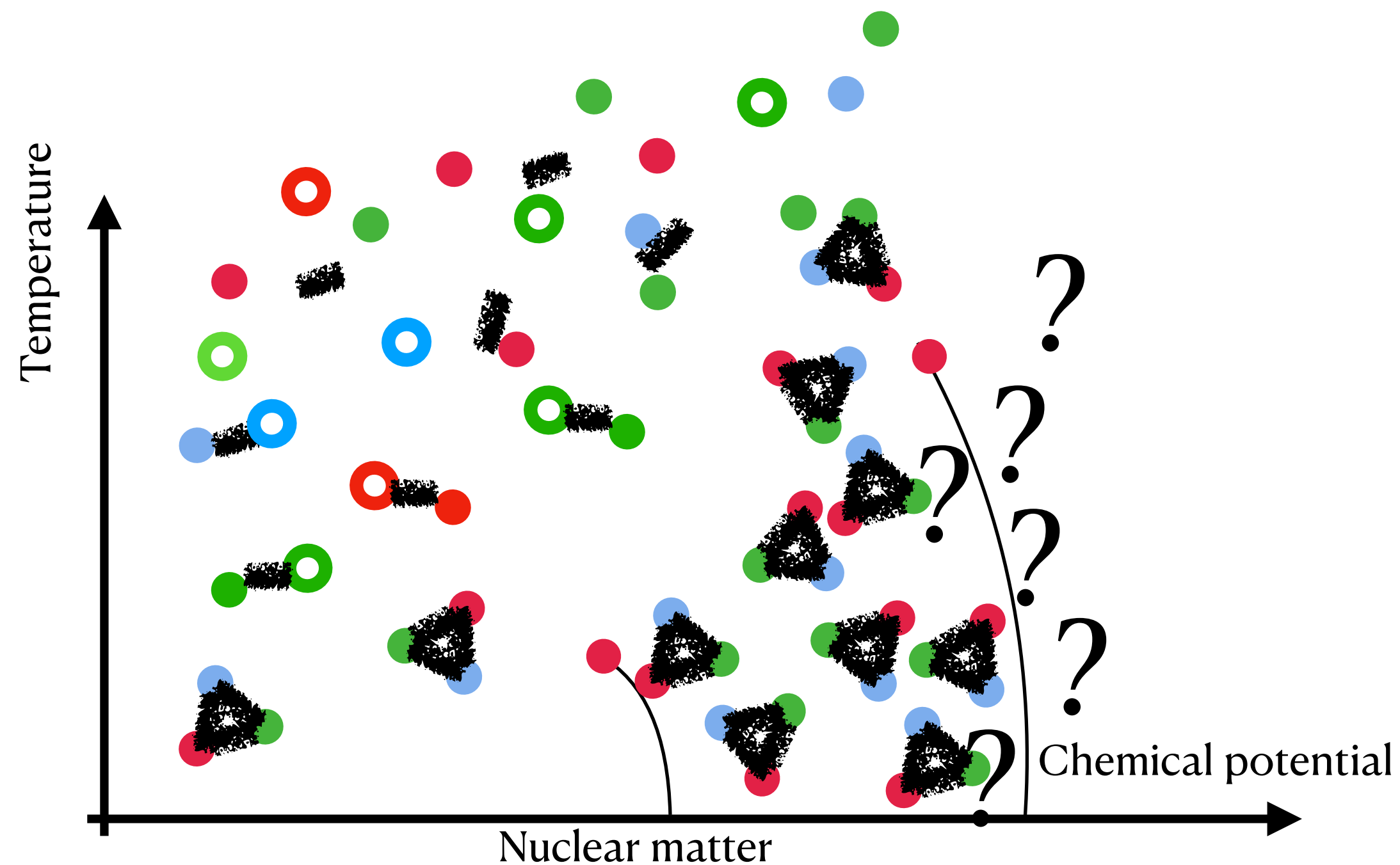
One observes critical fluctuations through critical opalescence:



[\[https://www.doitpoms.ac.uk/\]](https://www.doitpoms.ac.uk/)

Scattering of light on density fluctuations of the size of the wavelength of light.

Introduction: QCD Critical Point



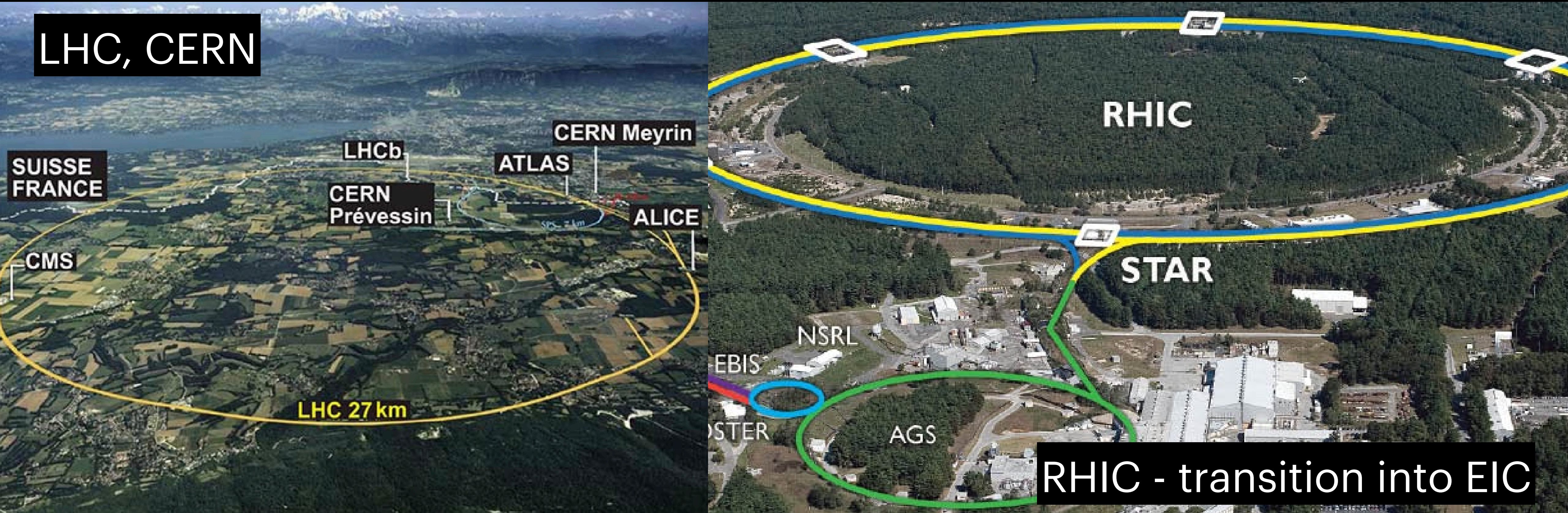
Finding Critical Point would imply existence of the first order phase transition

At the same time it is possible that critical point is easier to find by looking at signatures of criticality in a system.

Such signals might include fluctuations and correlations of conserved charges.

Strongly interacting matter is poorly understood.
We know that it has deconfined and confined phases.
But we know little of when and how they occur.

QCD is studied with Heavy-Ion Collisions

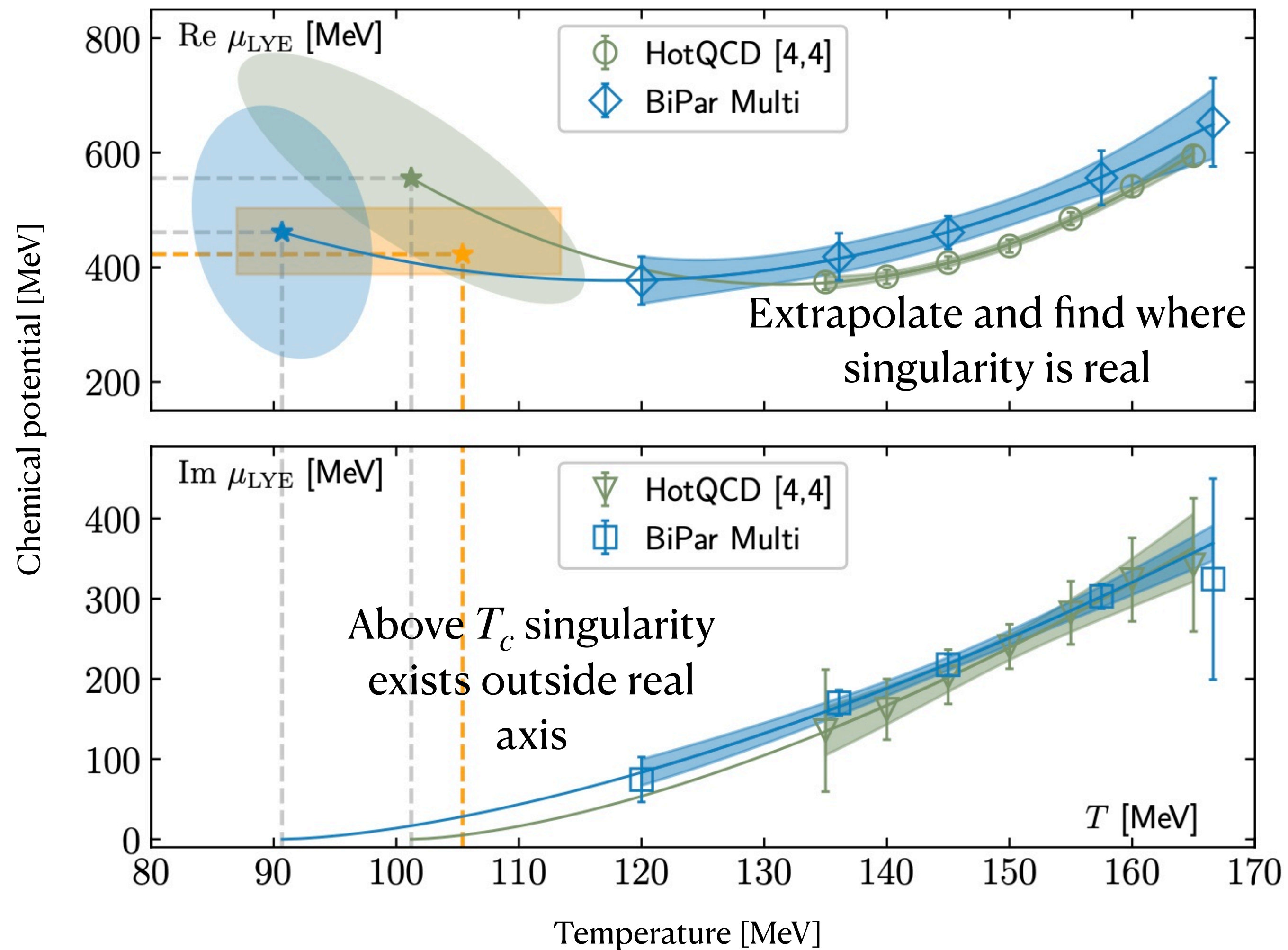
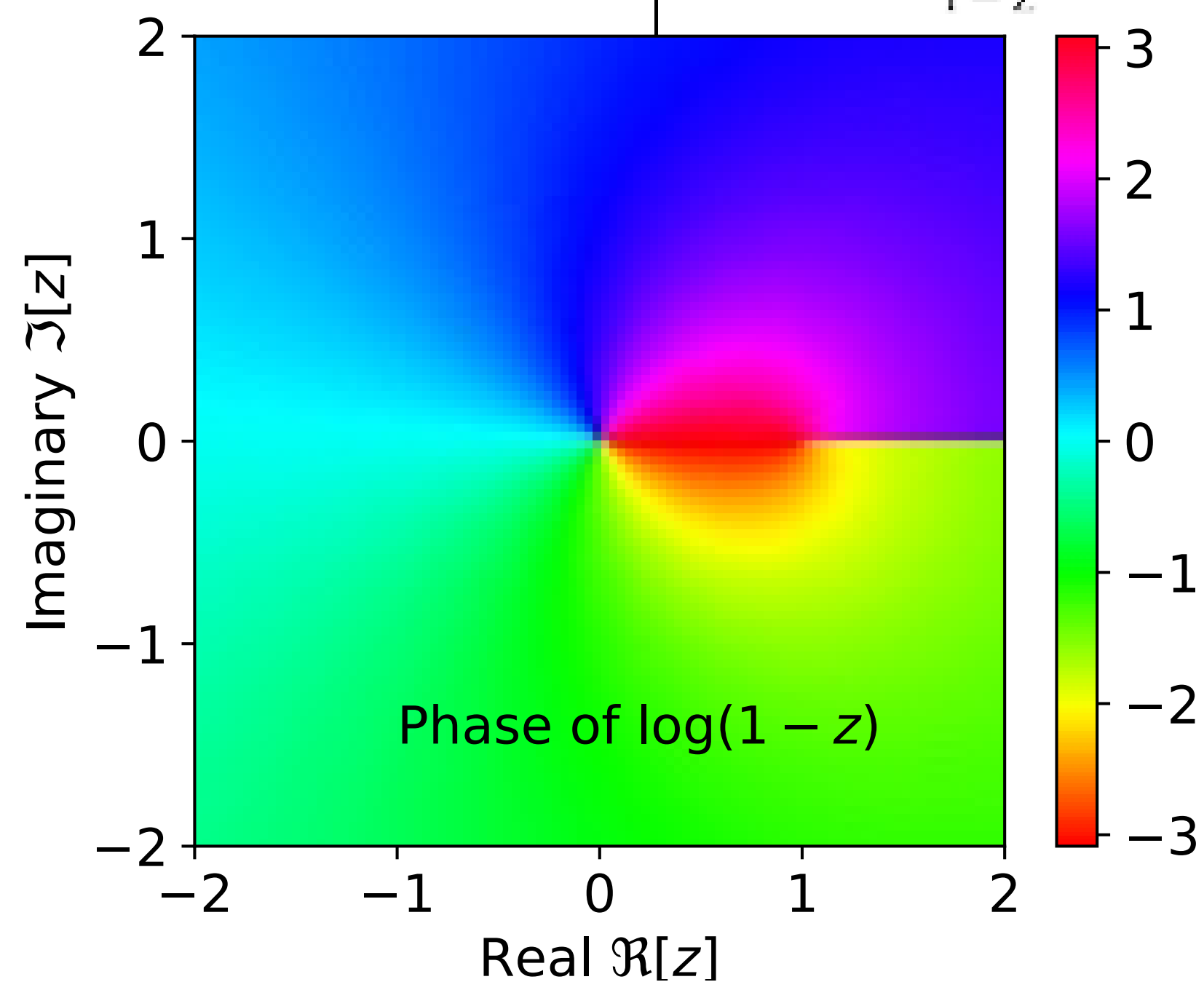
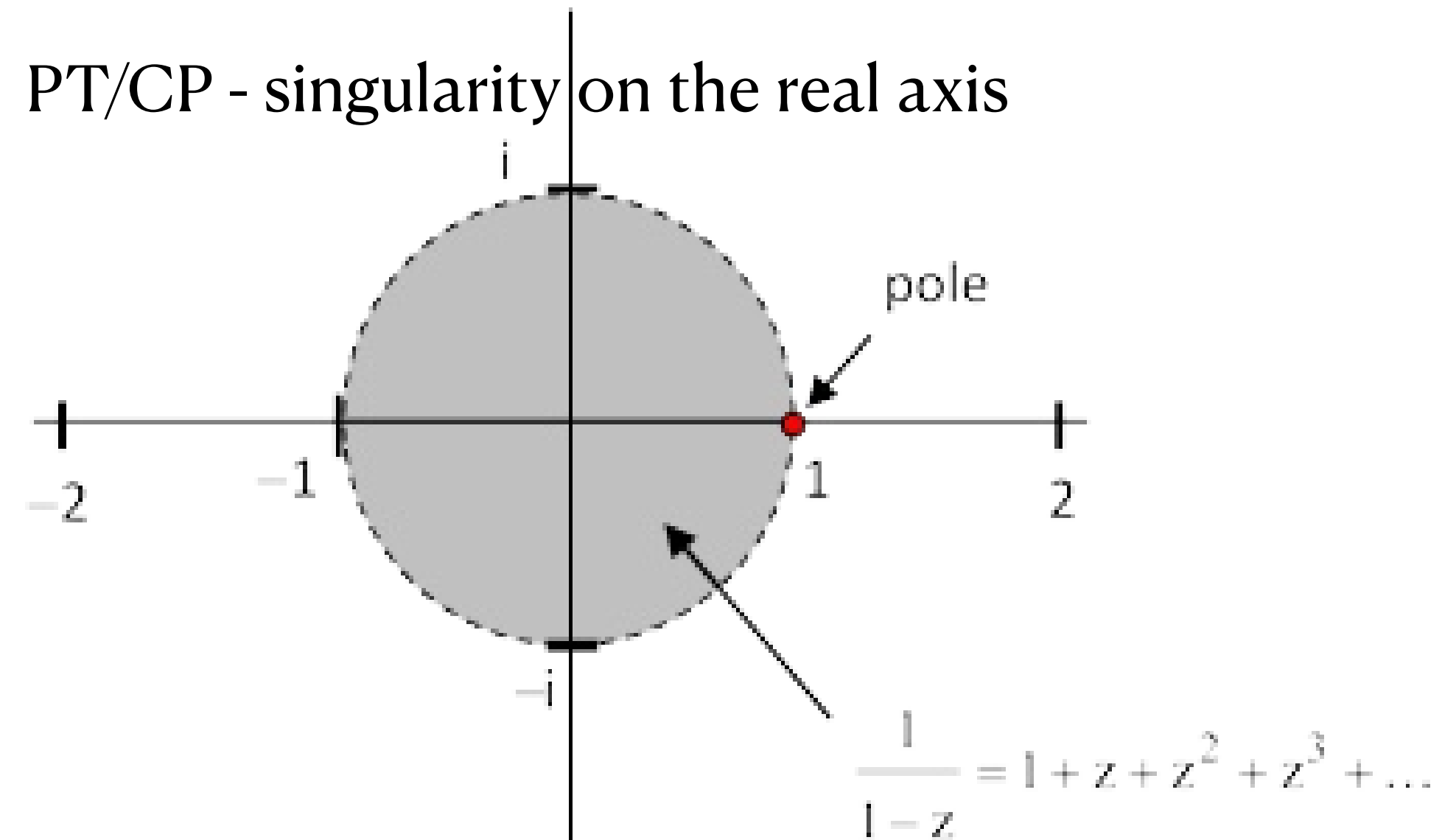


HIC create bulk of strongly interacting matter in the laboratory allowing to study its properties using various experimental observables

First principle IQCD

Calculate $P(T, \mu)$ on computer

PT/CP - singularity on the real axis

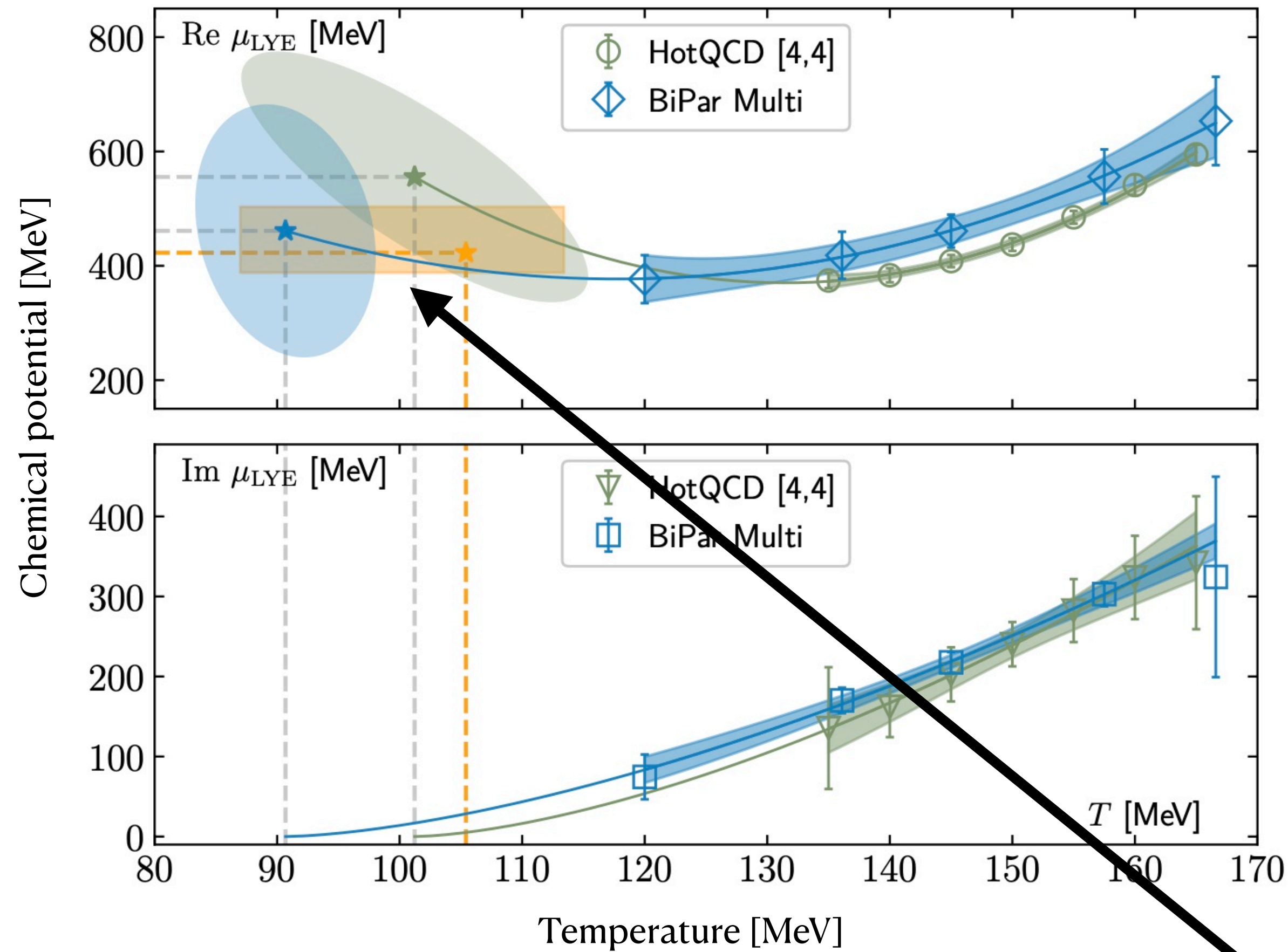


[D.A. Clarke et al. (Bielefeld-Parma), [arXiv:2405.10196](#)]

also [Gokce Basar, [arXiv:2312.06952](#)]

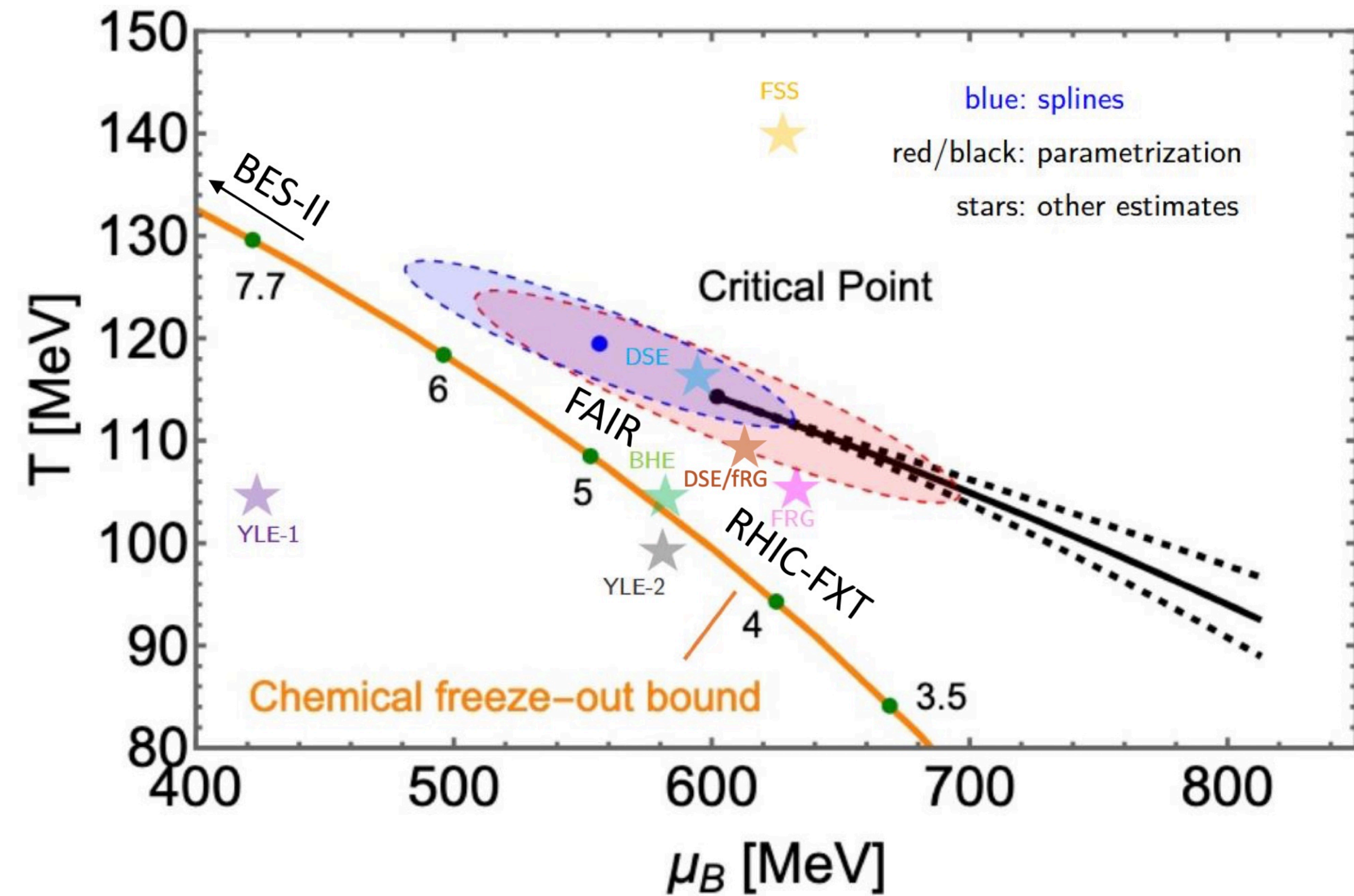
QCD Energy Scan

Extrapolations from lattice simulations at zero density:



[D.A. Clarke et al. (Bielefeld-Parma), arXiv:2405.10196]
also [Gokce Basar, arXiv:2312.06952]

Prediction for heavy-ion freeze-out and CP:

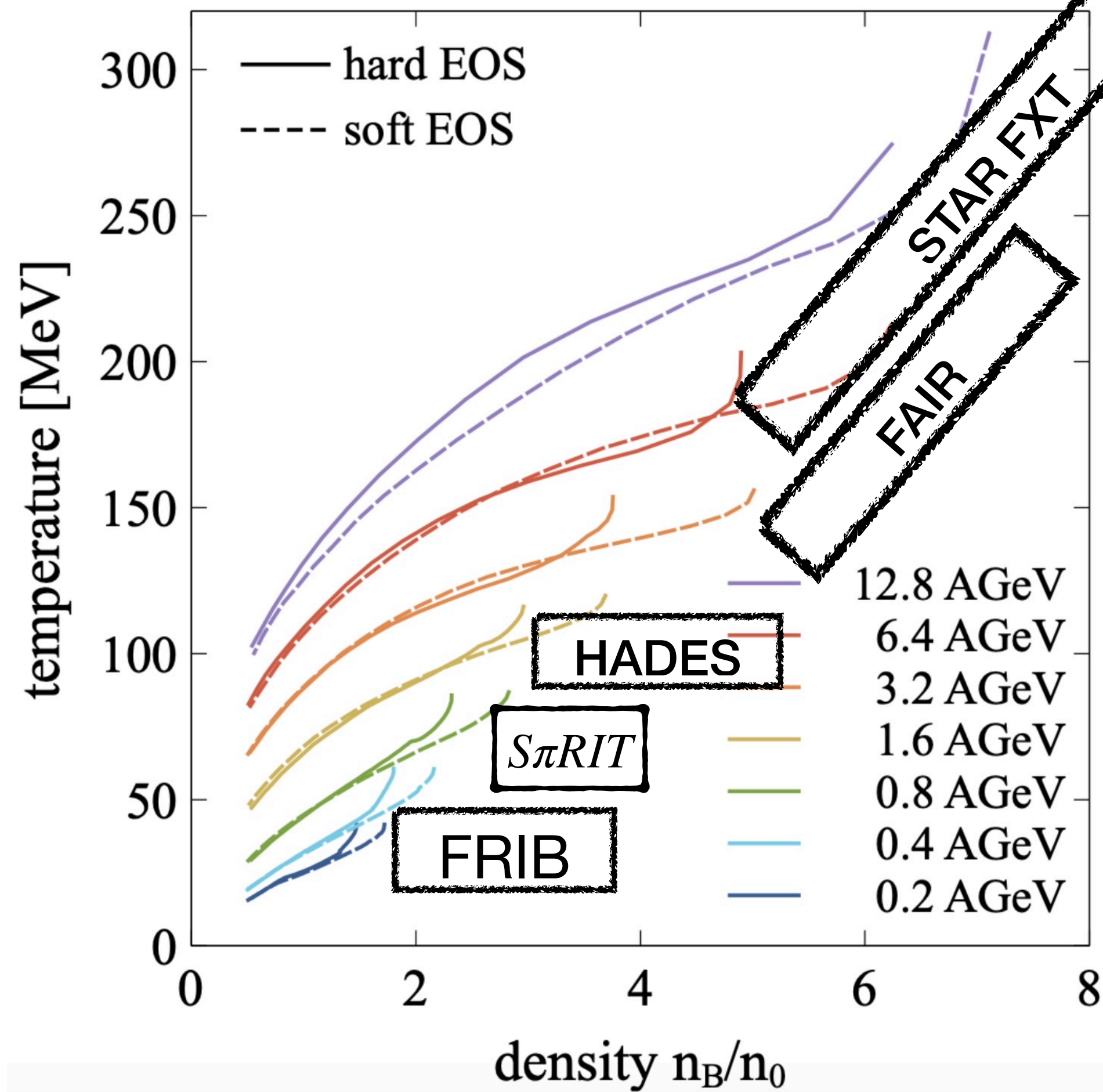


[Hitansh Shah, QM2025]

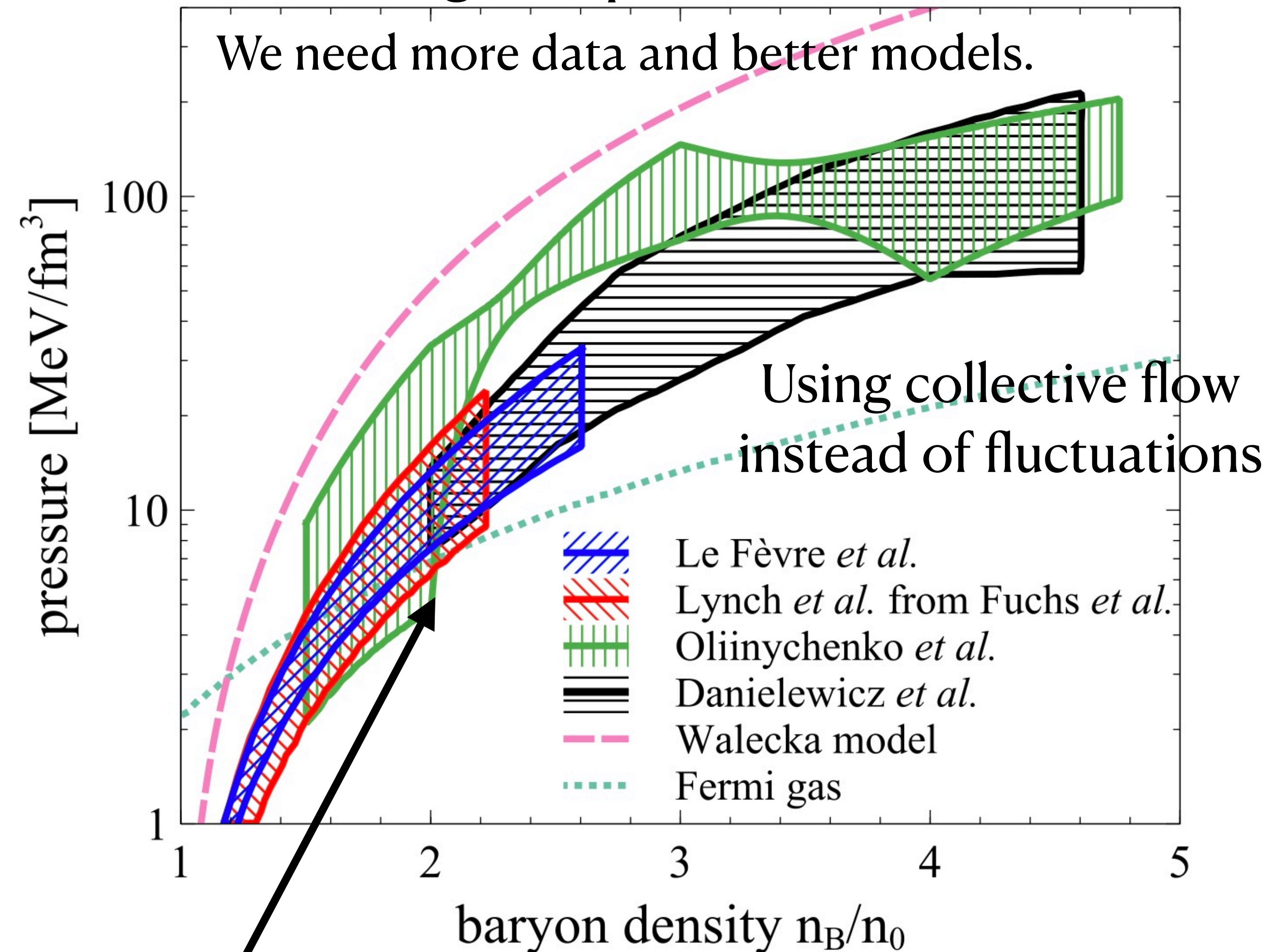
Critical point is expected at $\frac{\mu_B}{T} = 4 - 6$

Dense Nuclear Matter Equation of State

By going to lower energies - critical point.
Third(?) major push in energy scan



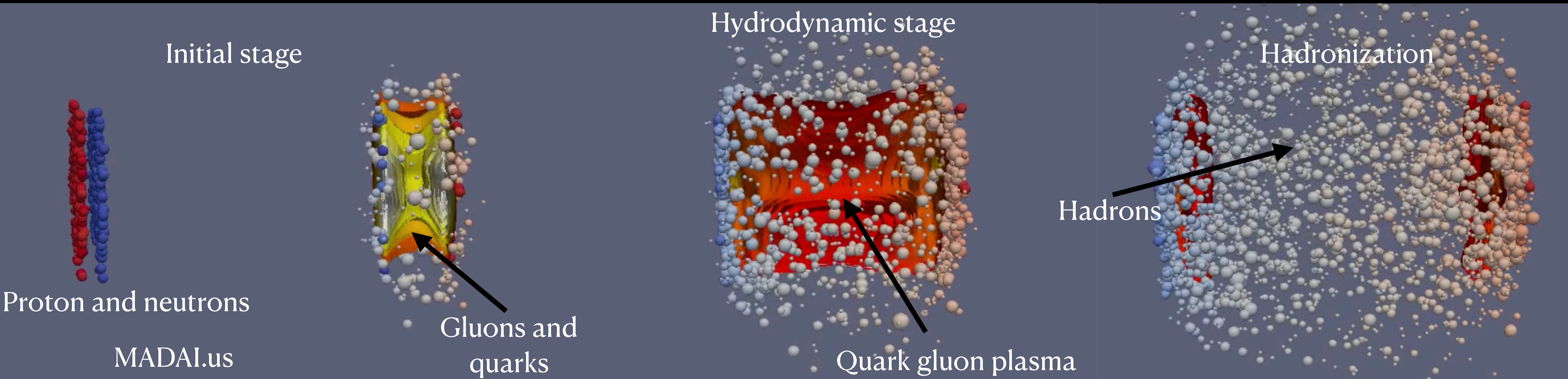
Equation of state can be constrained
using transport simulations



baryon density n_B/n_0
[Prog. Part. Nucl. Phys. **134**, 104080 (2024)]

Is there a room for a phase transition?

Stages of Heavy-Ion Collisions ($\sqrt{s} > 7.7 \text{ GeV}$)



Profiles of conserved
charges:

Energy-momentum, baryon,
electric, strange

Collective expansion and
fluctuations of thermal medium.
Out-of-equilibrium/memory
effects.

Quarks form hadrons.
Correlations and fluctuations of
charges freeze-out.

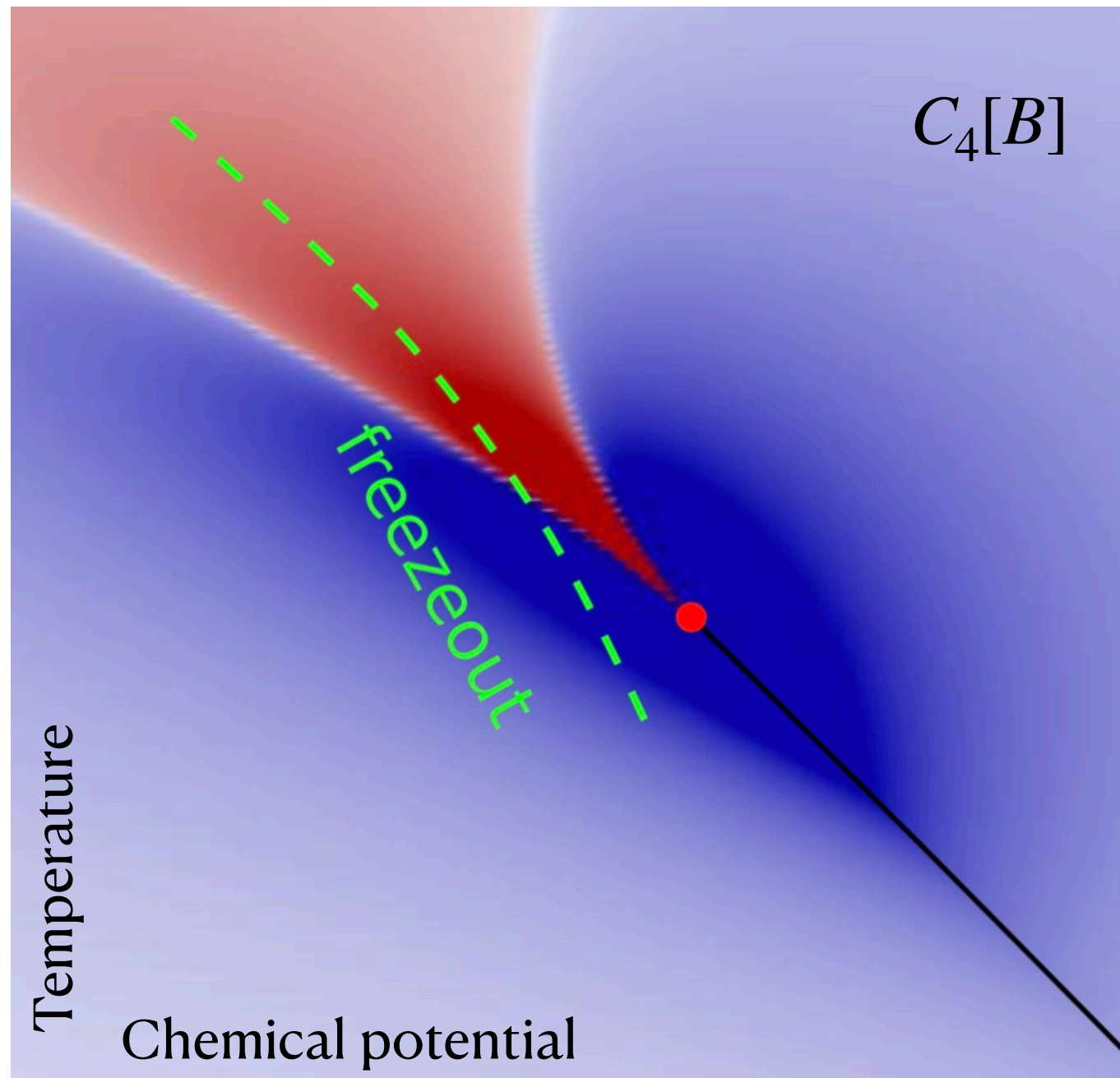
Can fluctuations during initial stages survive until freeze-out?

Susceptibilities

$$\frac{P}{T^4} = \frac{1}{VT^3} \ln \mathcal{Z}(T, V, \hat{\mu}_u, \hat{\mu}_d, \hat{\mu}_s) = \sum_{i,j,k=0}^{\infty} \frac{\chi_{ijk}^{BQS}}{i!j!k!} \hat{\mu}_B^i \hat{\mu}_Q^j \hat{\mu}_S^k,$$

- Fluctuations of baryon charge are connected to the equation of state as cumulants:

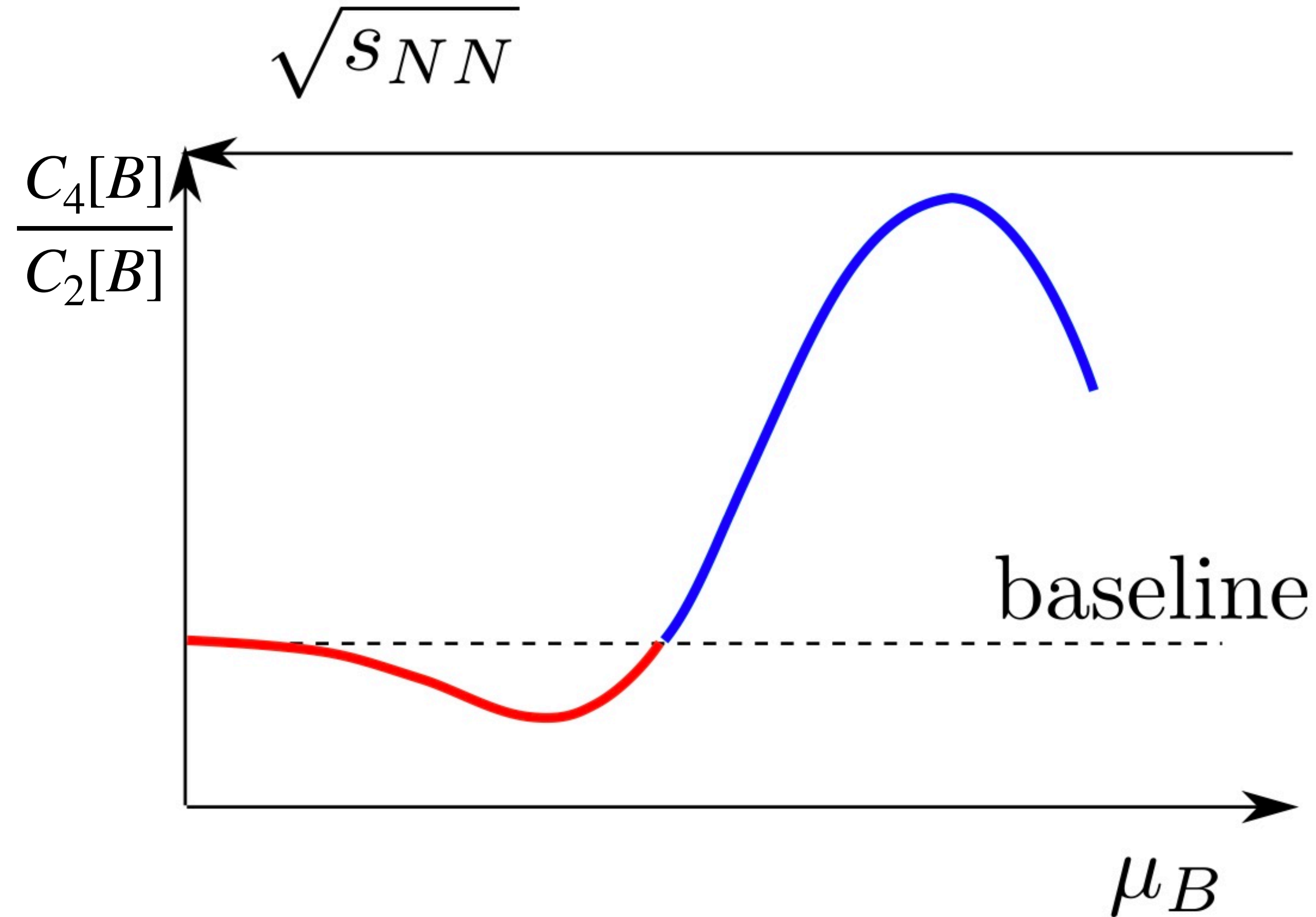
$$C_n[B] = \chi_n^B VT^3$$



[M. Stephanov, Phys. Rev. Lett. **102**, 032301]

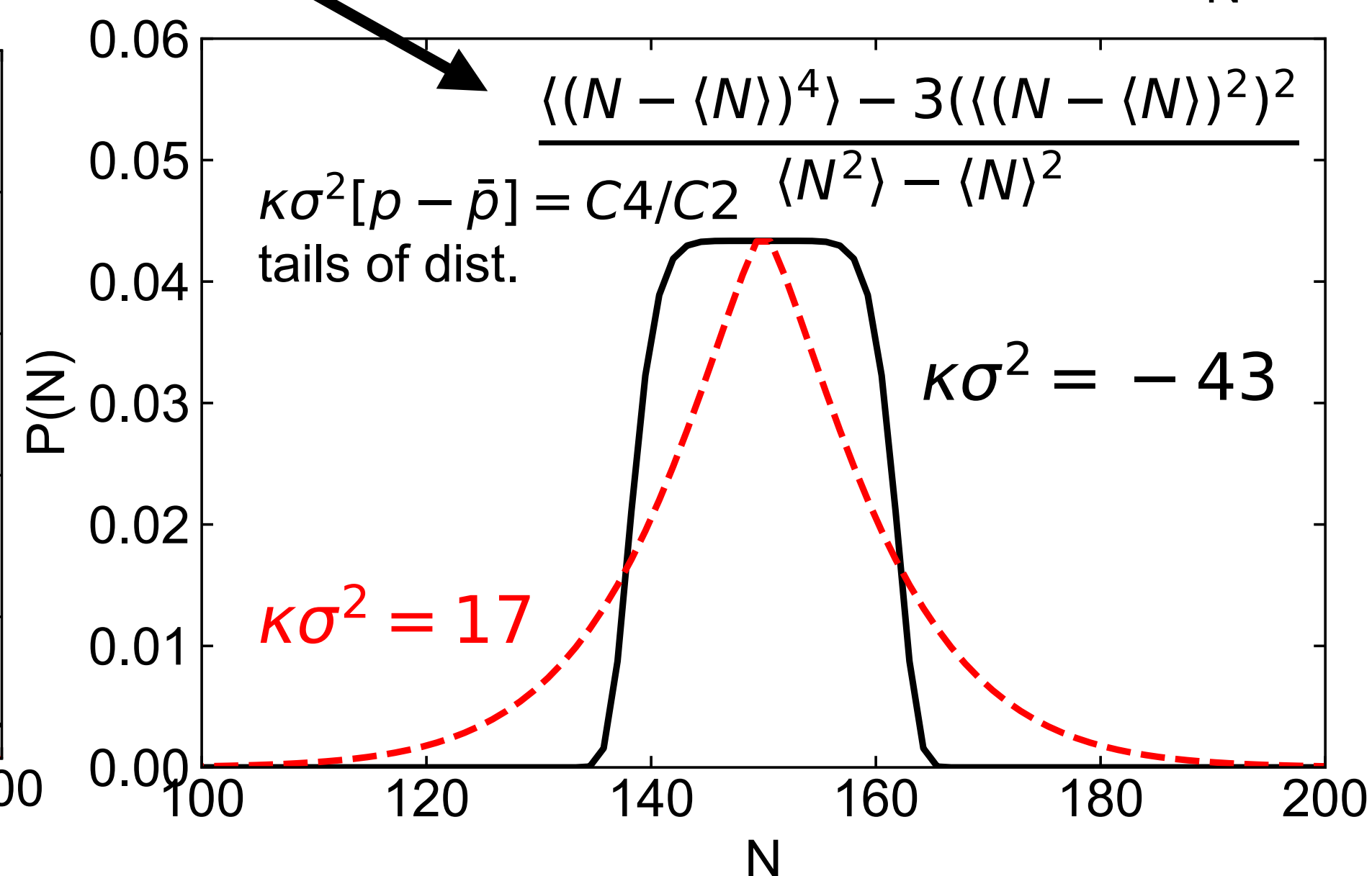
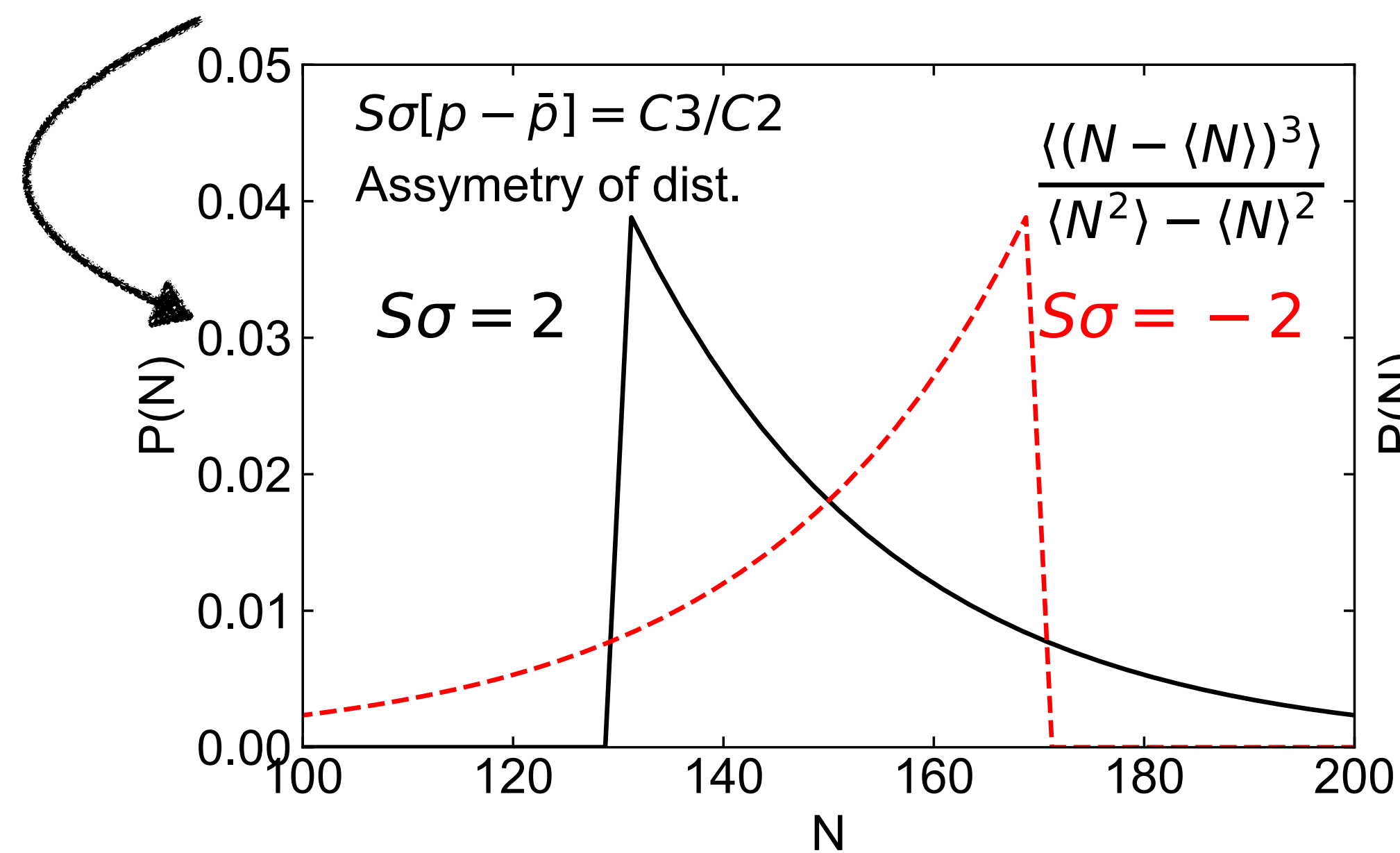
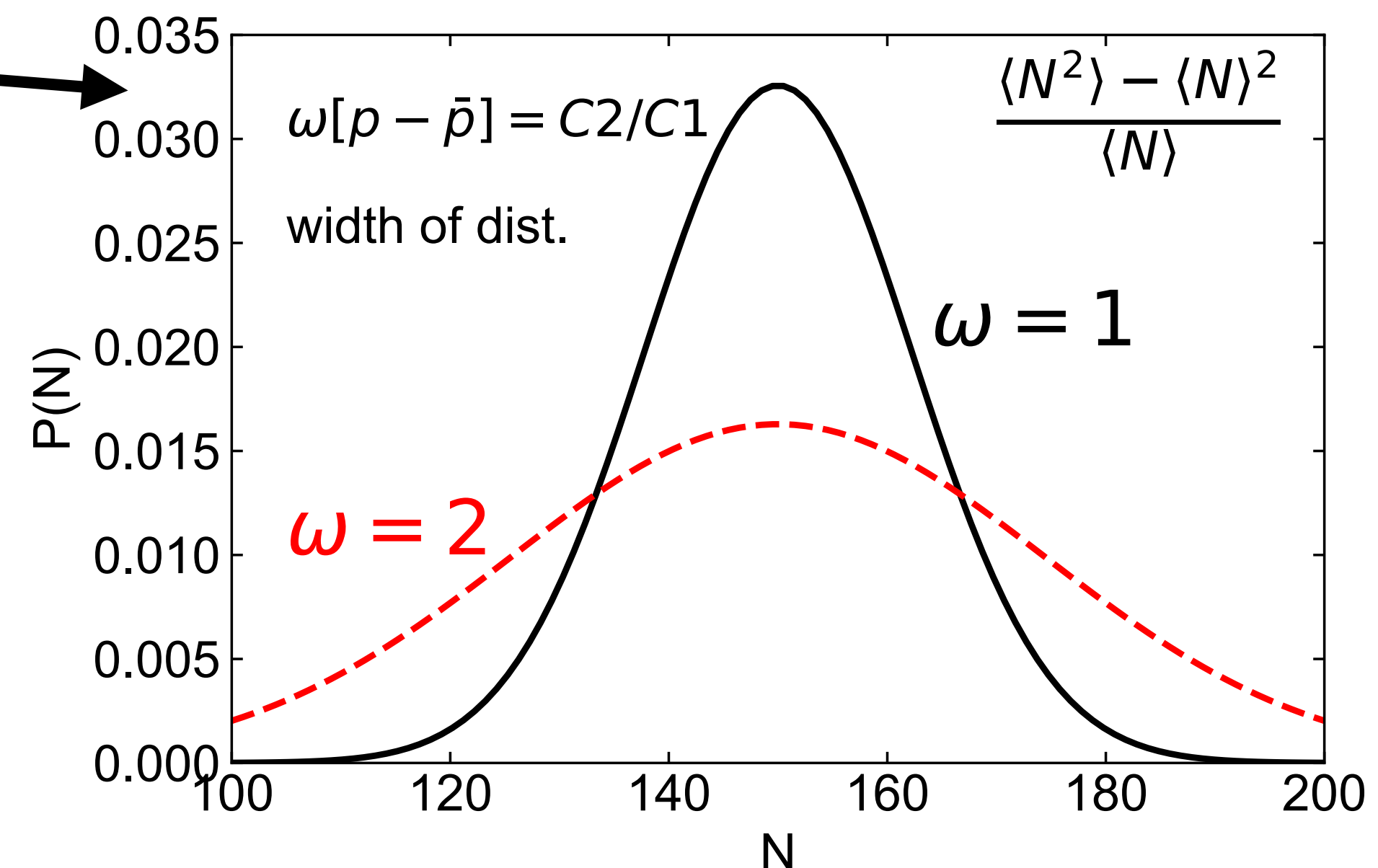
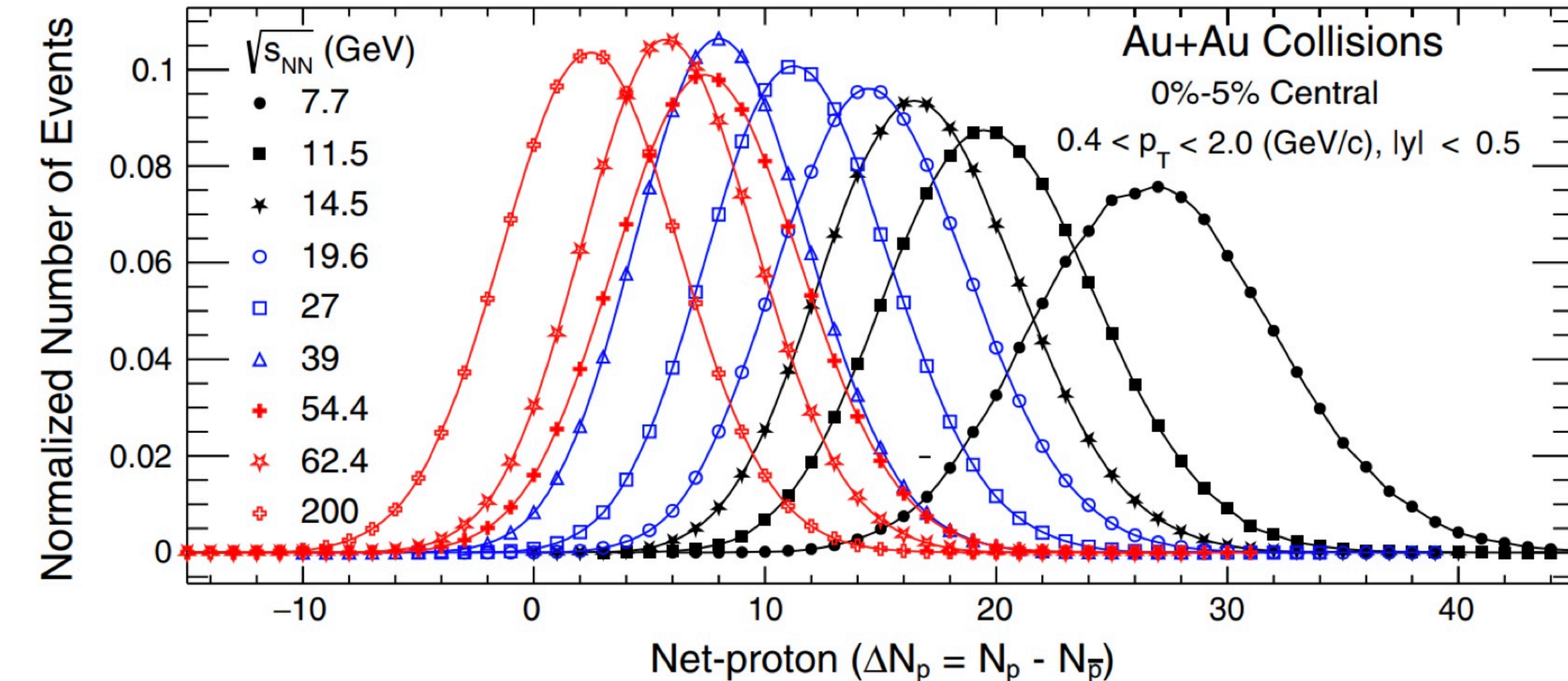
[Adam Bzdak, Shin'ichi Esumi, Volker Koch, Jinfeng Liao, Mikhail Stephanov, Nu Xu, Physics Reports 853 (2020) pp. 1–87]

- Cumulant ratios are non-monotonic and show strong sensitivity to microscopic structure of theory/phase transitions.
- If equilibrium is an approximation what is conserved charge dynamics?

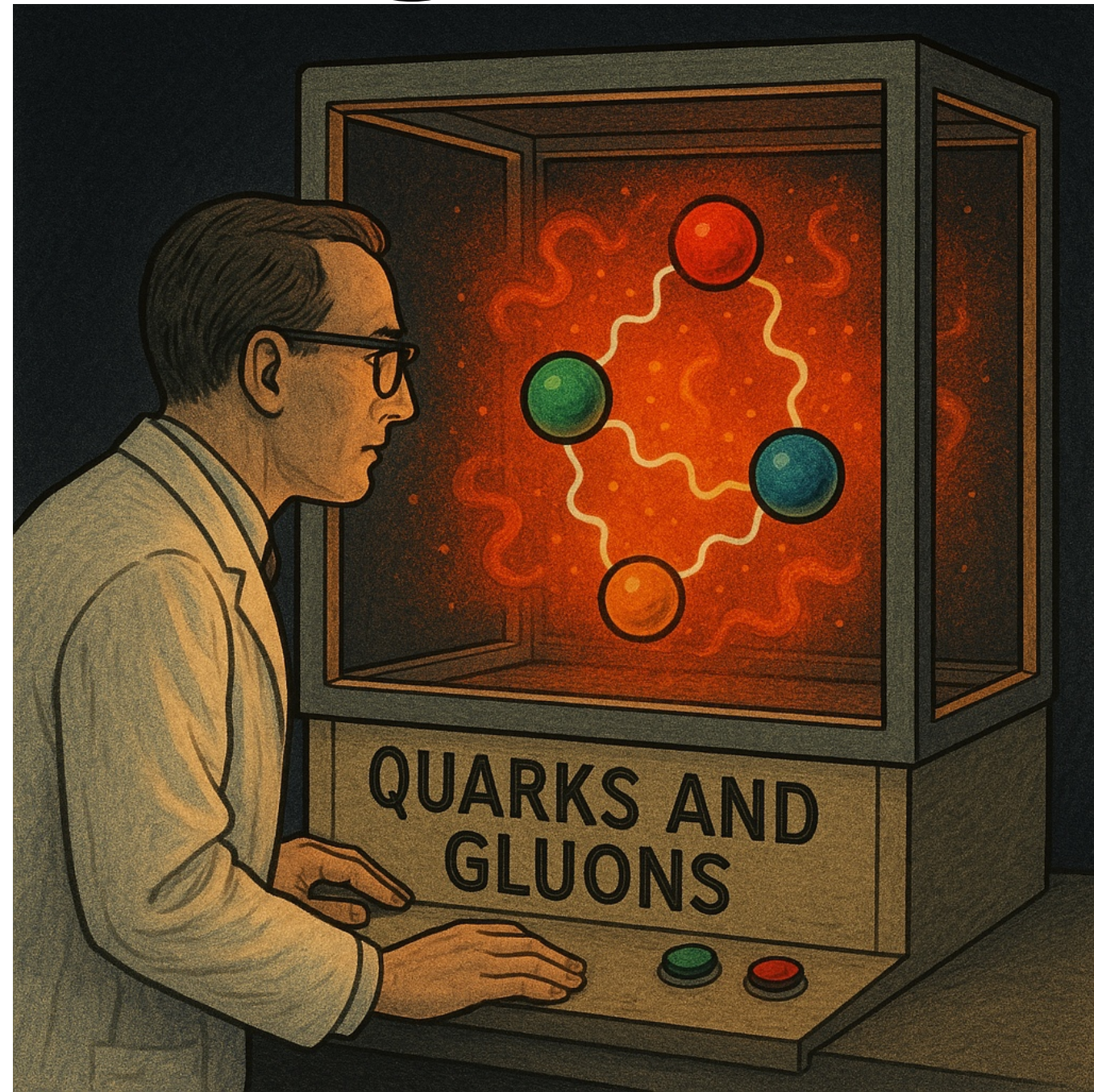


Cumulants of net-proton number

STAR Collaboration, Phys. Rev. Lett. 126, 092301 (2021)

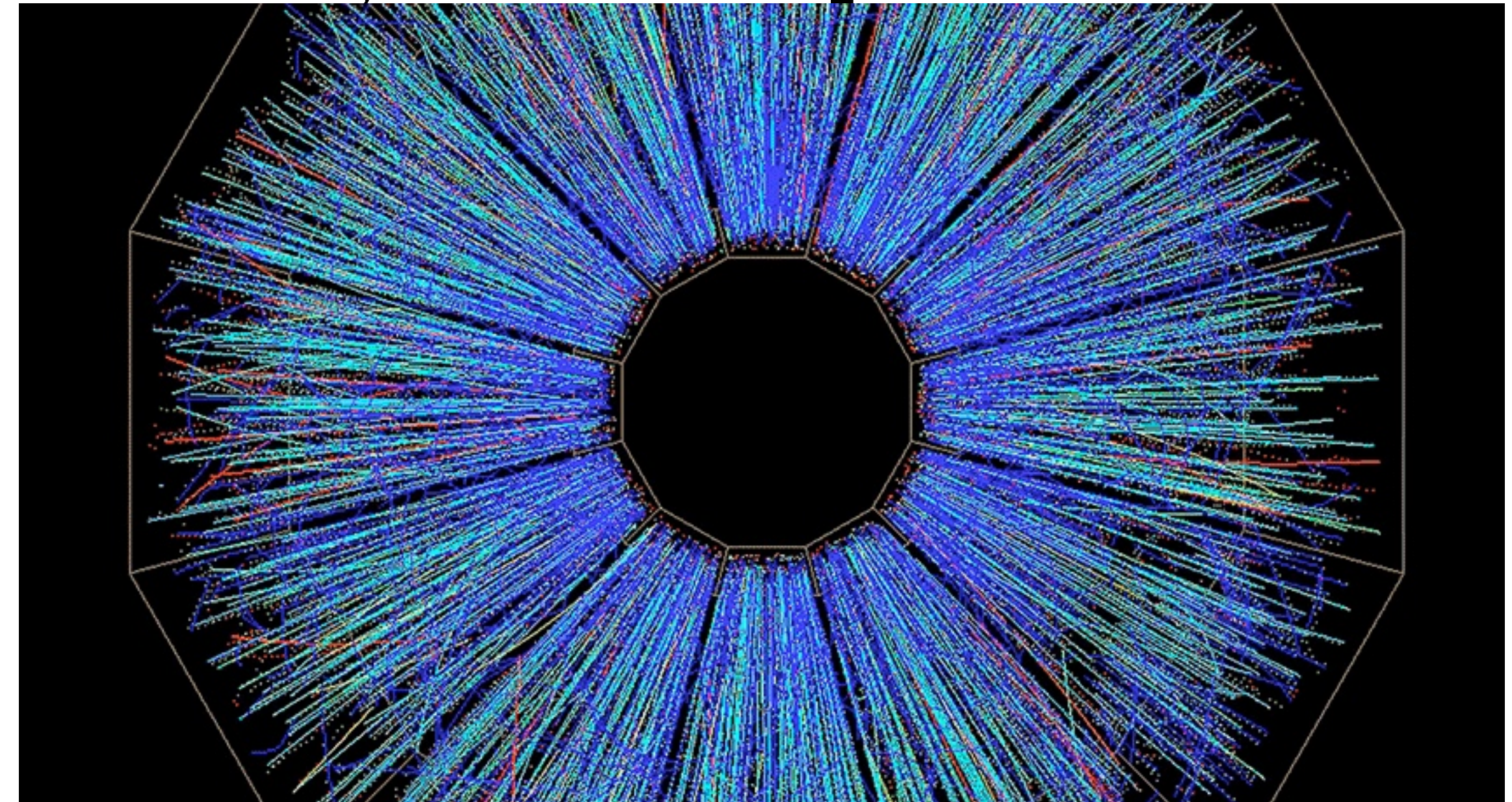


Challenges in comparing theory and experiment



Theory:

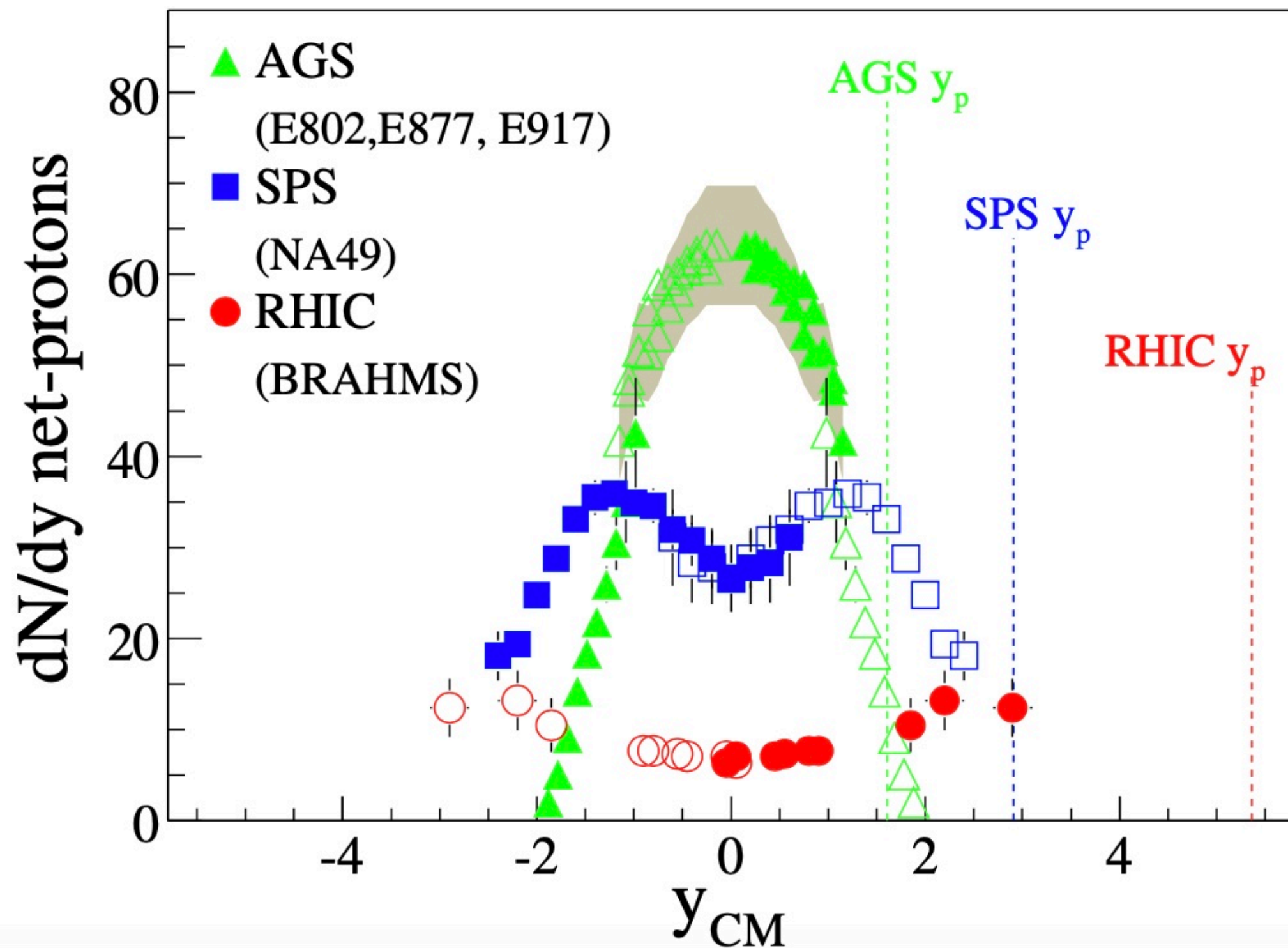
- Coordinate space
- Energy-momentum and baryon, electric, strange charges are conserved on average
- Matter at stationary equilibrium
- Ensemble is well controlled



[star.bnl.gov]

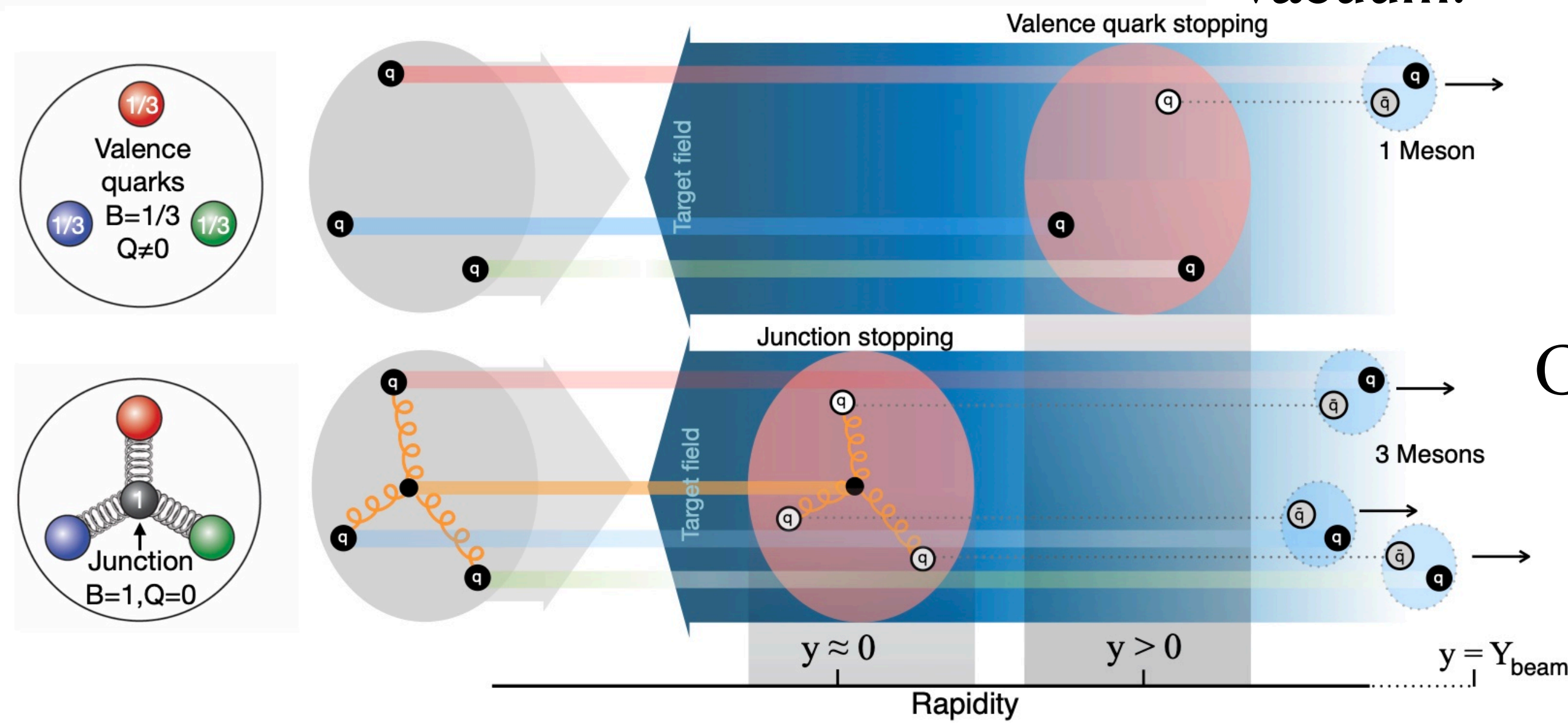
Experiment:

- Momentum distribution of charged particles
- Energy-momentum and baryon, electric, strange charges are conserved exactly
- Dynamic evolution
- Mixture of different event classes over various initial conditions



Baryon Stopping

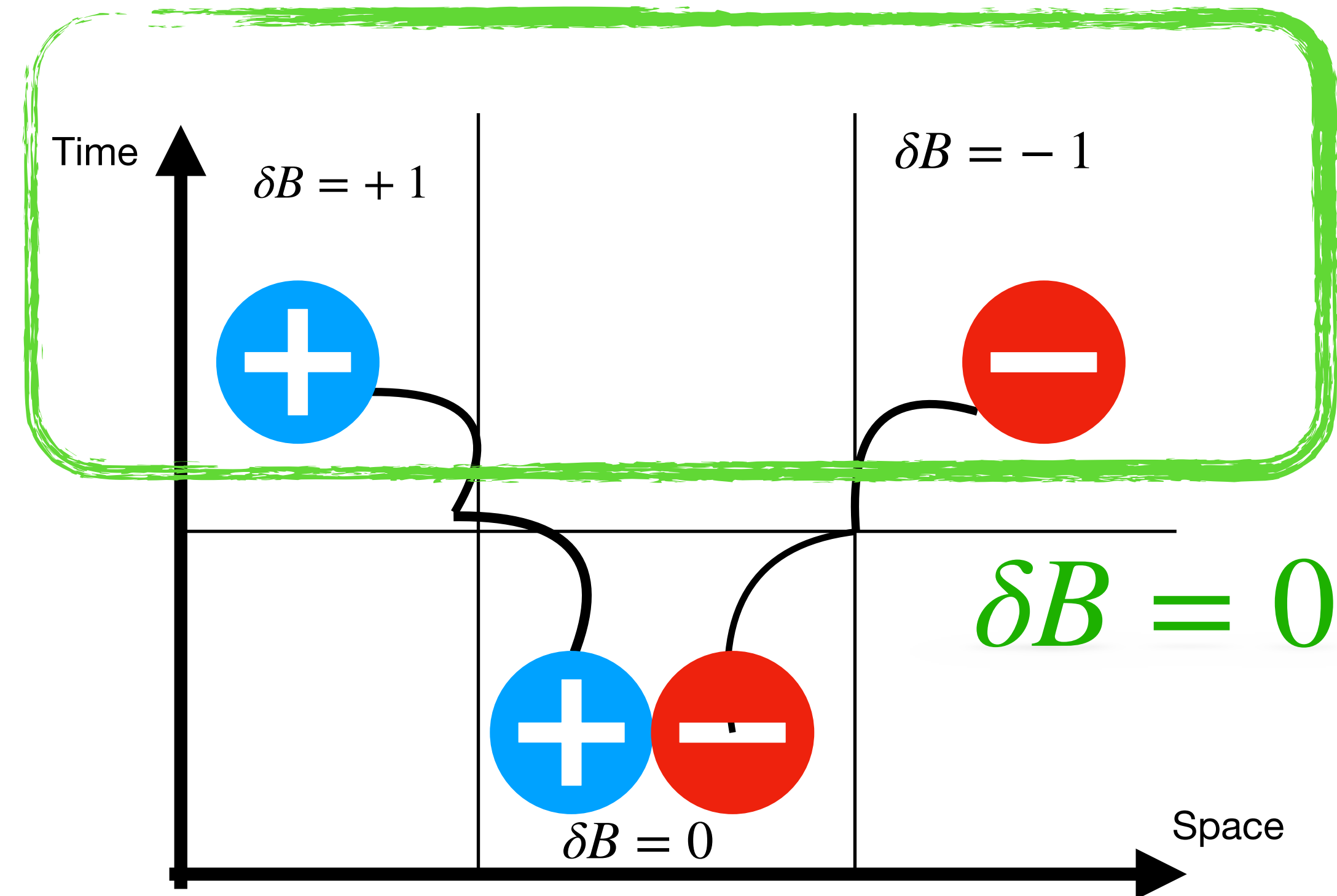
- At the very beginning baryon charge fluctuations are from stopping of proton and neutron, charge is transferred from beam rapidity to the mid-rapidity.
- Stopped charge is well approximated with net-protons.
- Qualitative change of rapidity spectra with energy - different physics!
- Very important at low energy (for fluctuations).
- Less important at high energy - charge produced in pairs from vacuum.



[STAR Collaboration, Do quarks or gluons carry baryon number?]

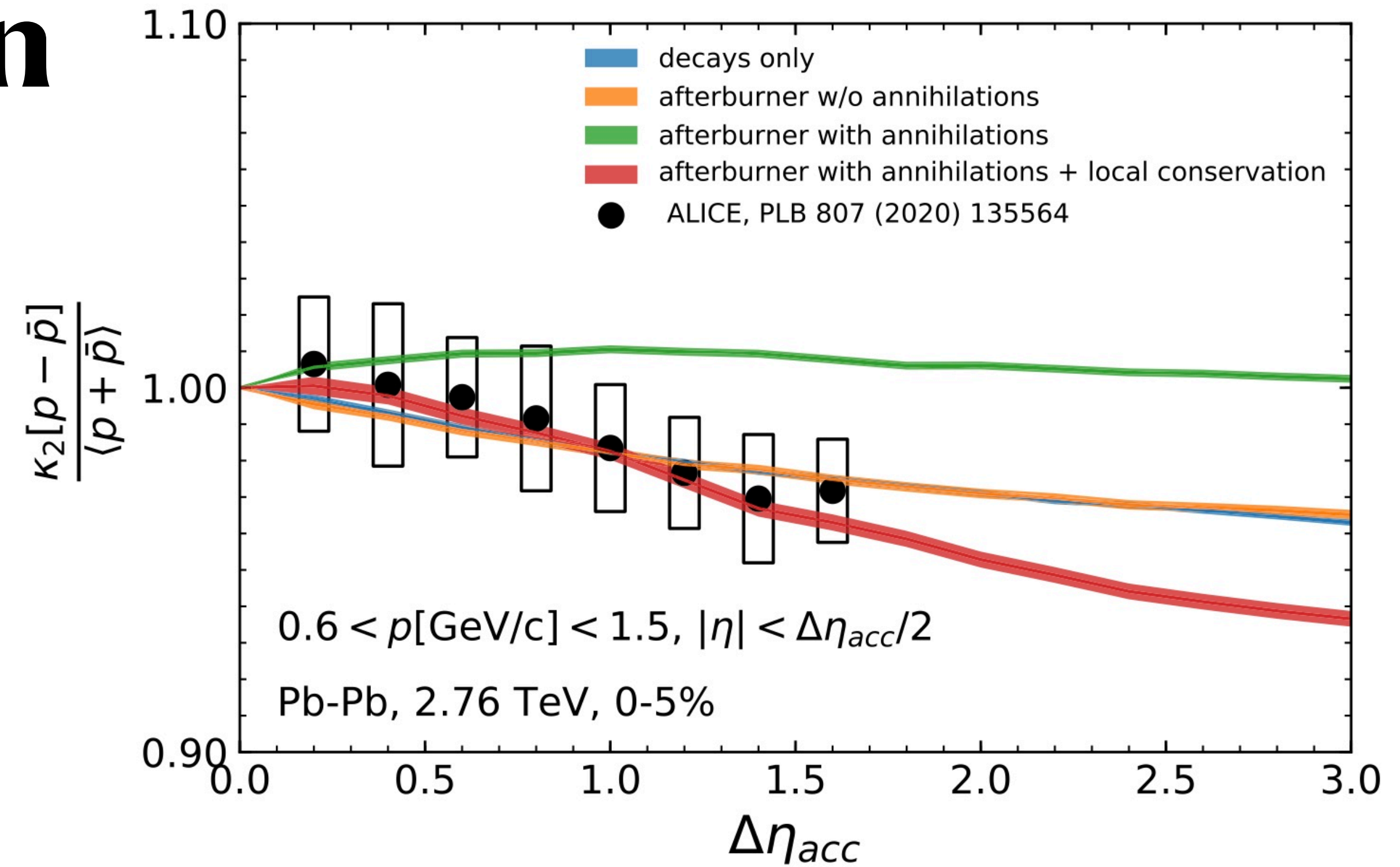
Can fluctuations be used to tell the difference?

Local charge conservation



[Scott Pratt, Rachel Steinhorst Phys. Rev. C **102**, 064906]
 [OS, J. Phys. G: Nucl. Part. Phys. 52 035106]
 [OS, Scott Pratt Phys. Rev. C **109**, 024910]
 [J. Parra et al., Physical Review Letters 135 (24), 242302]

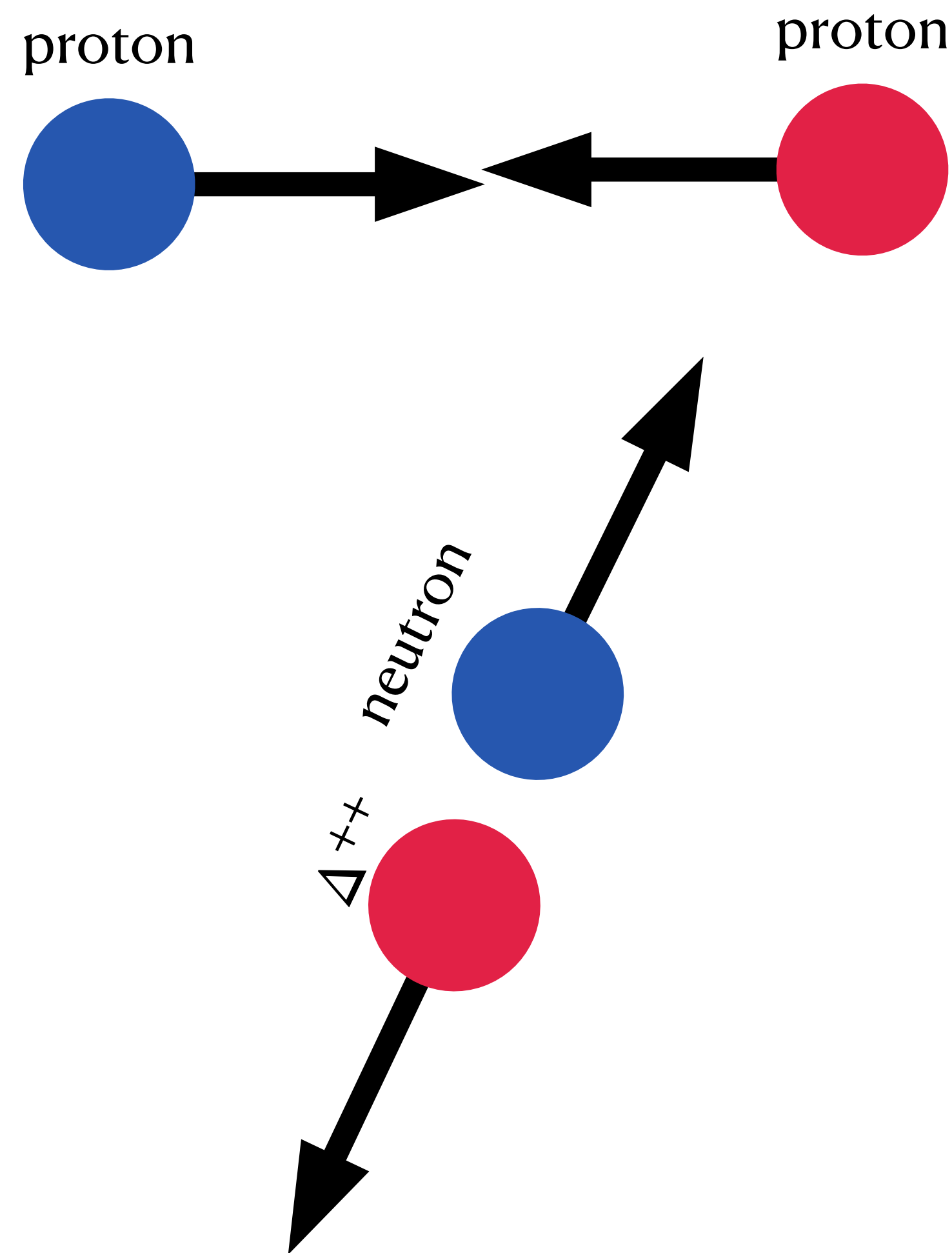
Local conservation important at high energies



[OS, et al. Physics Letters B 827 (2022) 136983]

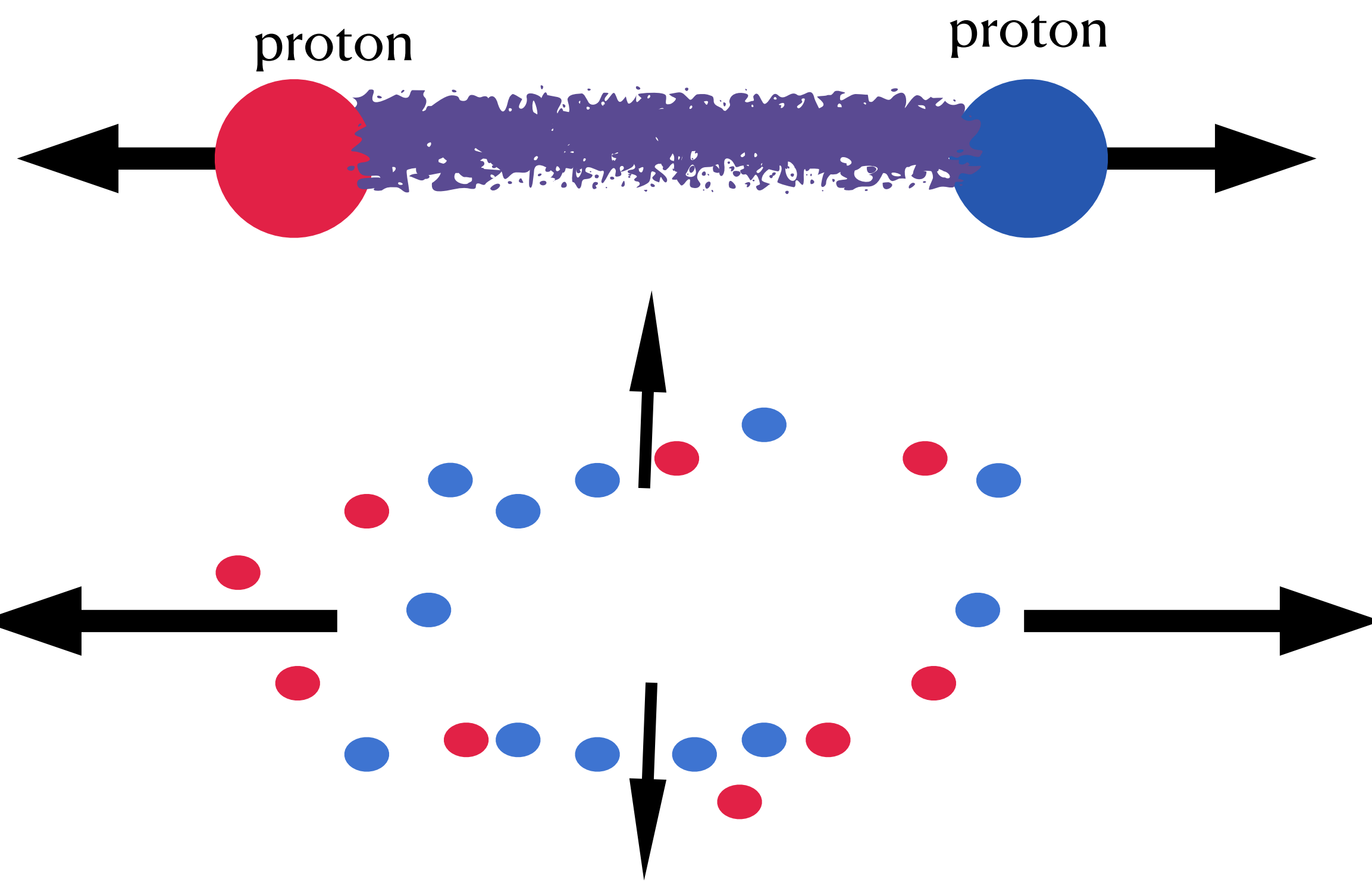
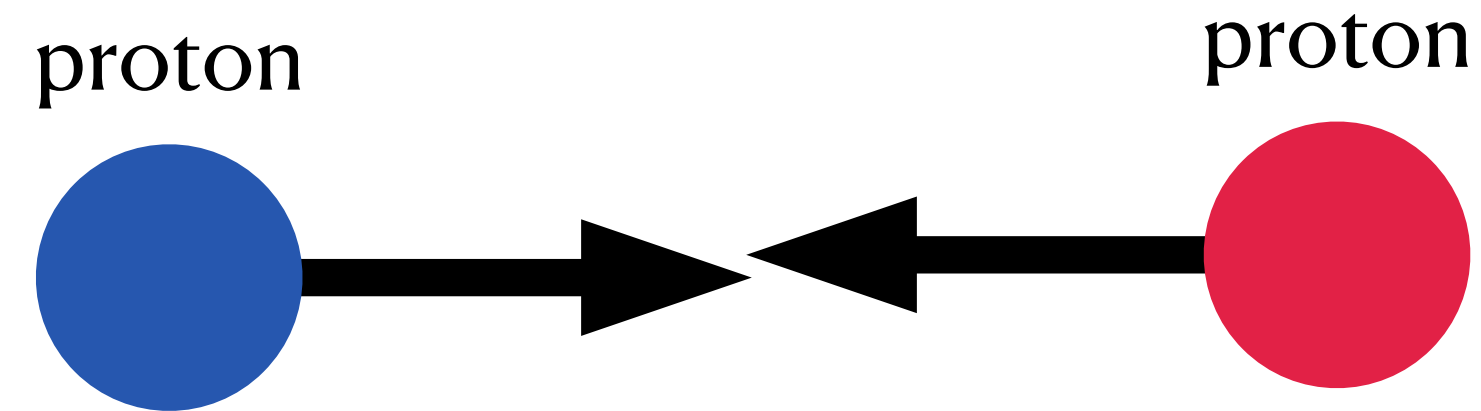
- Apart from stopped charge, particle anti-particle pairs can be produced dynamically in a collision.
- To affect fluctuations in a given bin only one of the particles must enter acceptance.
- Maximal spread of pairs depends on the properties of the medium and time.

Stopping: Resonances



- At lower energies incoming nucleons can undergo inelastic $2 \rightarrow 2$ scattering forming baryon resonances.
- This leads to baryon charge distribution peaked around mid-rapidity.
- Baryon charge is carried by hadrons with integer charge.
- Neutral particles decaying into two charged has the effect similar to pair production.
- Cannot describe high multiplicity events at high energies that requires $2 \rightarrow N$ reactions

Stopping: Strings

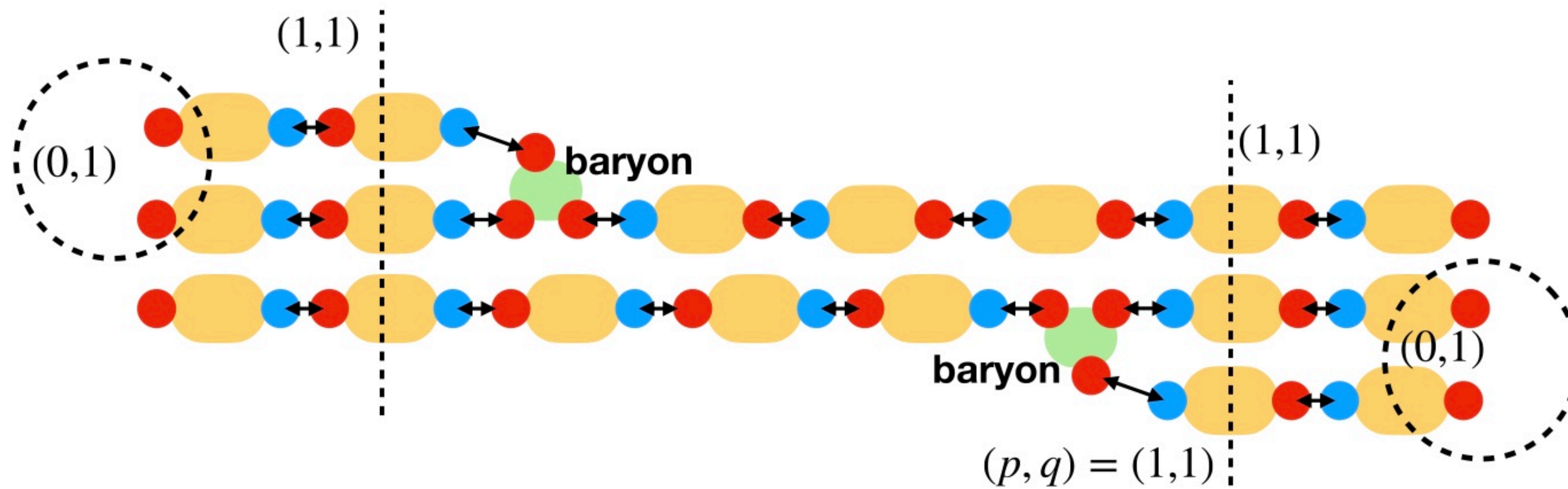


- At higher energies scattering can be described with strings stretched between quarks inside nucleons.
- This leads to baryon charge distribution being significantly wider.
- Baryon charge is carried by hadrons with integer baryon number?
- At very high energies one can consider baryon junction.

Stopping: Baryon junctions vs. String melting

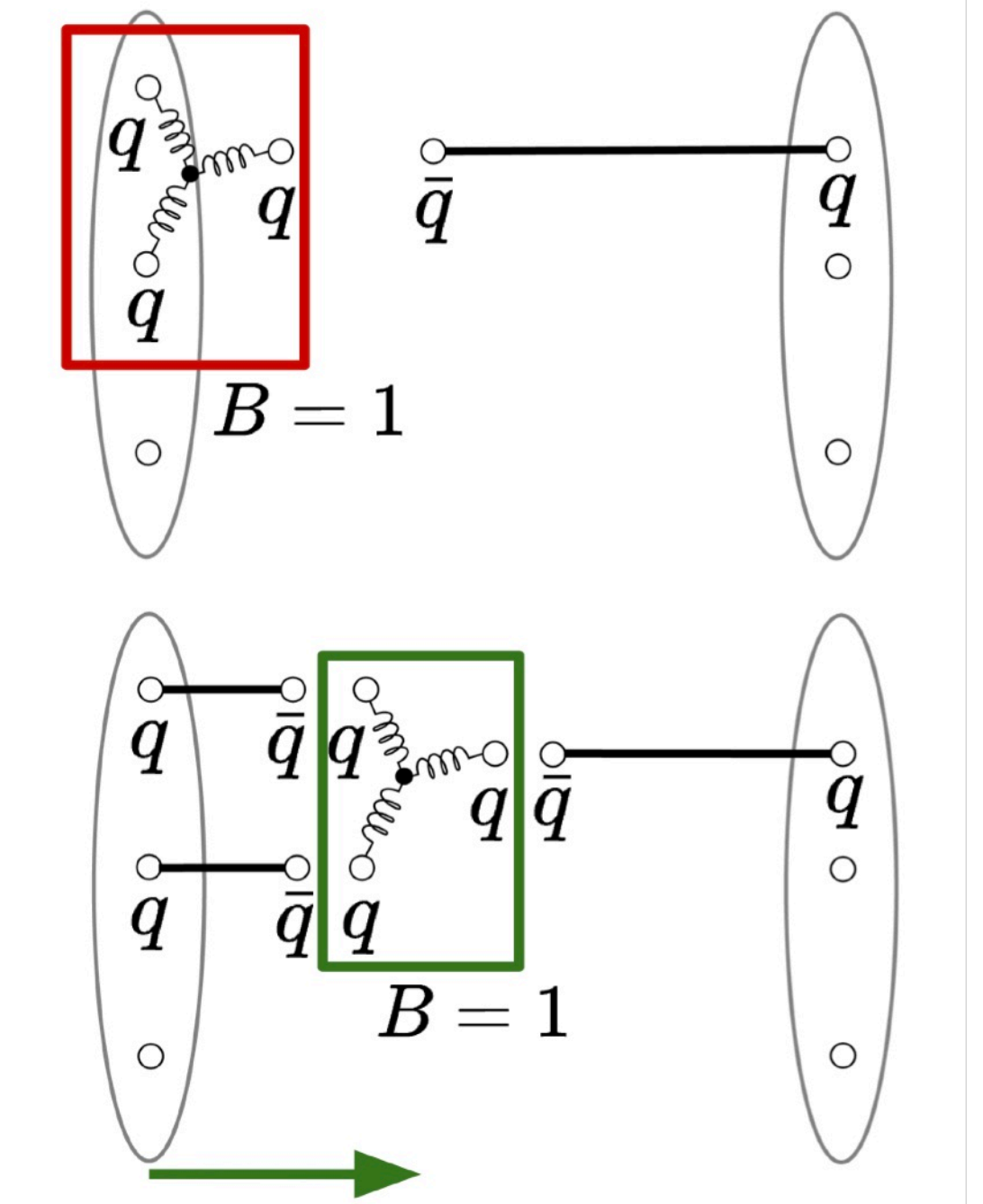
[G. Pihan, QM2024 Talk]

[D. Kharzeev, Phys. Lett. B 378, 238 (1996)]



[S.Pratt, Phys. Rev. C 109, 044910]

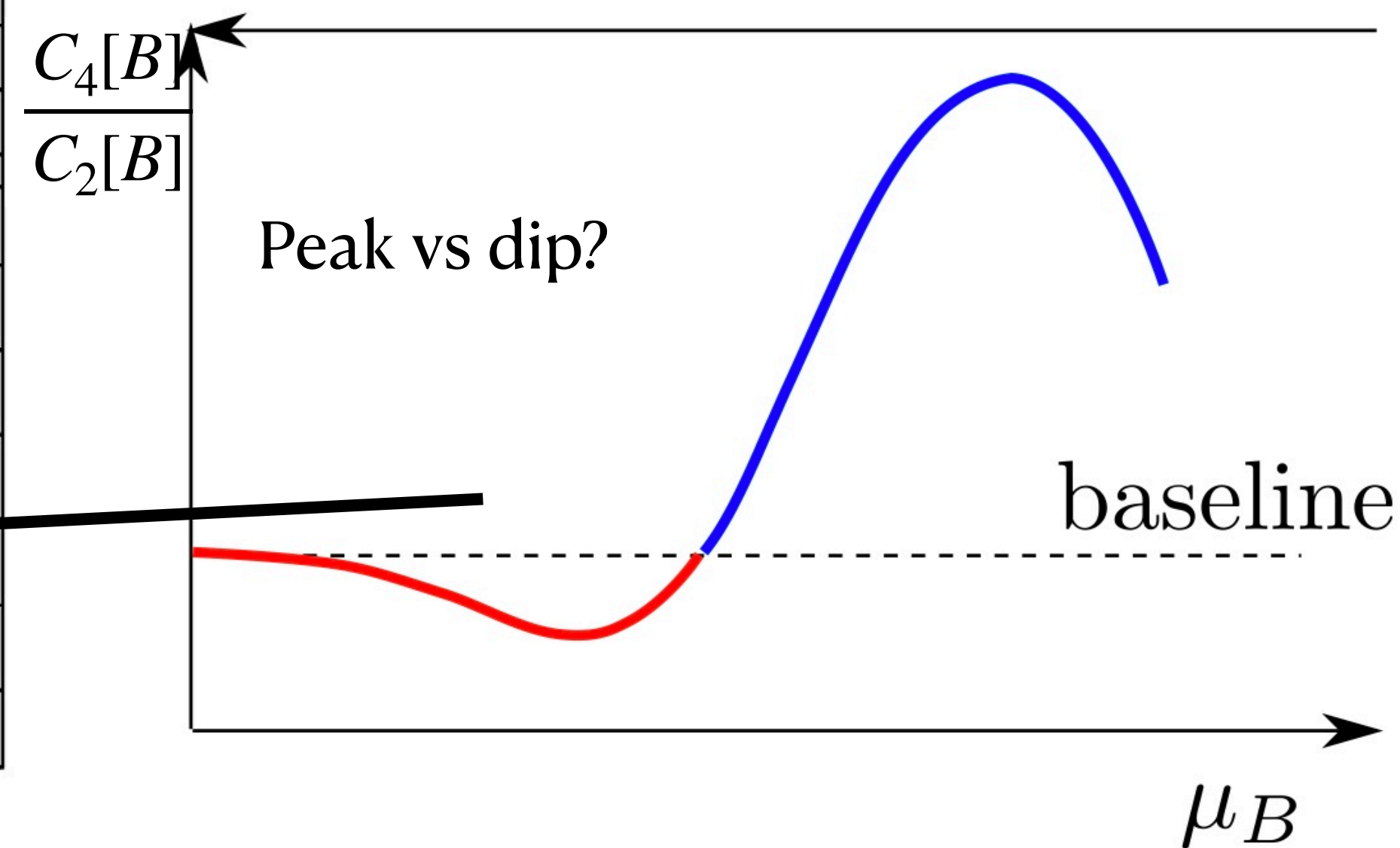
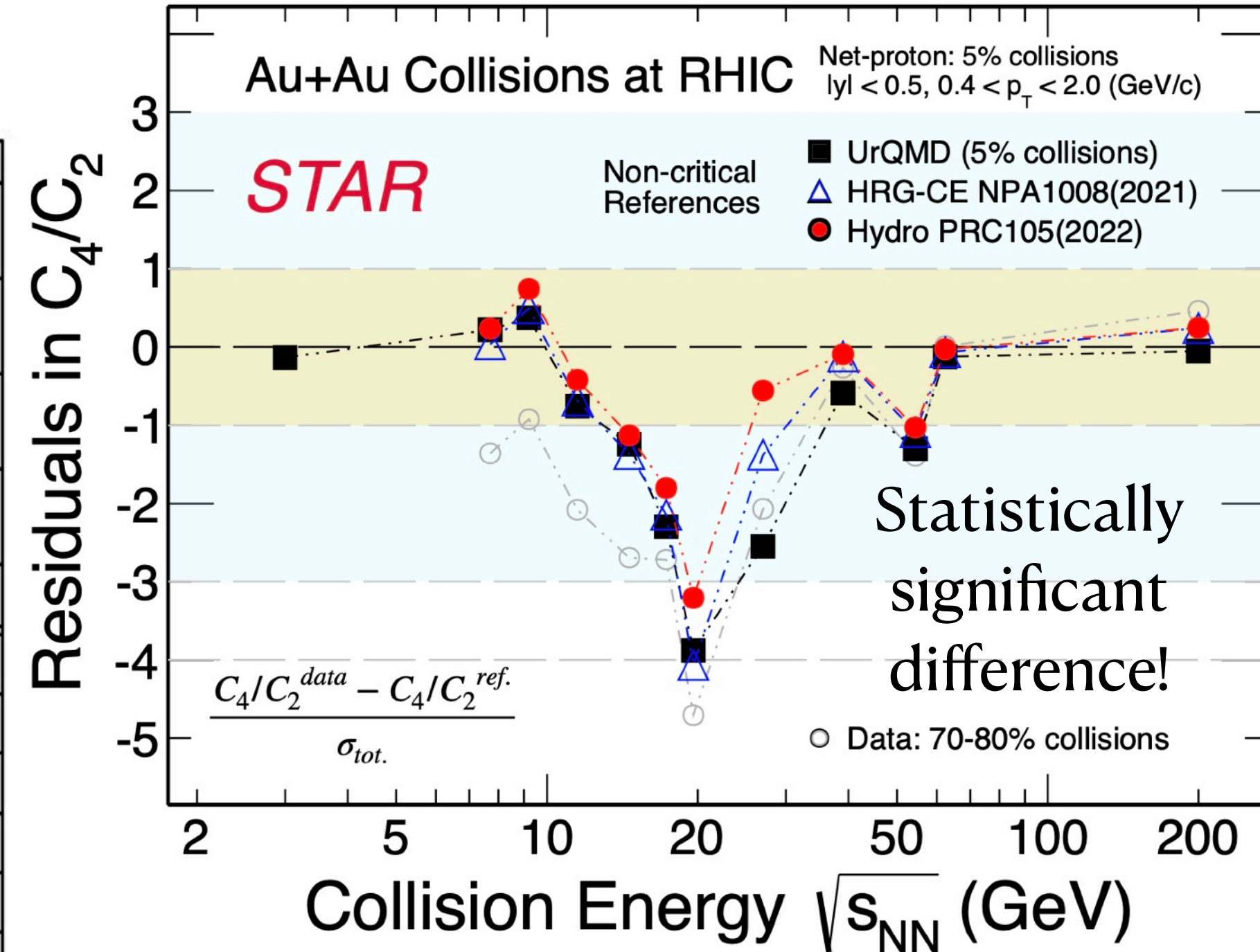
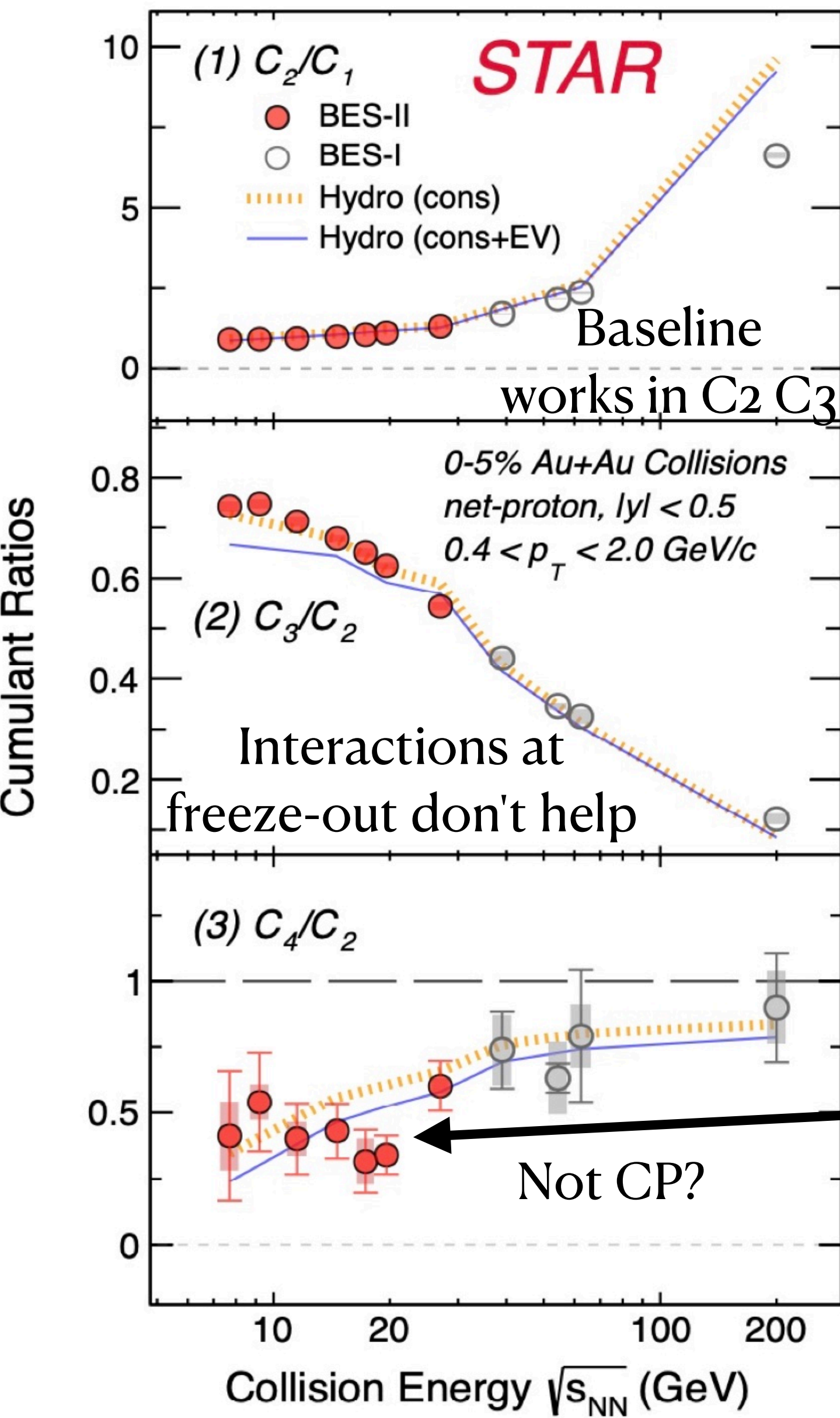
- Junctions transfer baryon charge in units of a single baryon.
- Are AuAu collisions different from pp? Are quarks still move as a baryon at very high energy?
- Does string melting introduce individual quarks as charge carriers?



Models vs Experiment

[A. Pandav, CPOD2024]

Net-proton cumulant ratios



- Non-interacting baselines fail in 20 GeV range.
- This difference is seen only in kurtosis.
- Hard to include interactions.
- Not where IQCD predicts.

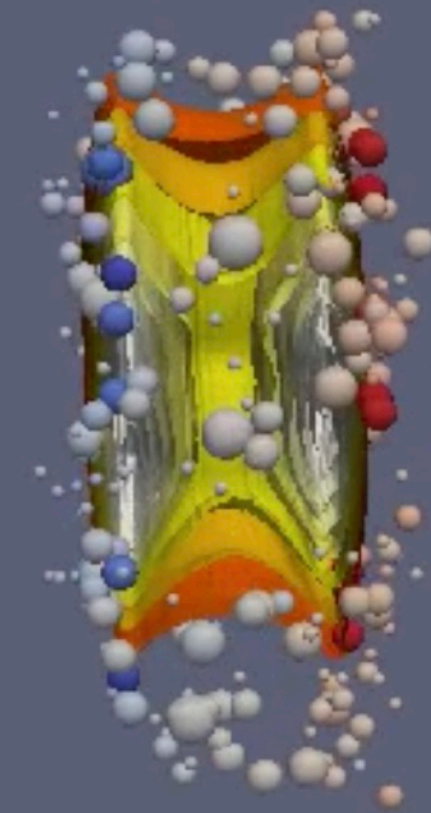
History of each event is important

Initial stage is ignored in baseline.

Non-trivial behaviour

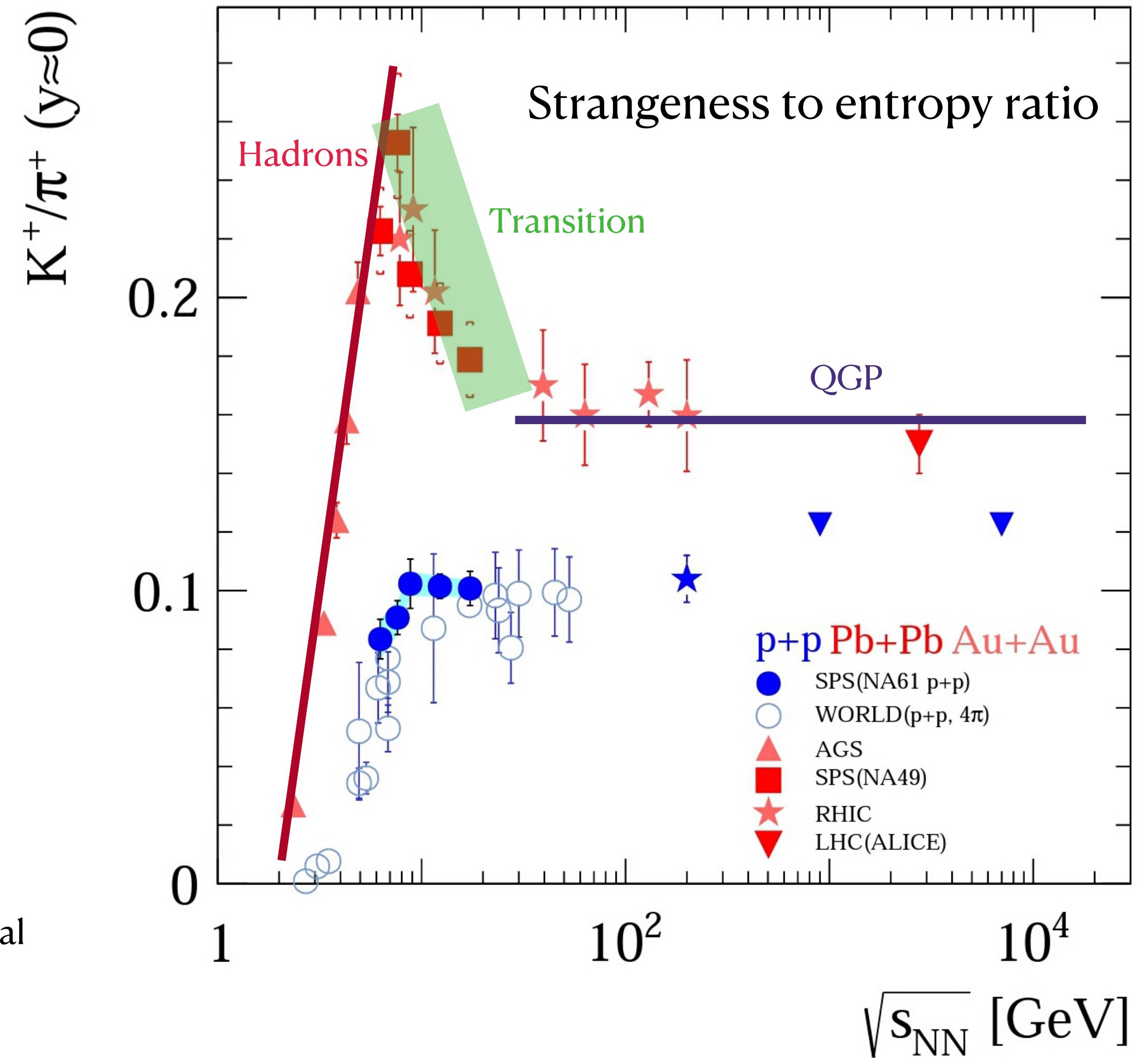
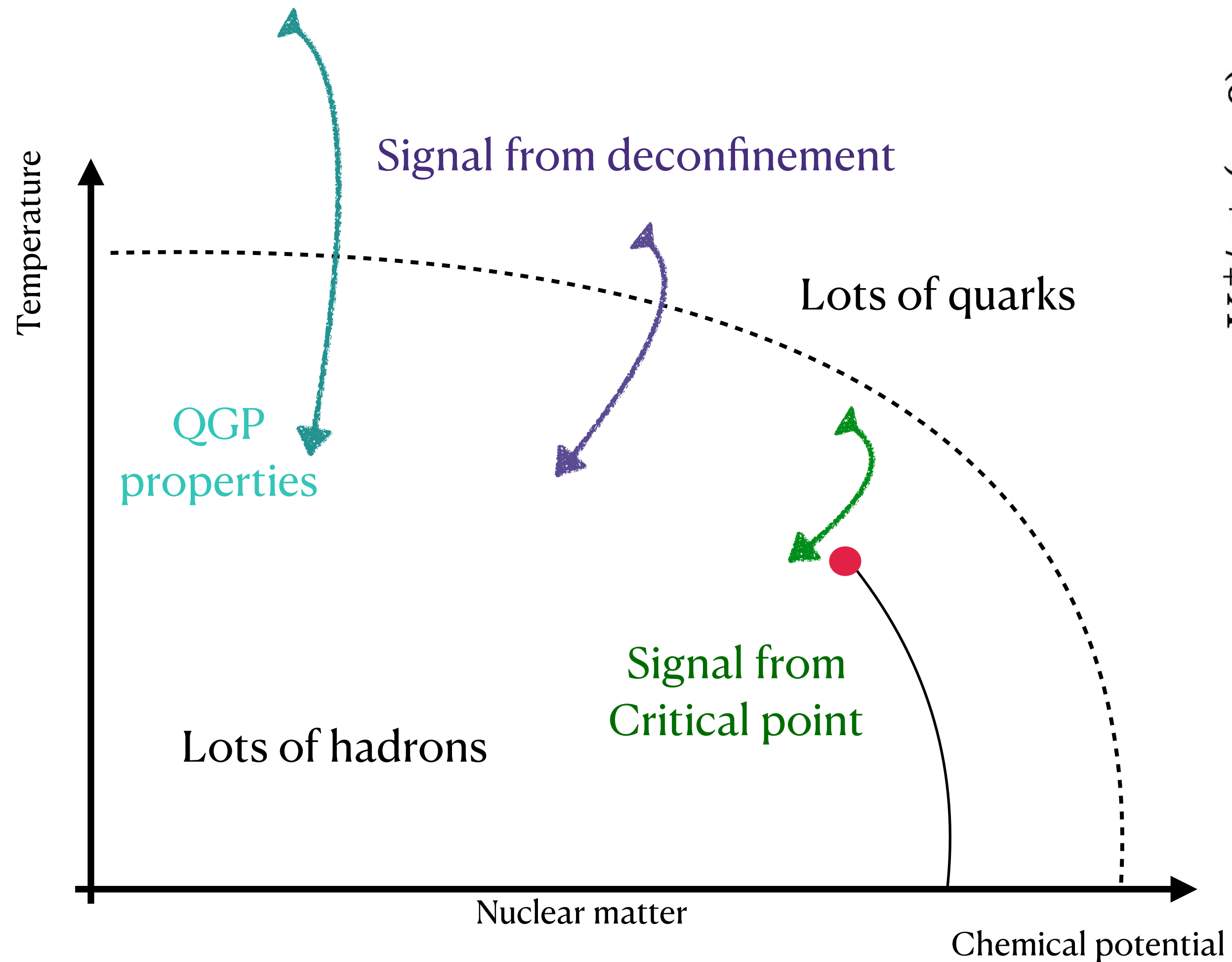
Onset of QGP formation.

String melting / recombination.



Local conservation is not accounted for

Initial state effects in fluctuations: Onset of deconfinement



[Marek Gazdzicki, Mark Gorenstein, Peter Seyboth, Acta Phys.Polon.B42:307-351,2011]

QGP/HRG fluctuations

1: Ideal gas of baryons and anti-baryons

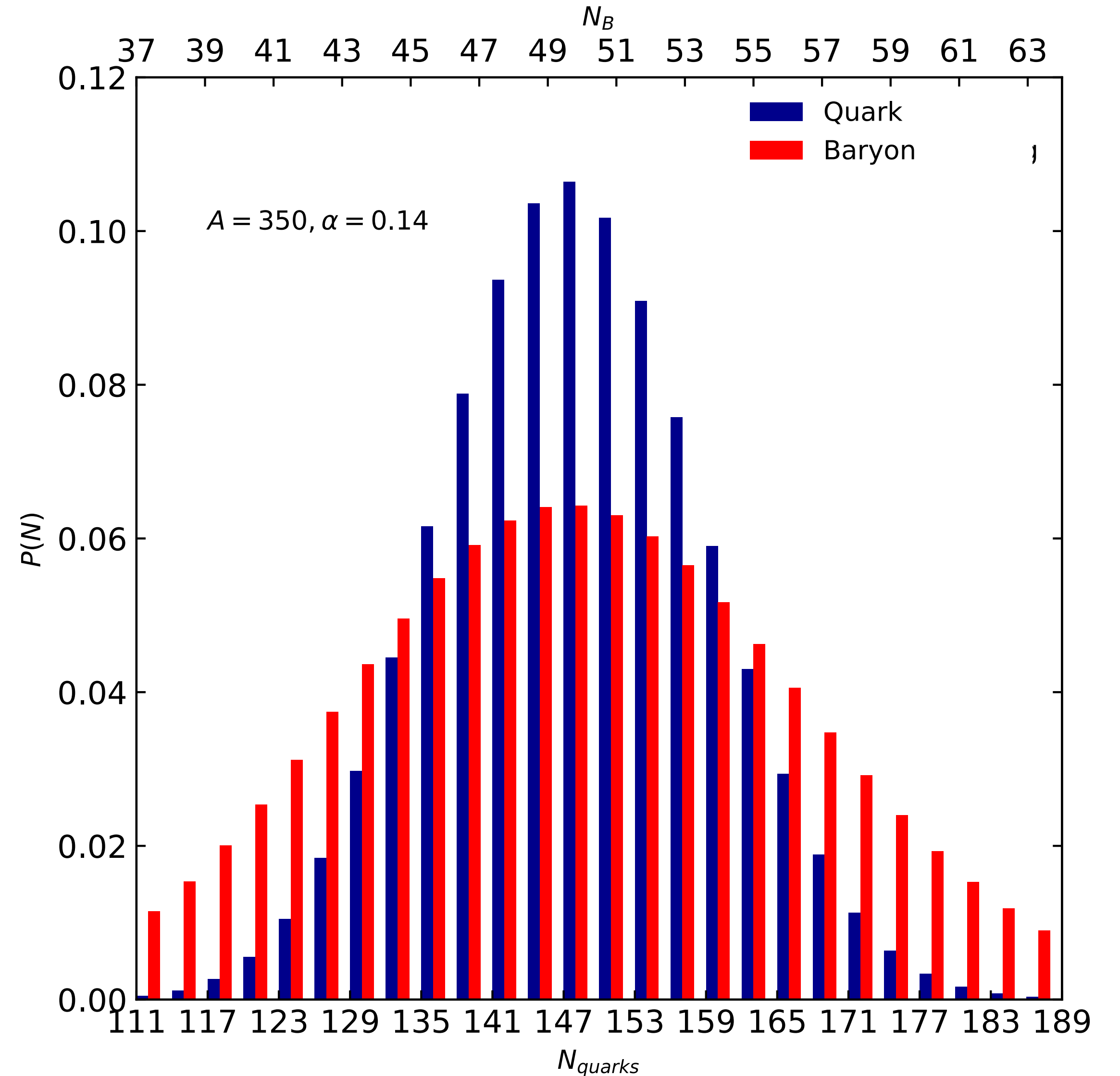
$$P(N_B) = \frac{\langle N_B \rangle^{N_B}}{N_B!} \exp^{-\langle N_B \rangle} \frac{\langle N_{\bar{B}} \rangle^{N_{\bar{B}}}}{N_{\bar{B}}!} \exp^{-\langle N_{\bar{B}} \rangle}$$

2: Ideal gas of quarks and anti-quarks

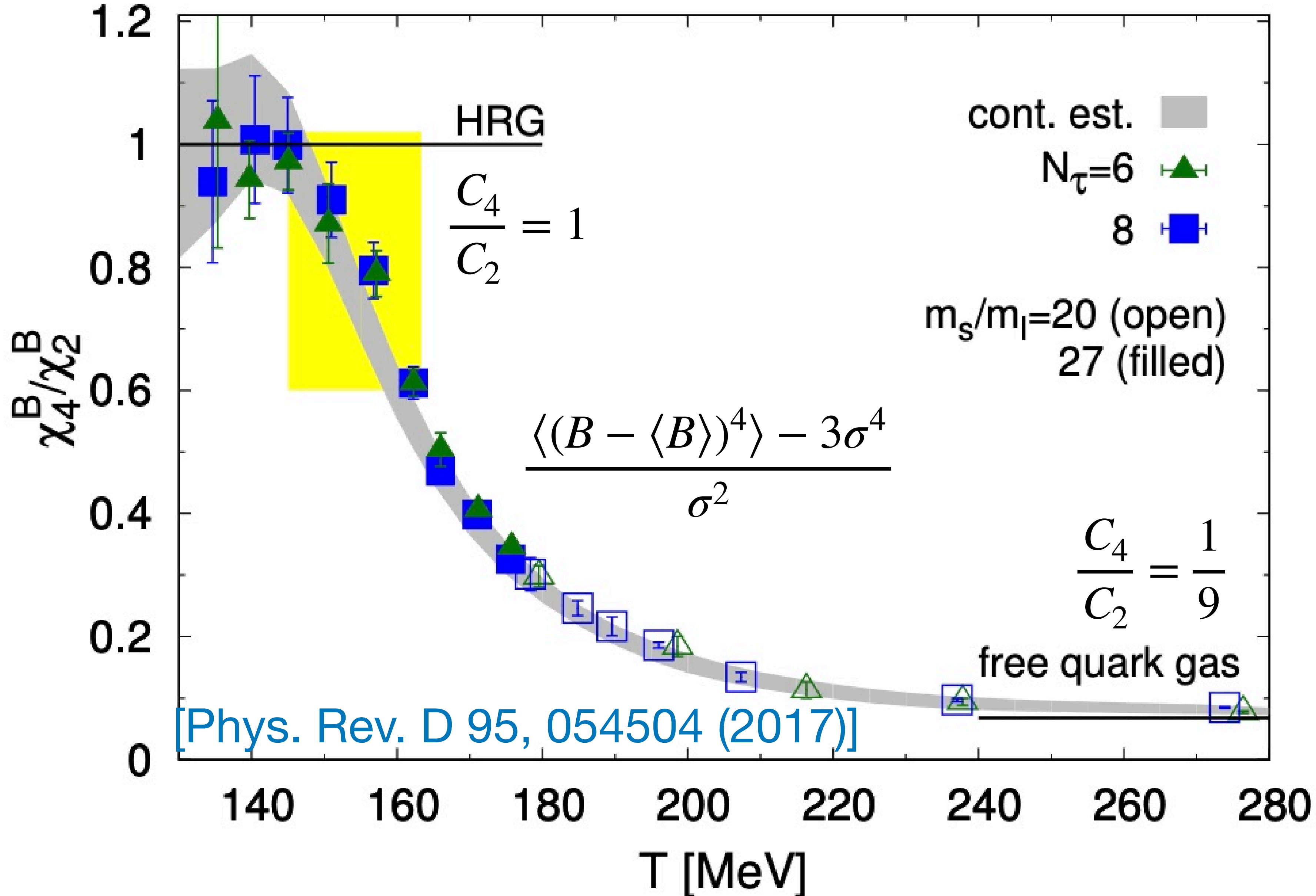
$$P(n_q) = \frac{\langle n_q \rangle^{n_q}}{n_q!} \exp^{-\langle n_q \rangle} \frac{\langle n_{\bar{q}} \rangle^{n_{\bar{q}}}}{n_{\bar{q}}!} \exp^{-\langle n_{\bar{q}} \rangle}$$

In other words distribution of quarks less wide and has weaker tails compared to baryons

(in ideal non-degenerate gas limit)



Kurtosis of baryon charge



$$C_4 = N_B + N_{\bar{B}},$$

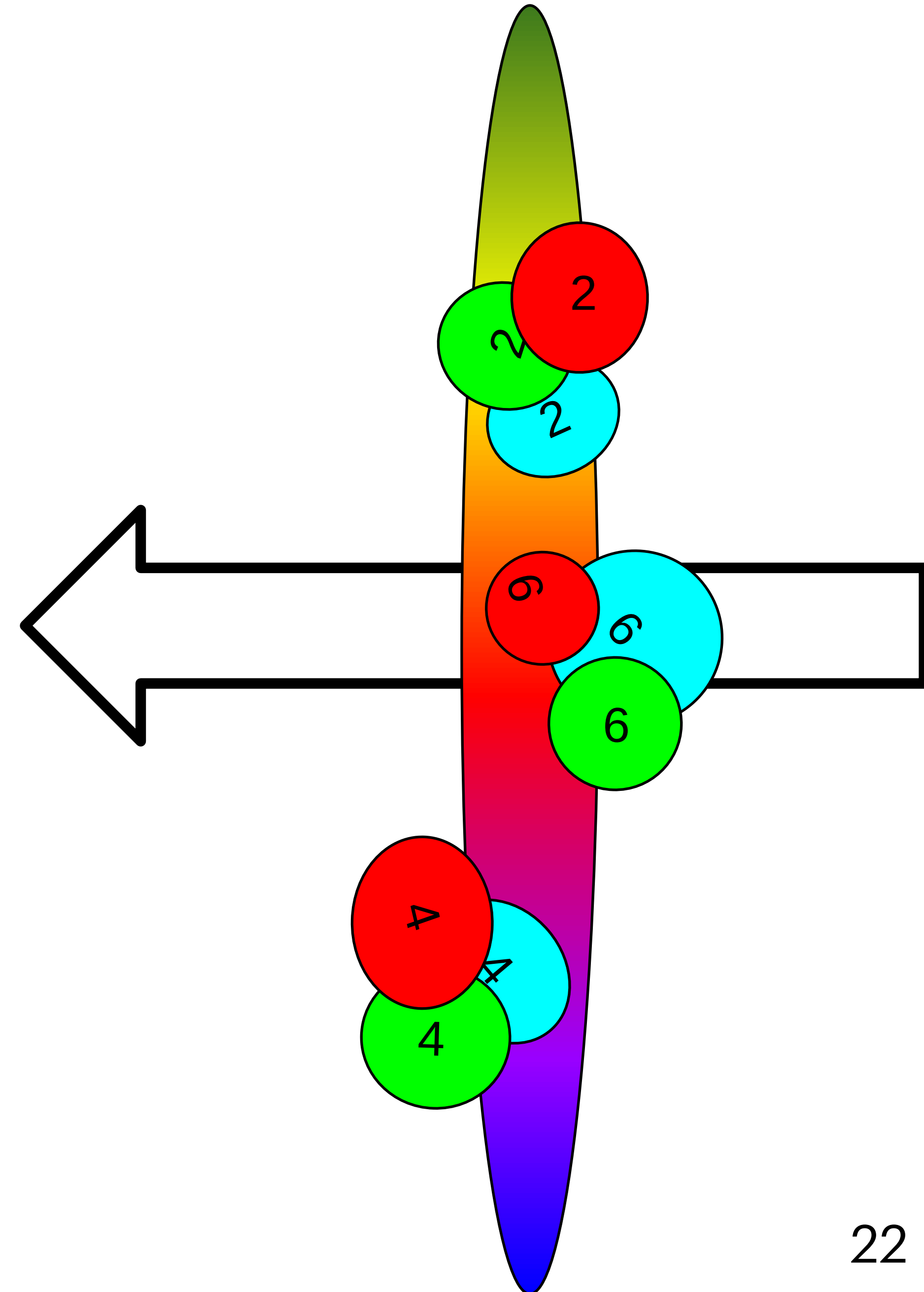
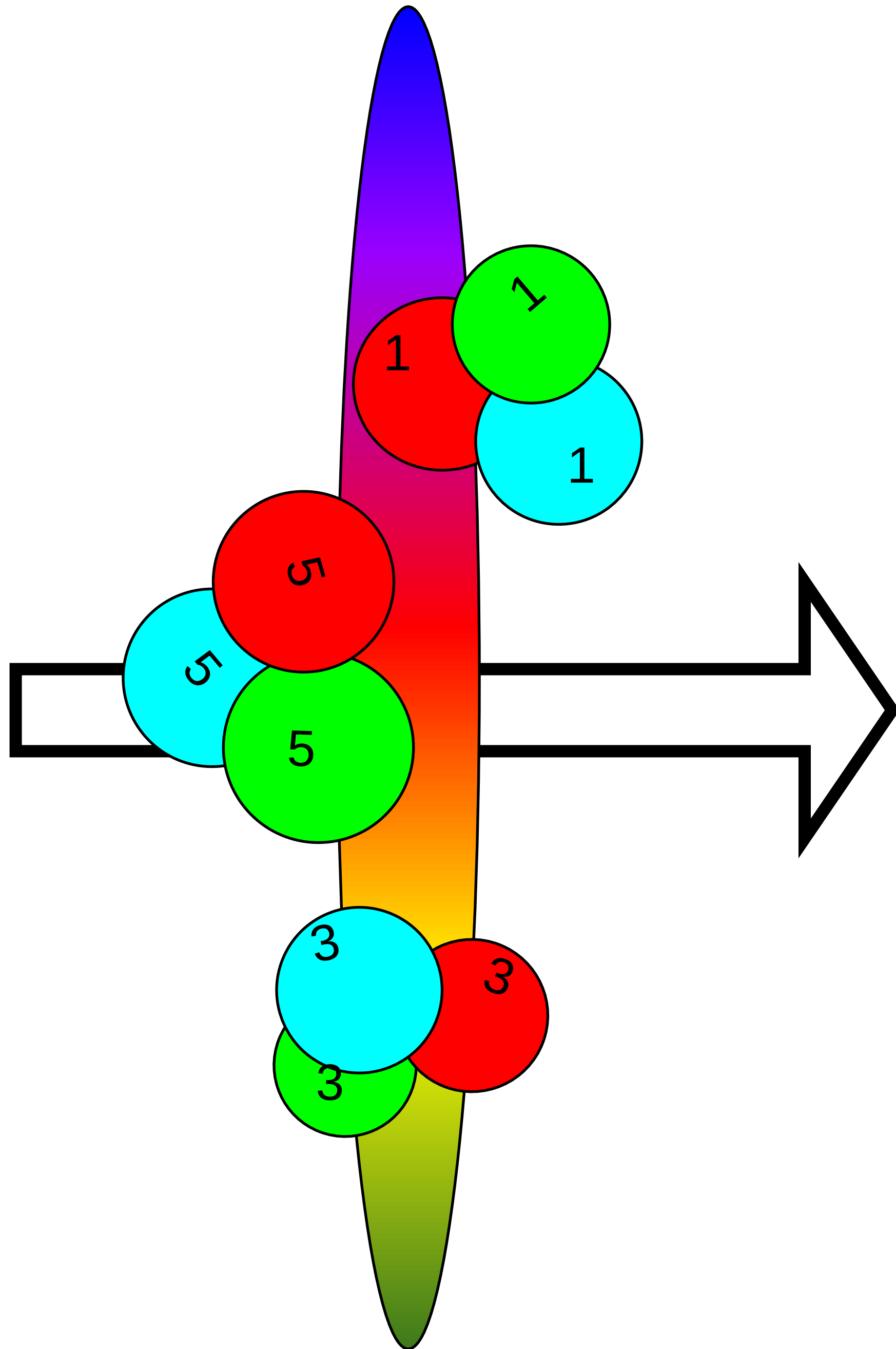
$$C_2 = N_B + N_{\bar{B}},$$

$$C_4 = \frac{1}{81} (n_q + n_{\bar{q}}),$$

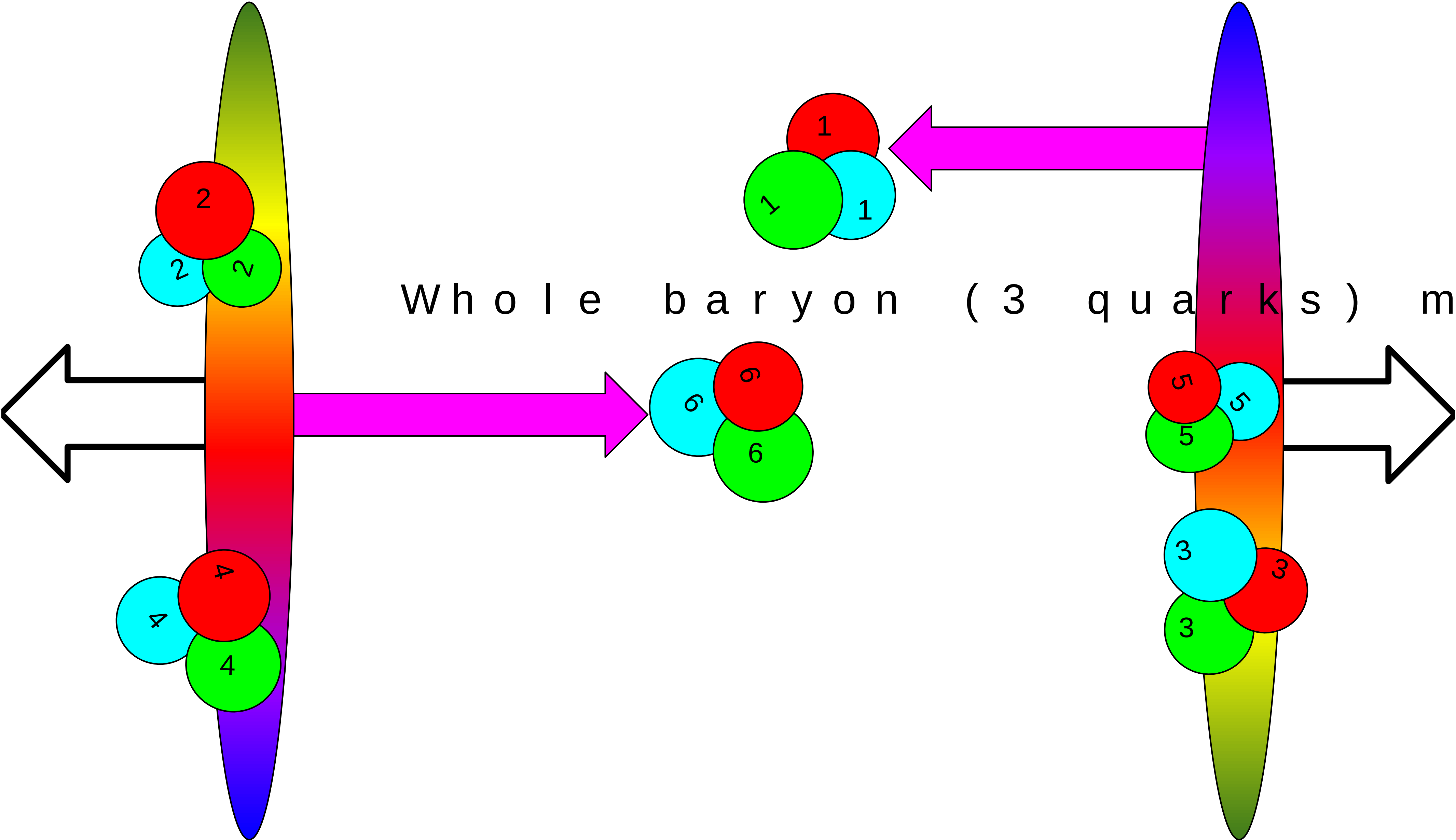
$$C_2 = \frac{1}{9} (n_q + n_{\bar{q}}),$$

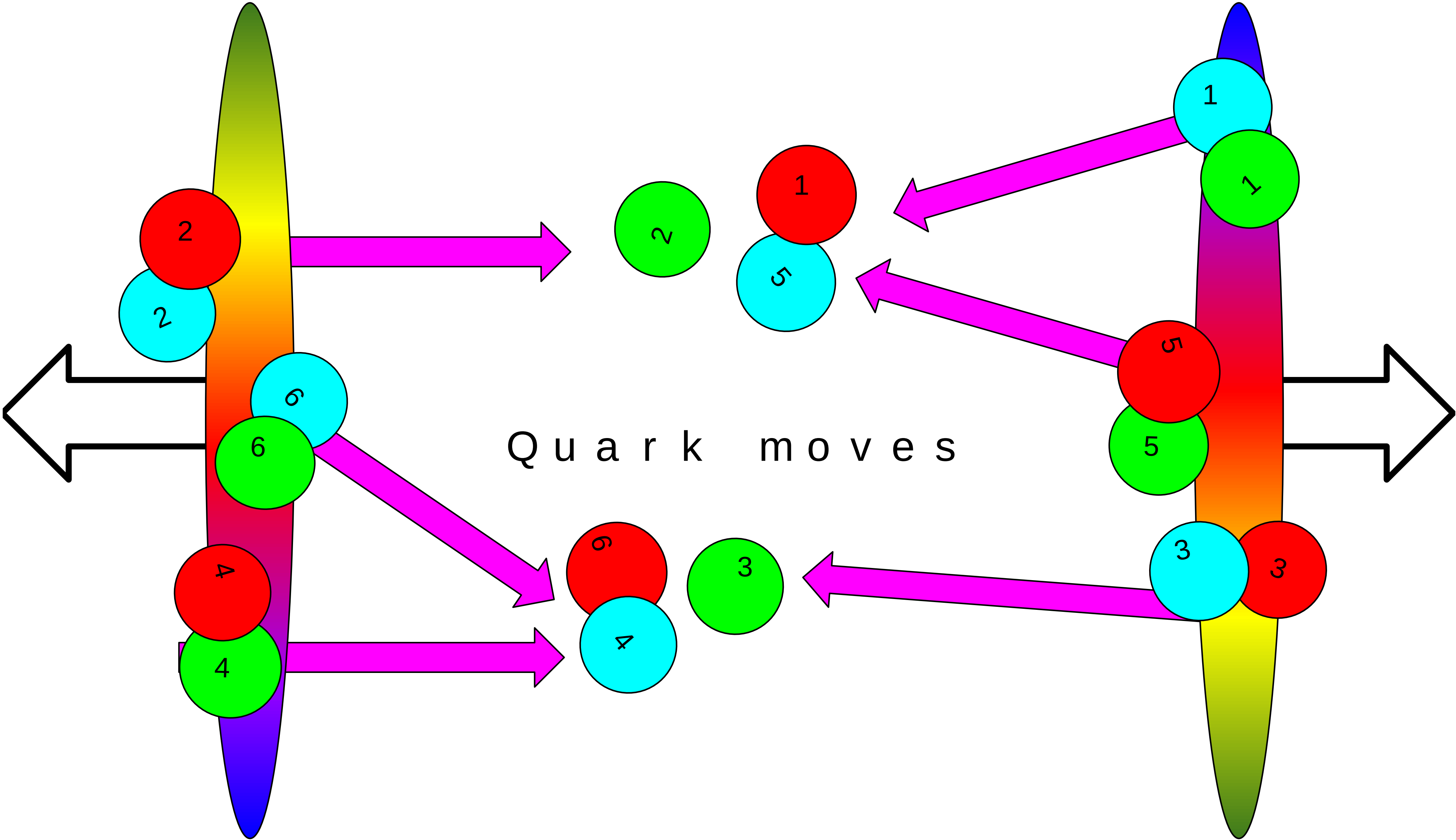
This suppression was predicted but never observed.

B e f o r e c o l l i s i o n

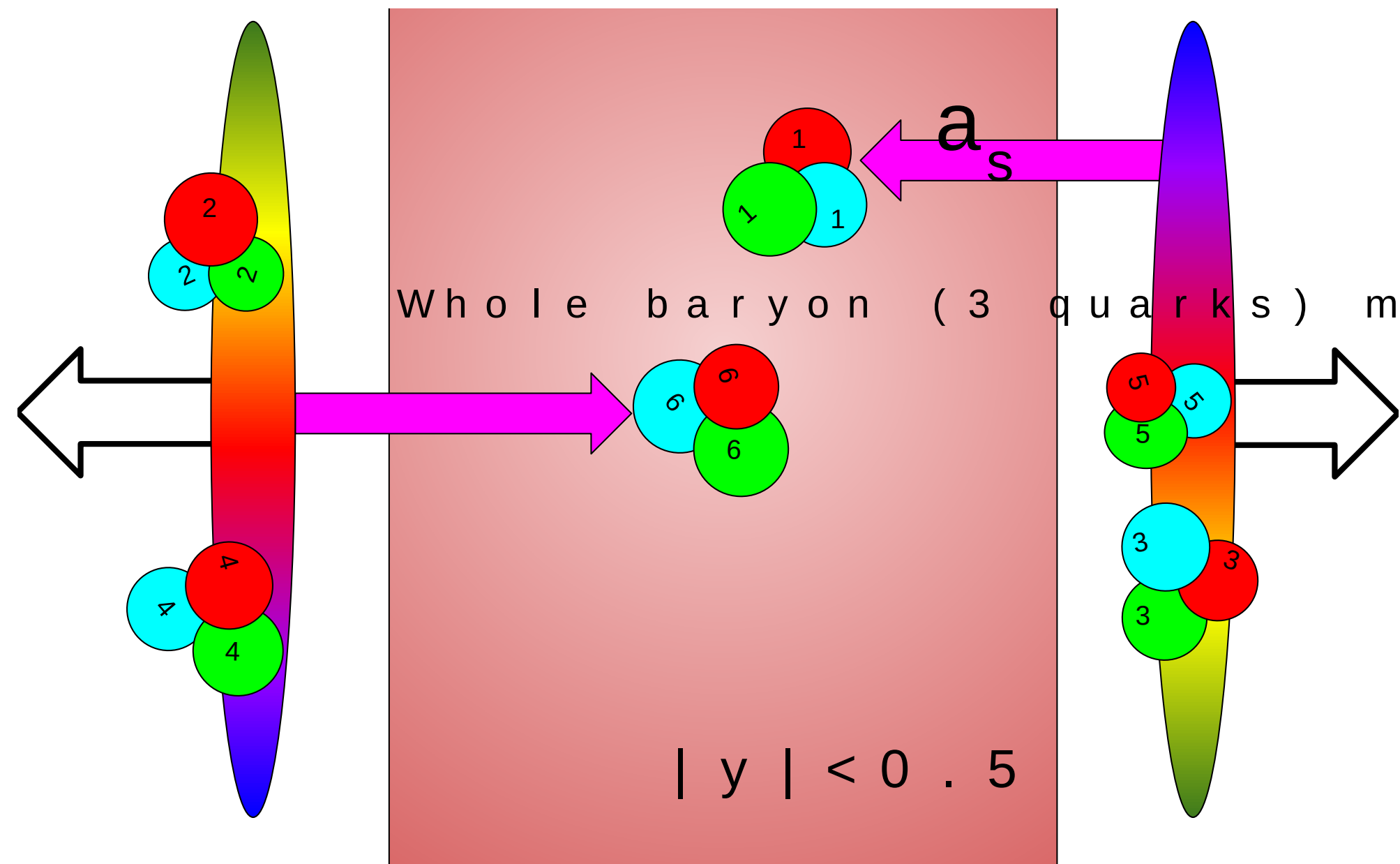


B a r y o n s t o p p i n





Case 1: Baryon stopping



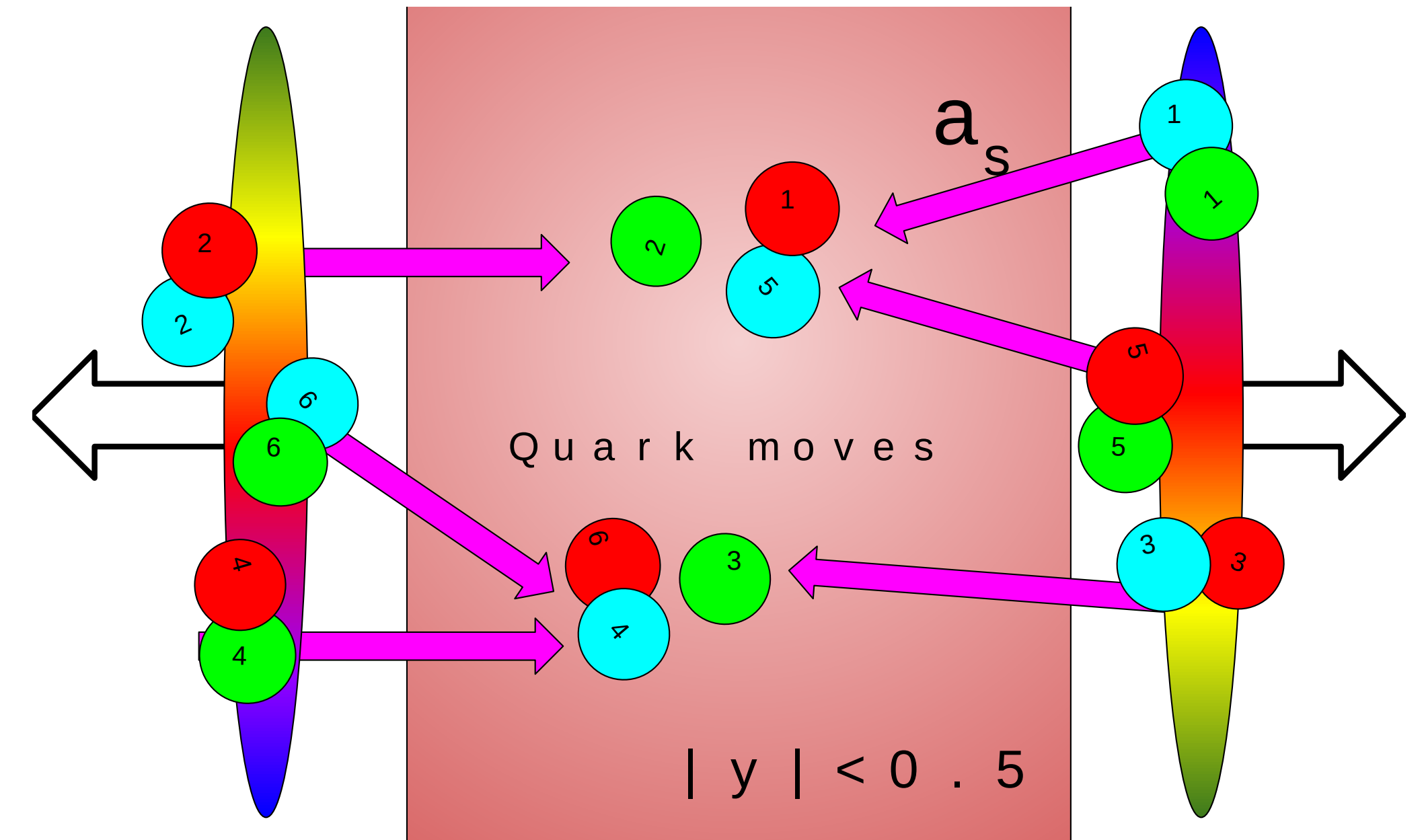
$$C_1^{stopping} = \alpha_s B_s,$$

$$C_2^{stopping} = B_s \alpha_s (1 - \alpha_s),$$

$$C_3^{stopping} = B_s \alpha_s (1 - 2\alpha_s),$$

$$C_4^{stopping} = B_s \alpha_s (1 - \alpha_s) (1 - 6\alpha_s (1 - \alpha_s))$$

Case 2: Quark stopping



$$C_1^{stopping} = \alpha_s B_s,$$

$$C_2^{stopping} = \frac{1}{3} B_s \alpha_s (1 - \alpha_s),$$

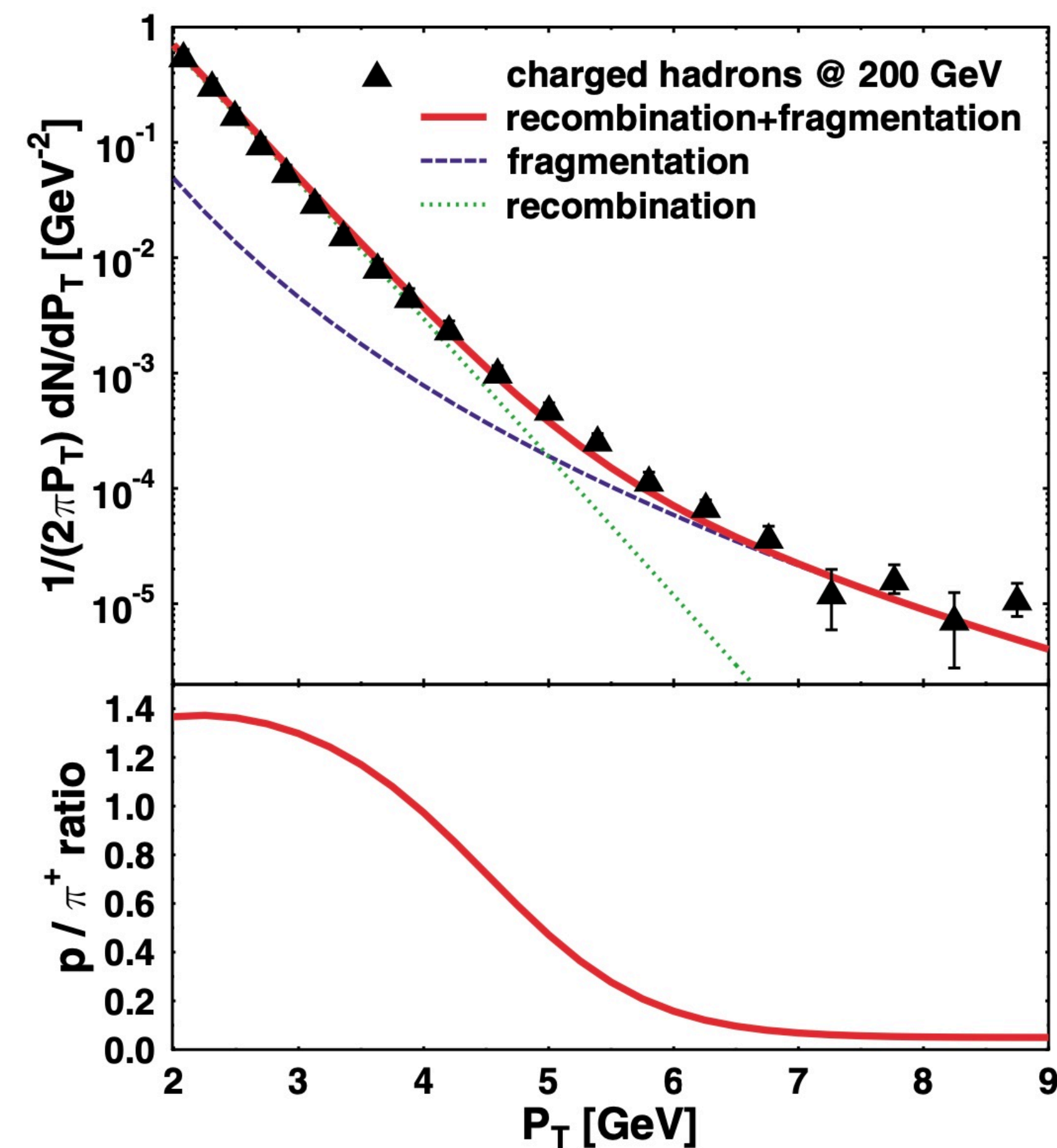
$$C_3^{stopping} = \frac{1}{9} B_s \alpha_s (1 - 2\alpha_s),$$

$$C_4^{stopping} = \frac{1}{27} B_s \alpha_s (1 - \alpha_s) (1 - 6\alpha_s (1 - \alpha_s))$$

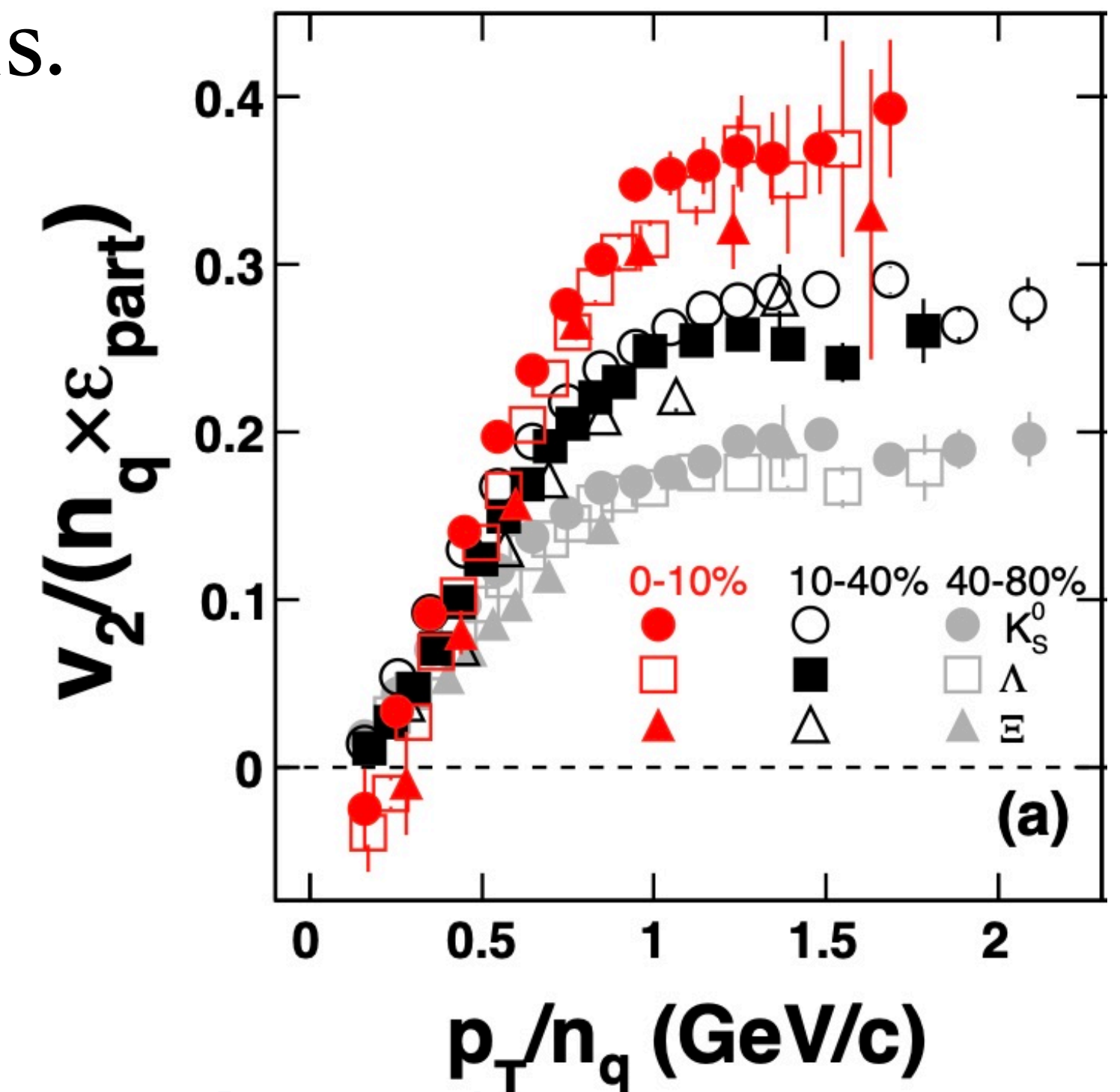
Huge suppression of fluctuations if the transition from baryons to quarks occurs

Recombination

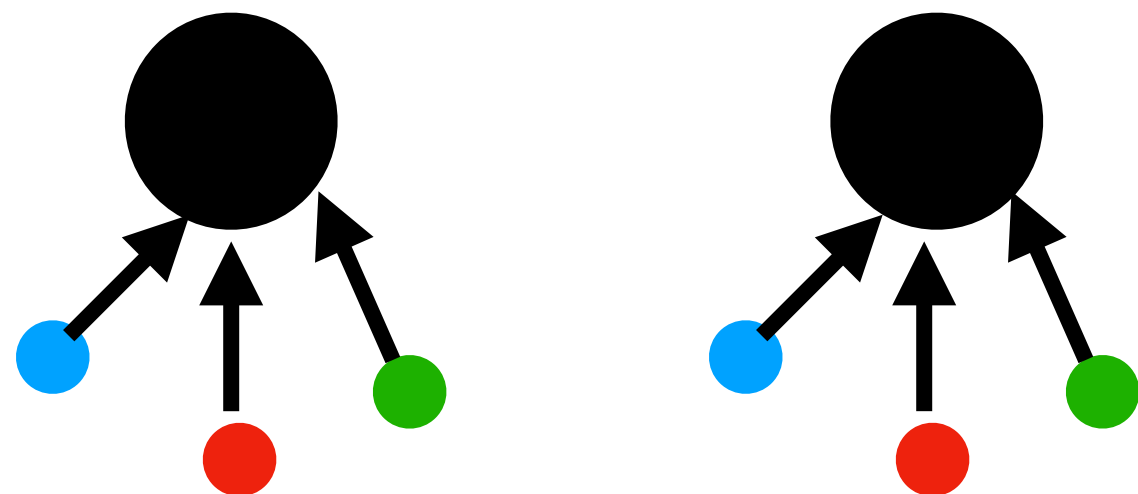
- Quark coalescence into hadrons.
- Number of constituent quarks scaling supports this picture.
- Alternatively at high transverse momentum one expects fragmentation of partons.
- Net-proton fluctuations originate from fluctuations of stopped quarks/baryons.
- Coalescence does not modify fluctuations.



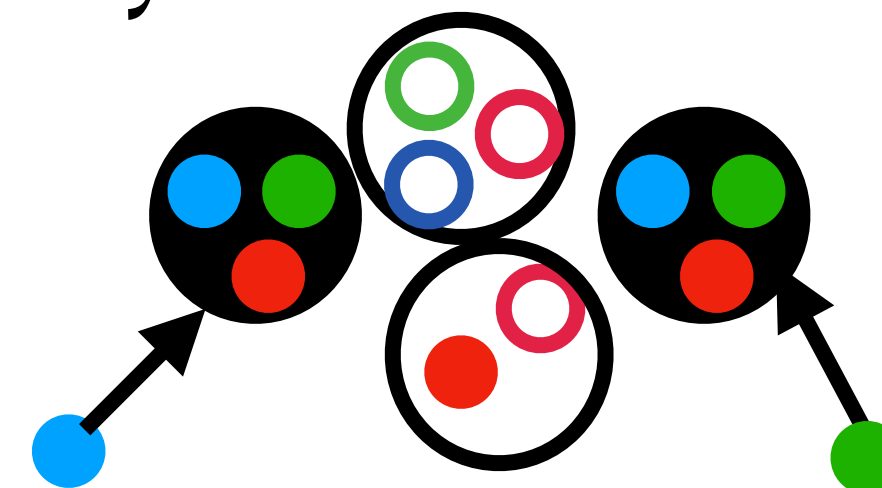
[R. J. Fries, S. A. Bass, B. Muller, C. Nonaka, Phys.Rev.Lett.90:202303,2003]



[STAR, Phys.Rev.C77:054901,2008]



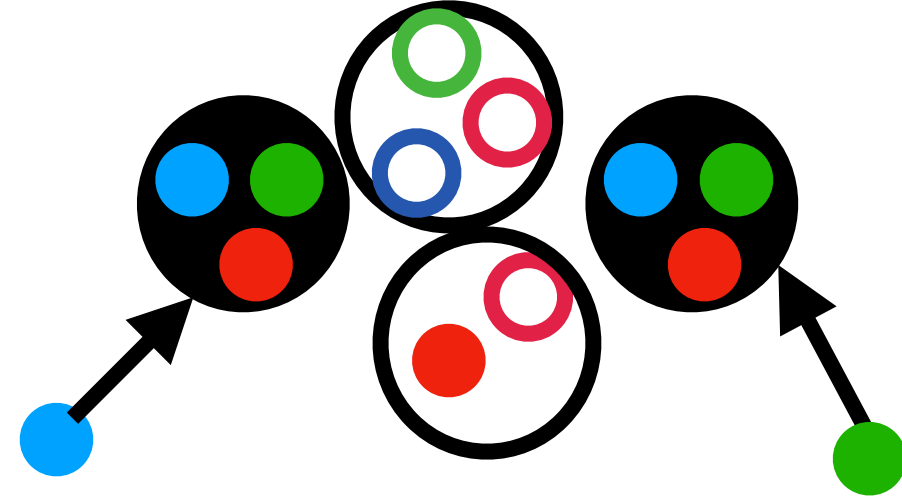
Fluctuations conserved on coalescence scale



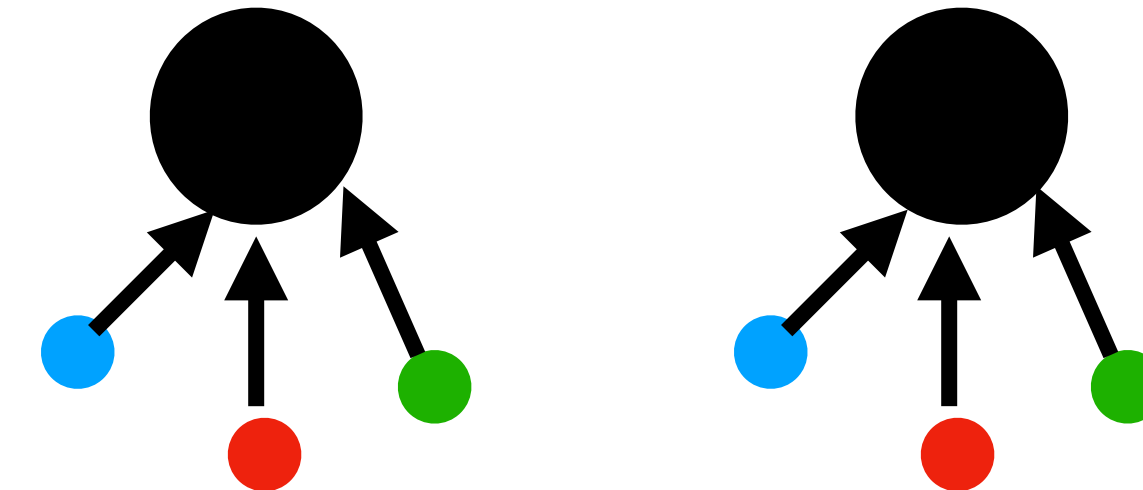
Anti-quarks from vacuum

Drastically change baryon charge distribution

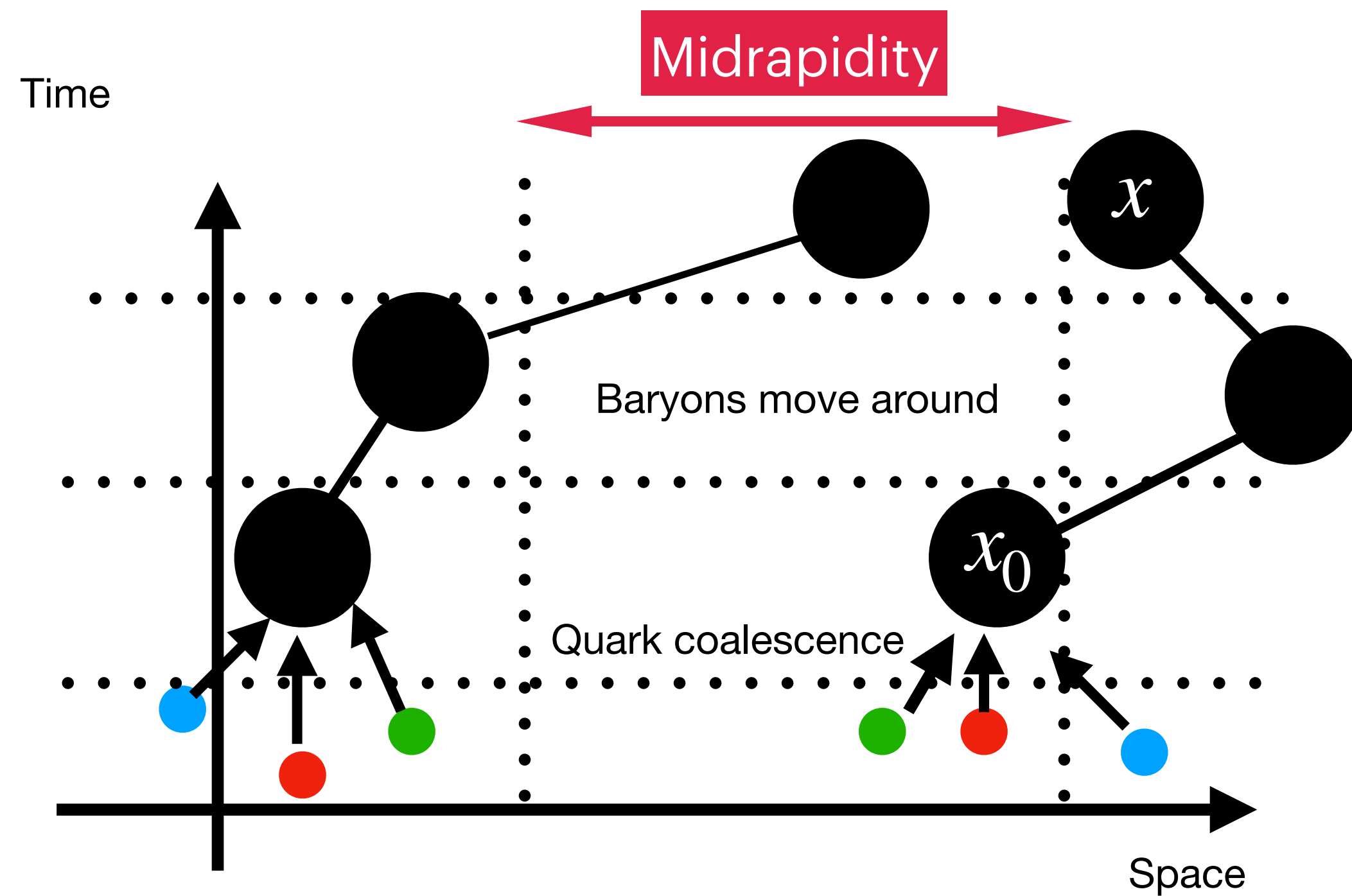
Post quark stage



Anti-quarks from vacuum
Drastically change baryon charge distribution

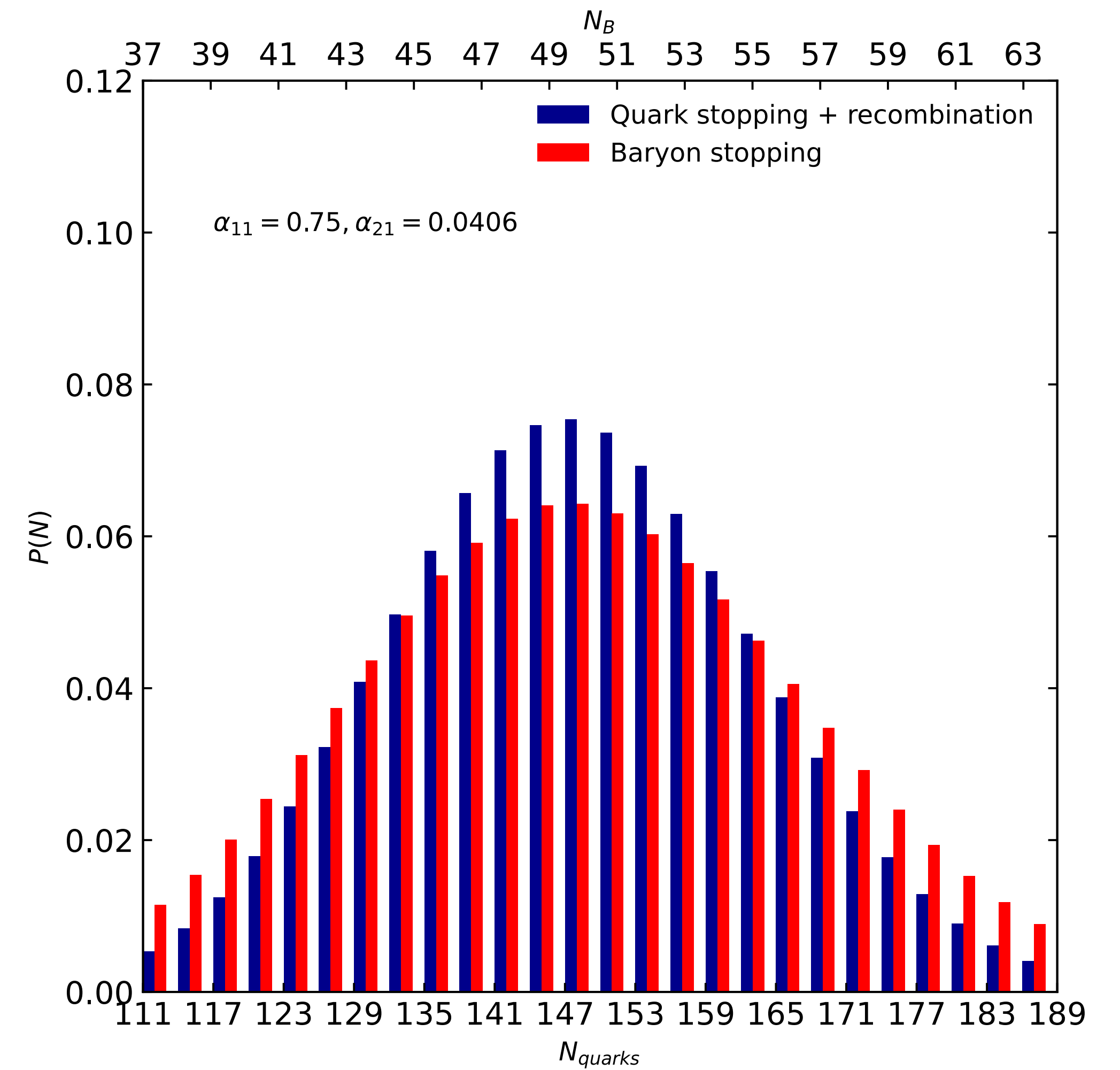
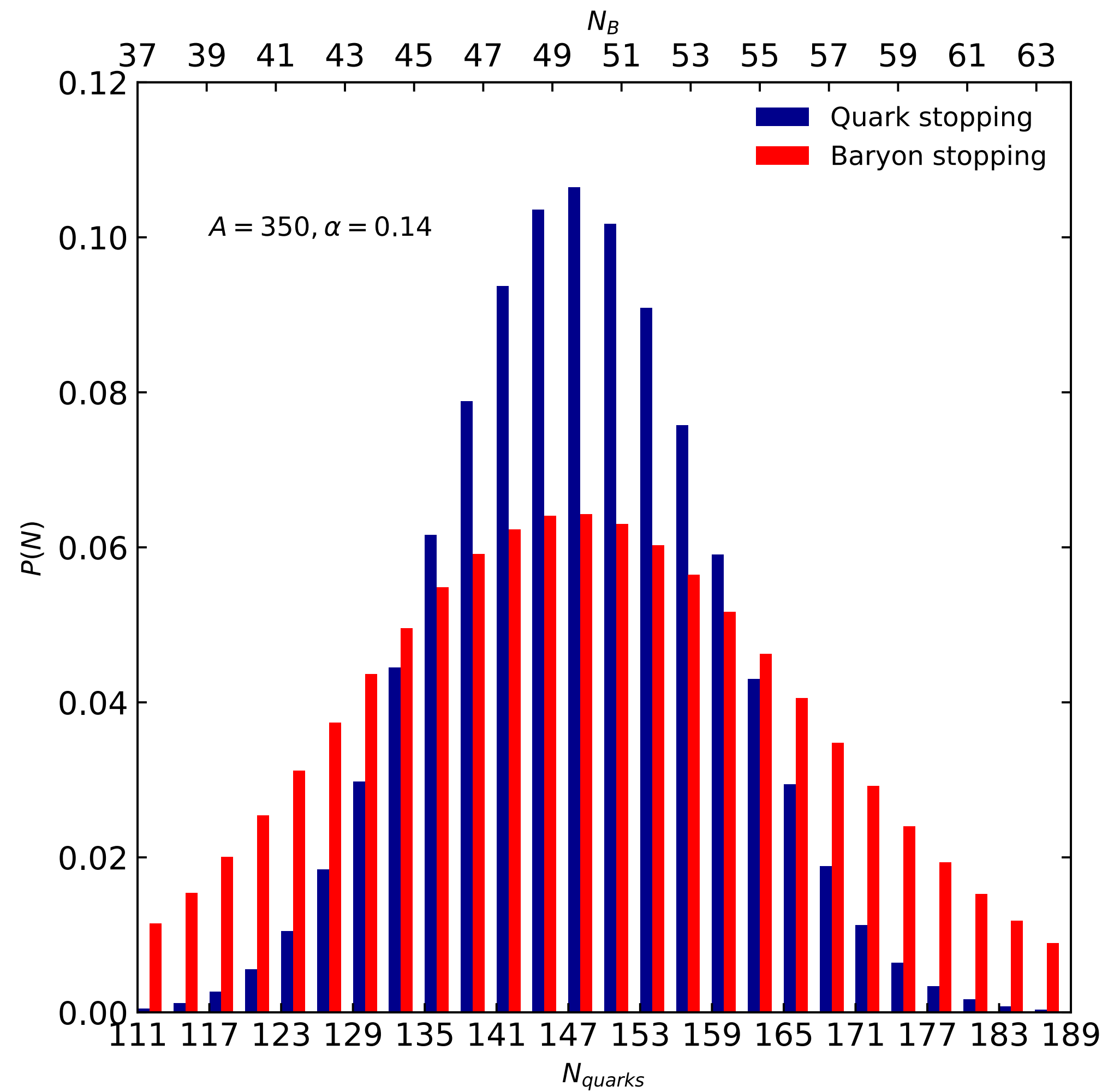


Fluctuations conserved on coalescence scale



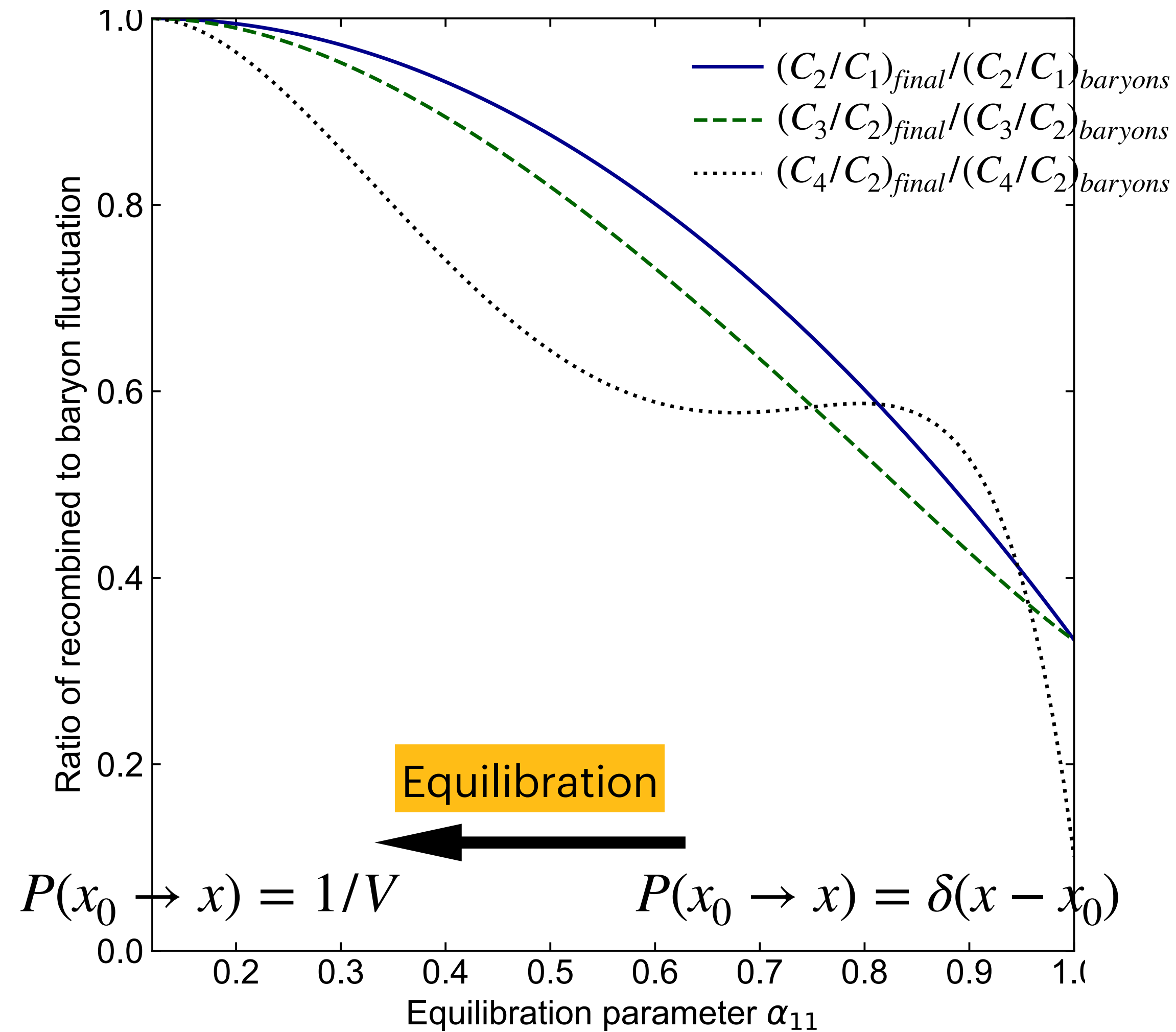
- After recombination baryons undergo diffusion/scattering/momentum smearing
- Because quarks now move in a group of three fluctuations should recalibrate towards hadron resonance gas values - global conservation baseline

Impact of final evolution stage



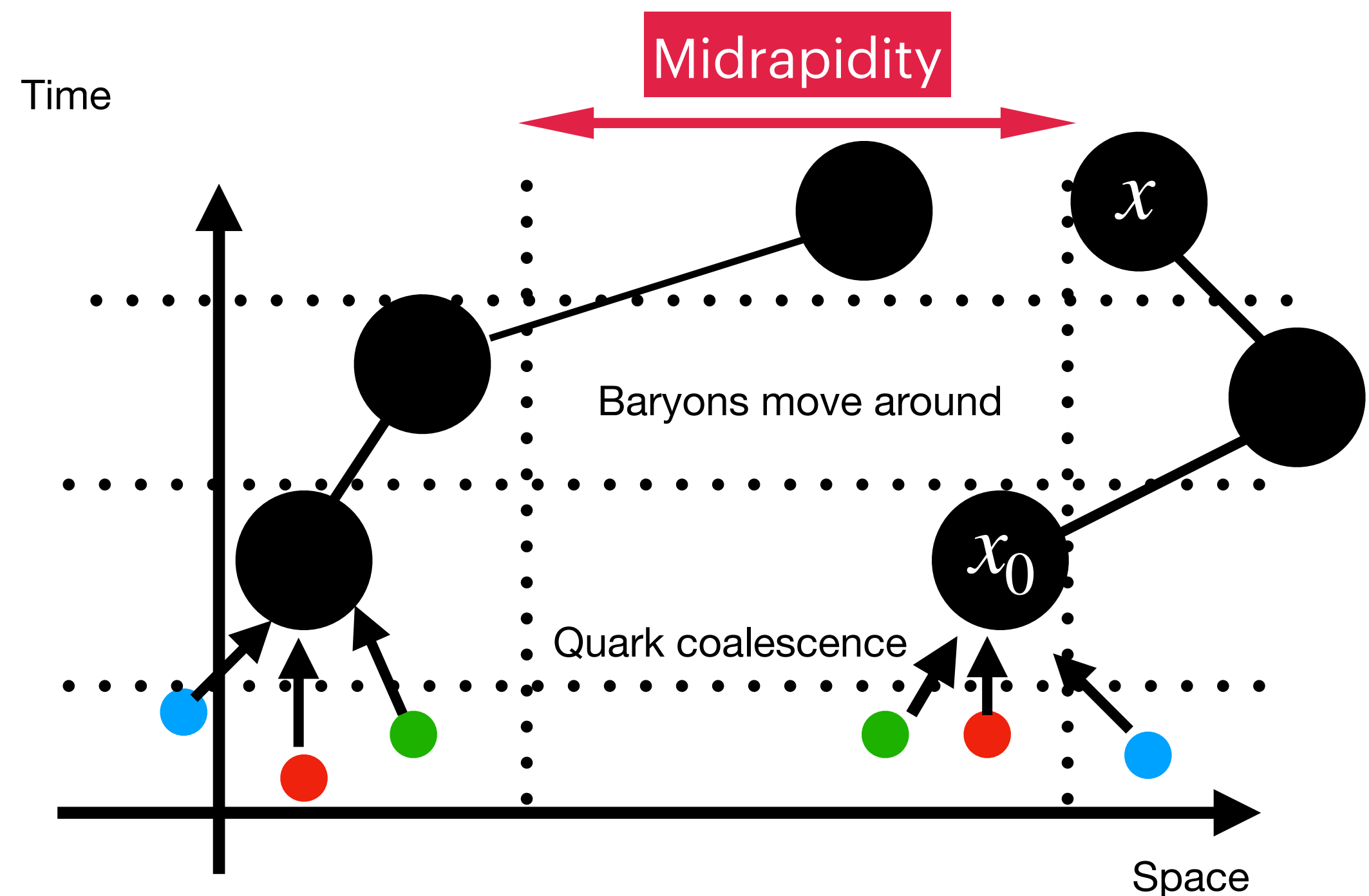
Final evolution reduces differences between stopping variants

Post quark stage

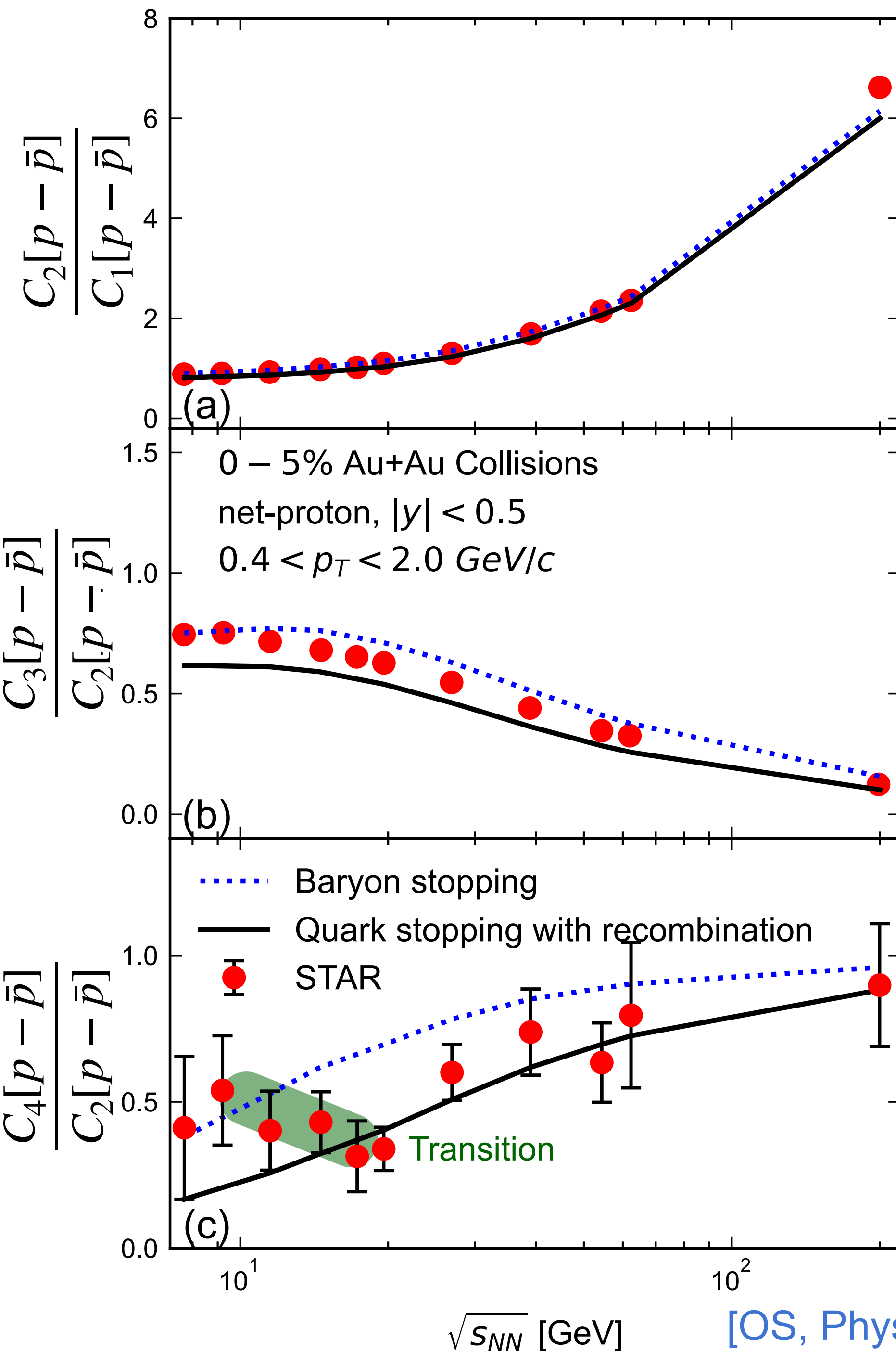


[OS, Physical Review C 111 (2), 024913]

- The equilibration happens faster in lower orders of fluctuations.
- As a consequence scaled variance and skewness can approach baseline values.
- Signal survives in higher cumulants such as kurtosis C_4/C_2 .



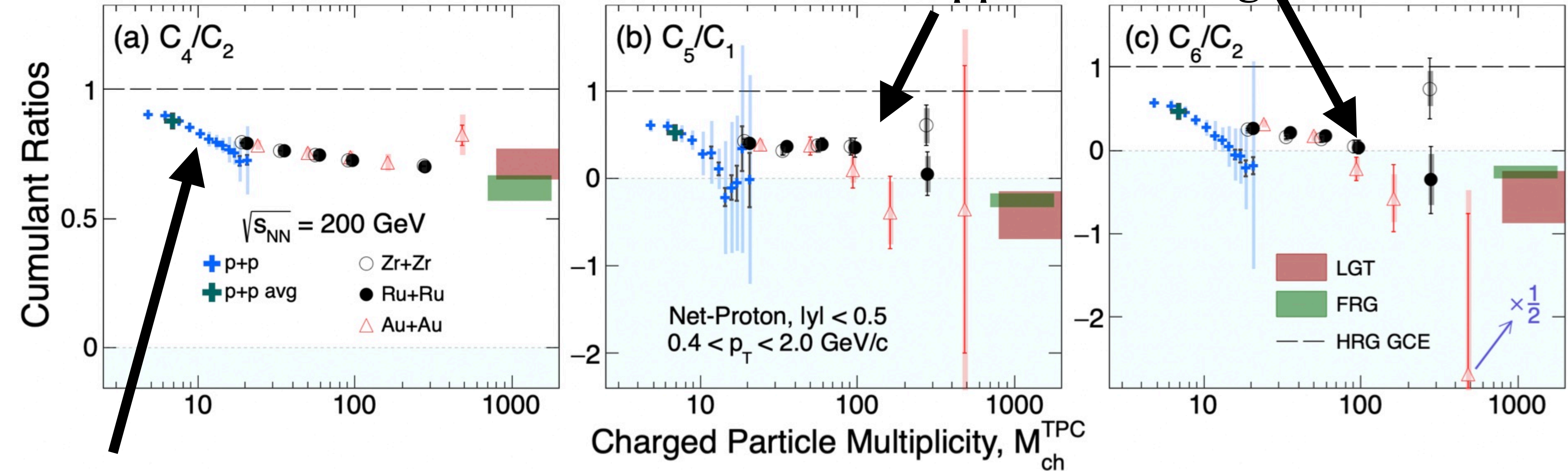
Data vs Model



- Global conservation baseline fails to capture trends seen in the data.
- Local conservation and fluctuations at stopping stage improve the description.
- Memory about initial state dissipates faster for low orders of cumulants.
- Recombination stage and subsequent hadron phase are crucial for modeling fluctuations following initial stage and hydrodynamic expansion.
- Signal should survive in kurtosis and higher cumulants.
- At high energies stopping happens on a quark level. Challenge to baryon junction picture?

Expectations for higher cumulants

More suppression in higher orders



QGP forms at high multiplicity [STAR, arXiv:2511.05043v1 [nucl-ex] 7 Nov 2025]

Require high statistics - new methods for numerical simulation are required

But so far one expects the general trend to hold in higher orders of fluctuations

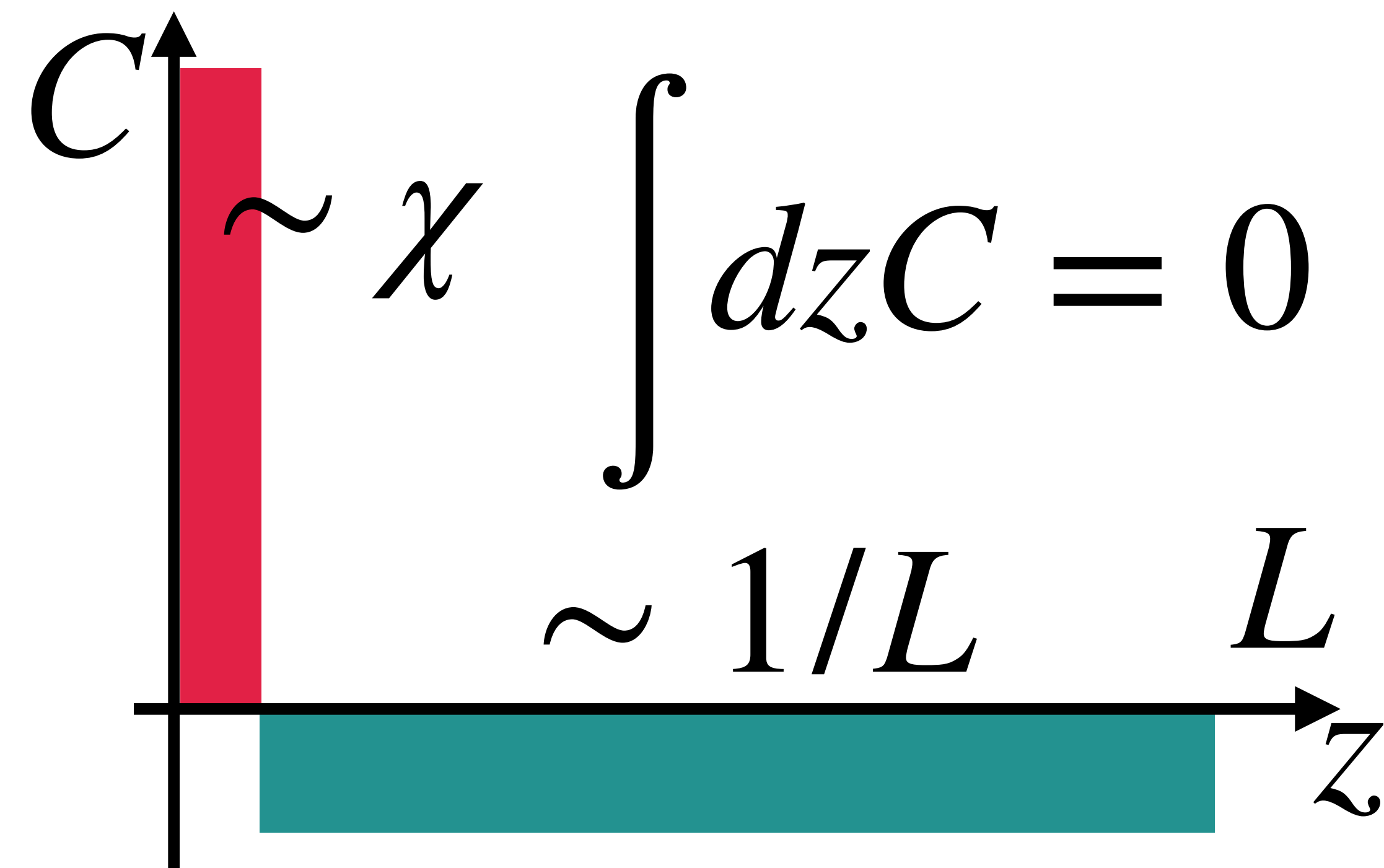
Local charge conservation

Correlations between particles in the liquid are on the order of inter-particle distance:

$$C_{AB}(\vec{r}_A, \vec{r}_B) = \langle \delta\rho_A(\vec{r}_A) \delta\rho_B(\vec{r}_B) \rangle = \chi_{AB} \delta(\vec{r}_A - \vec{r}_B)$$

If the charge is conserved:

$$C_{AB}(t, \vec{r}_A, \vec{r}_B) = \chi_{AB}(t) \delta(\vec{r}_A - \vec{r}_B) + C'_{AB}(t, \vec{r}_A, \vec{r}_B)$$

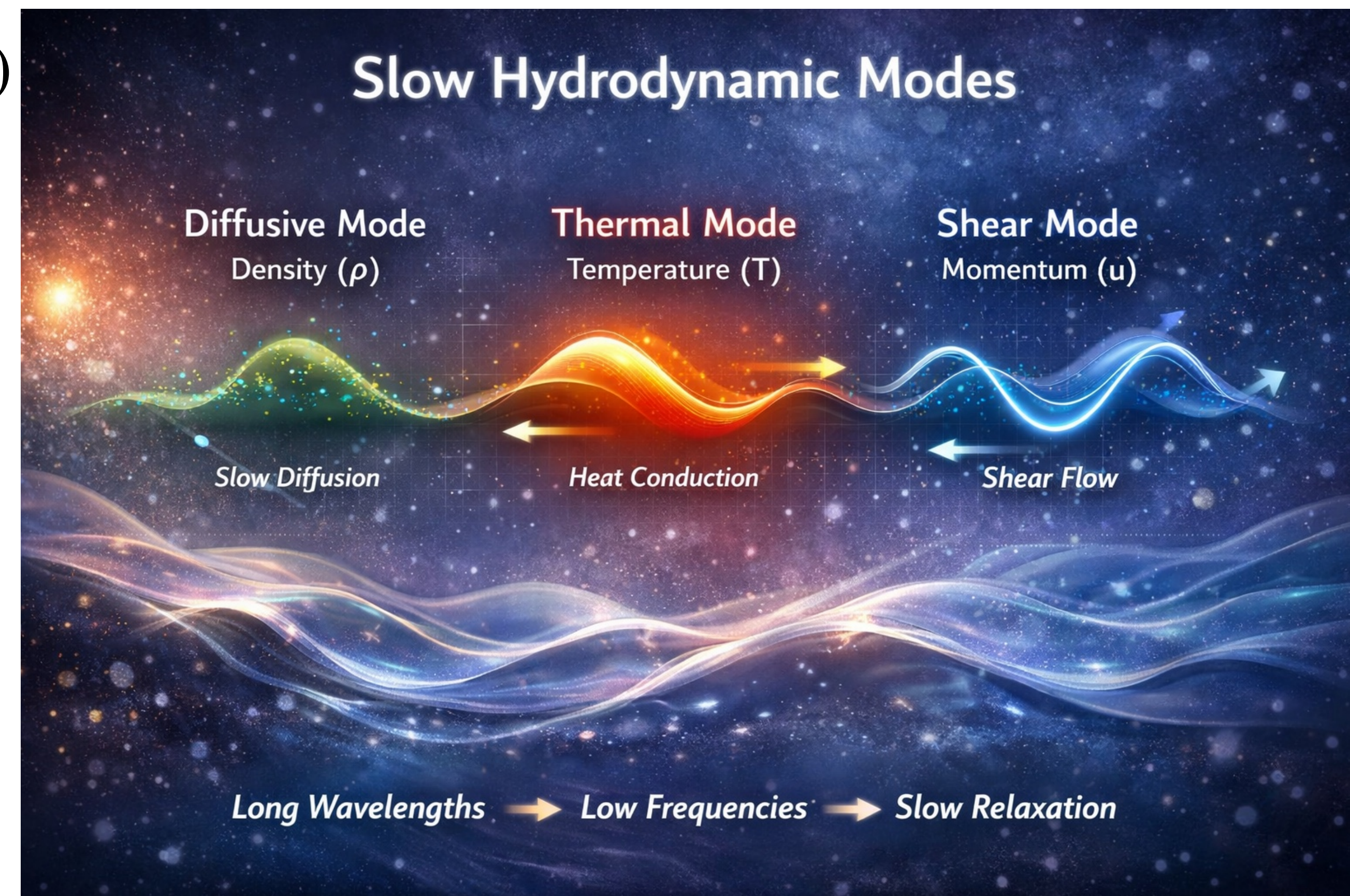


$$C \sim \chi \int dz C = 0$$

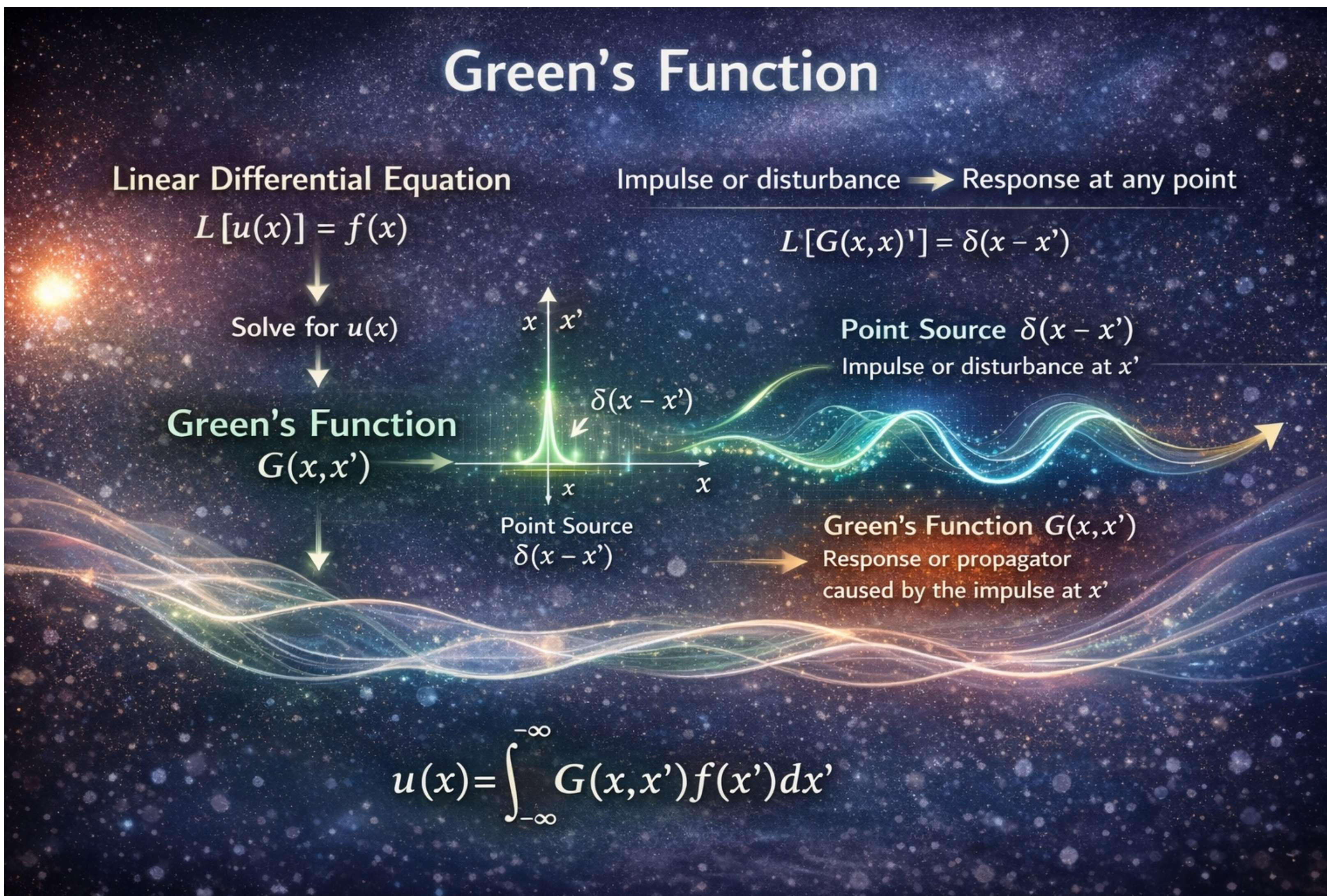
$$\sim 1/L$$

Conserved charges Q -
energy-momentum,
baryon, strange, electric

$$\chi_{AB} \equiv \frac{1}{V} \langle \delta Q_A \delta Q_B \rangle$$



Local charge conservation

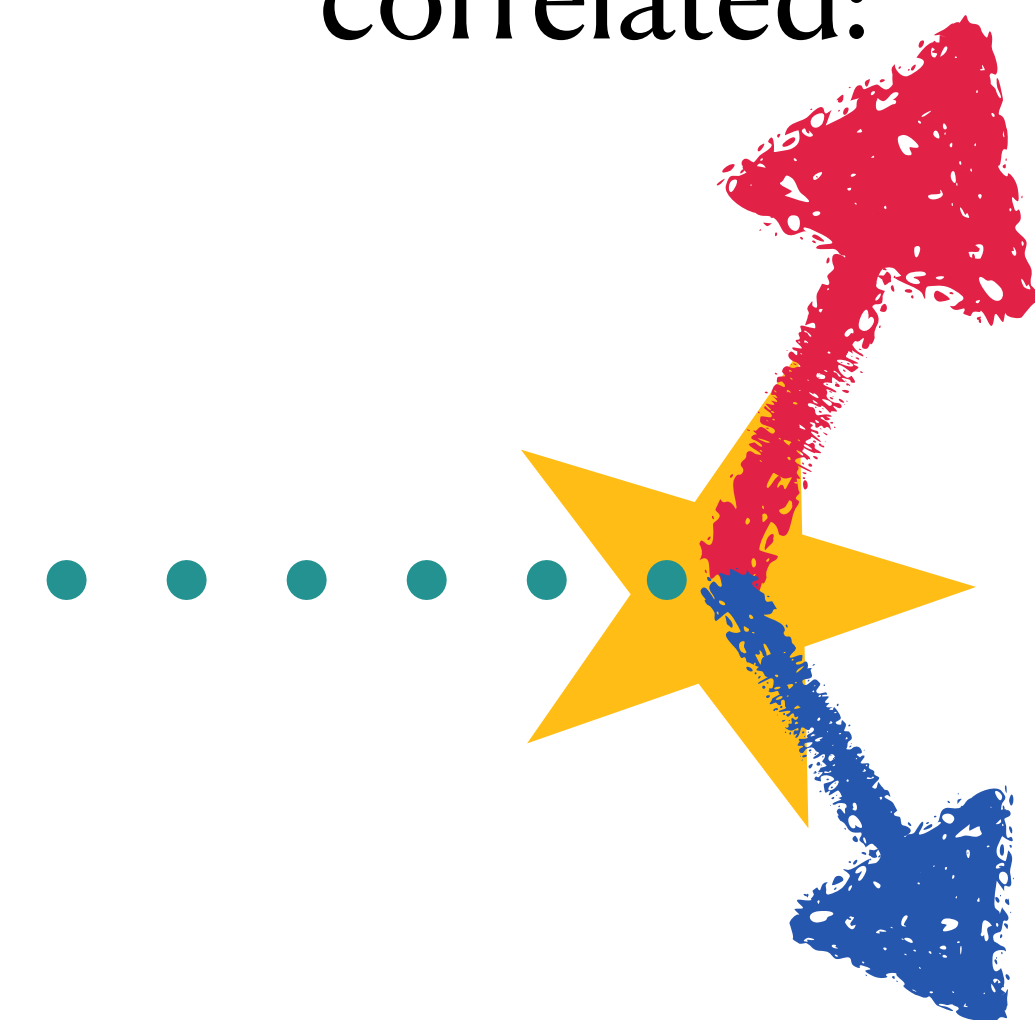


In the linear approximation
particle with charge A produced
from source with charge B:

$$\delta\rho_A = G_{AB}(\vec{r}_A, \vec{r}_B, t - t') J_B(t')$$

$$\langle \delta\rho_A \rangle = 0 = G_{AB}(\vec{r}_A, \vec{r}_B, t - t') \langle J_B(t') \rangle$$

But pairs charge/anti-charge
correlated:

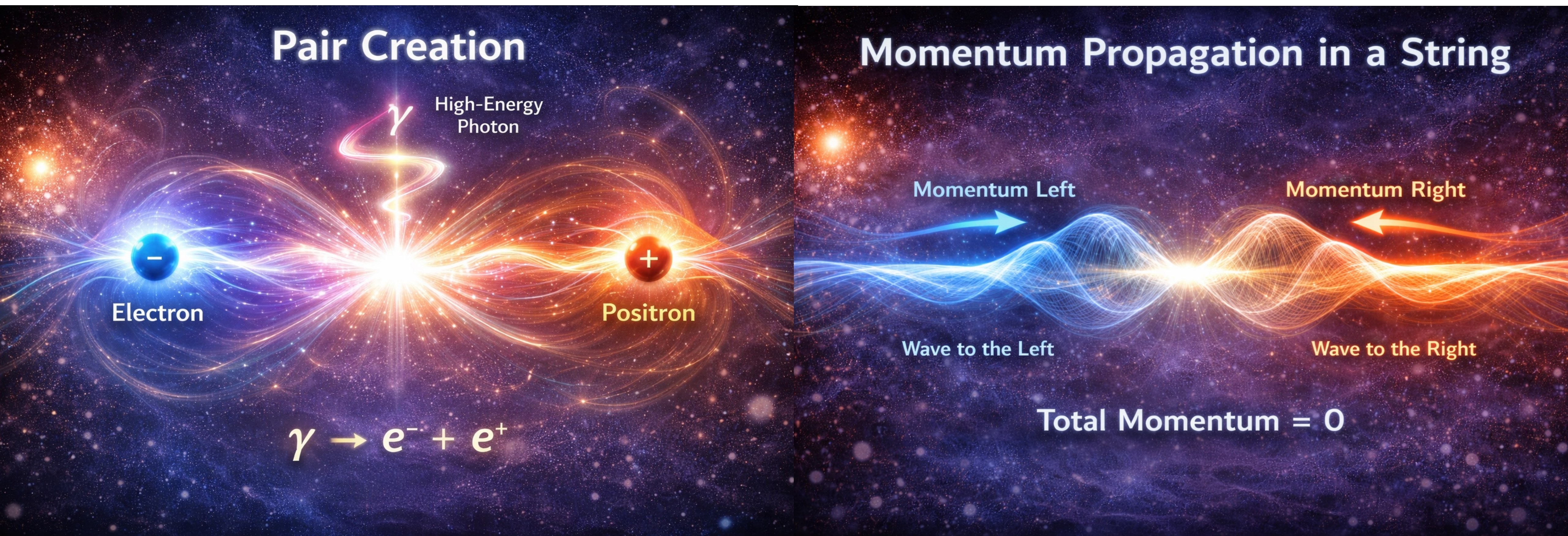


Local charge conservation

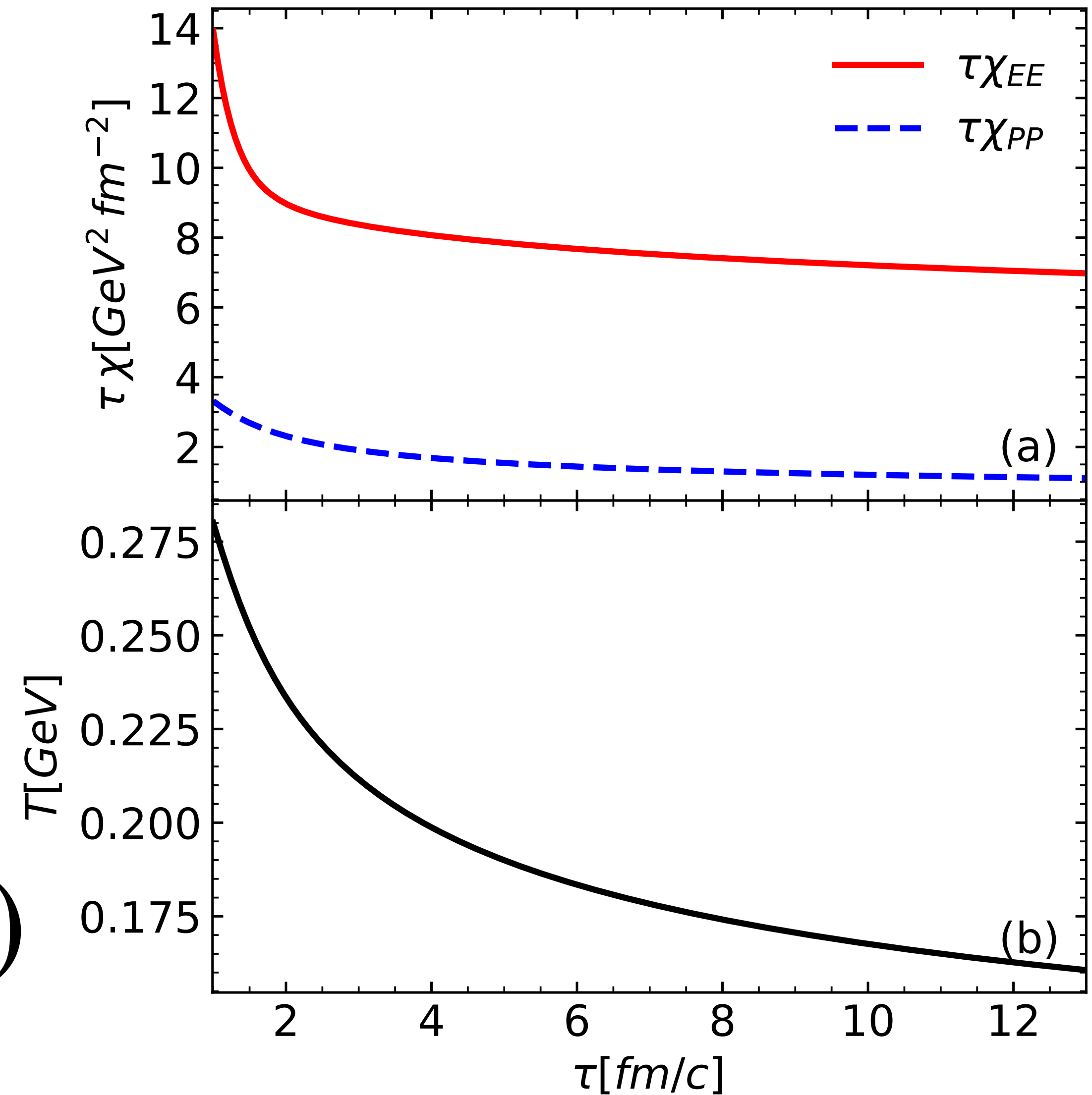
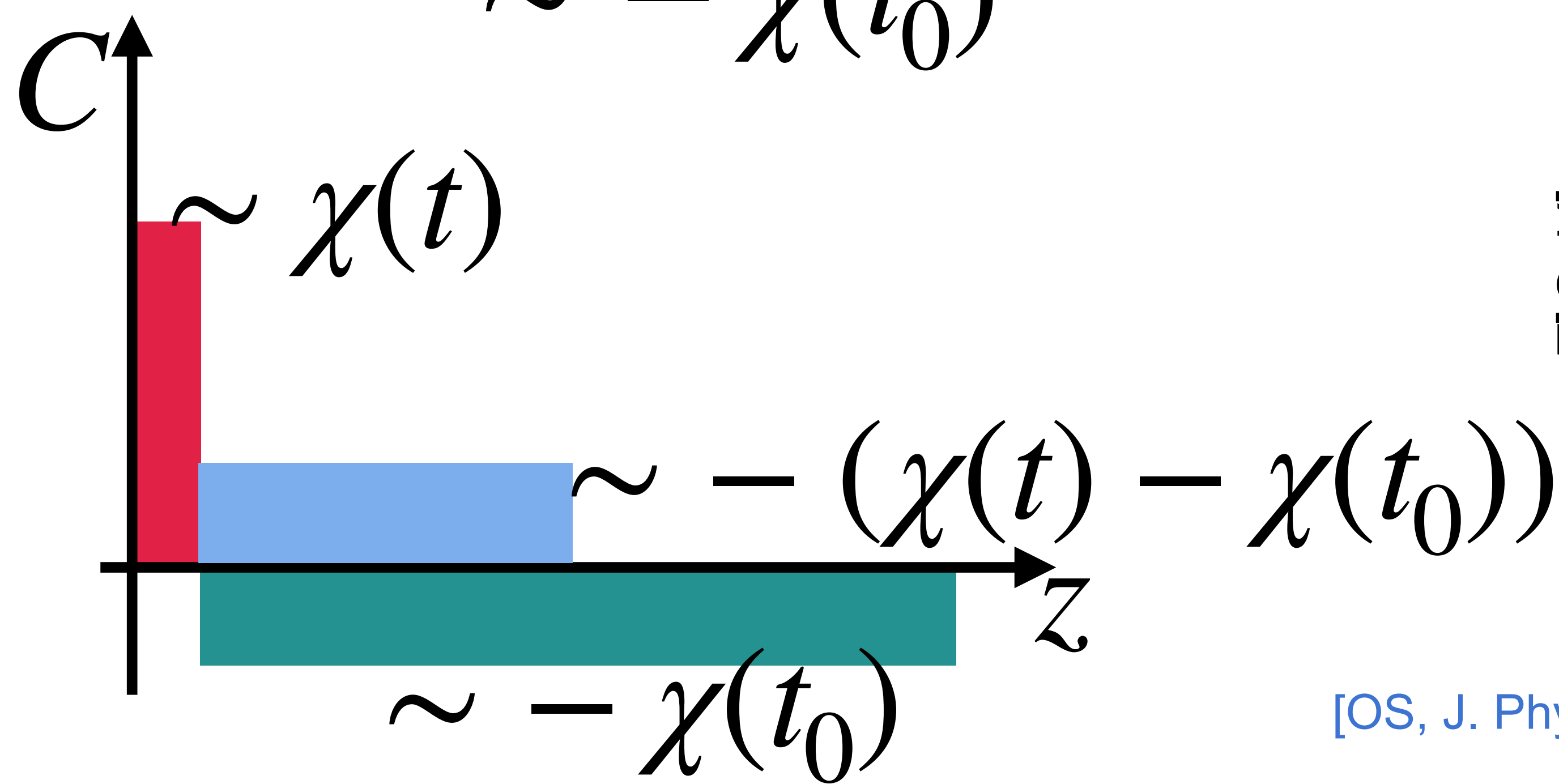
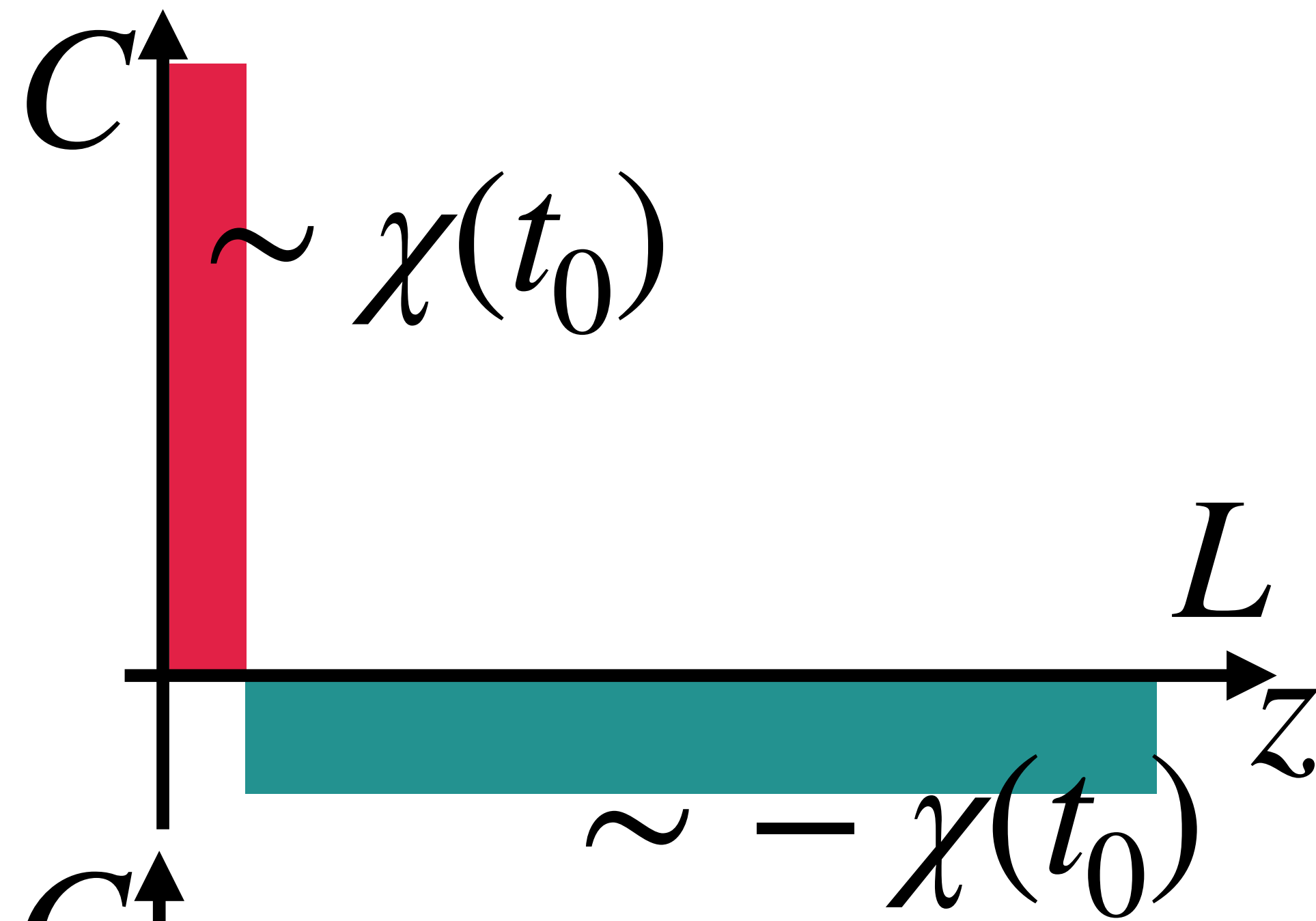
$$C' = \langle \delta\rho_A \delta\rho_B \rangle = G_{Aa} G_{Bb} \langle J_a J_b \rangle = G_{Aa} G_{Bb} \delta(\vec{r}_a - \vec{r}_b) \delta(t - t') (\partial_\mu u^\mu) \chi_{ab}$$

Correlation within the pair spreads as predicted by the Greens functions.

We are interested in long-range (balancing modes) and as such hydrodynamics can be used as the effective field theory

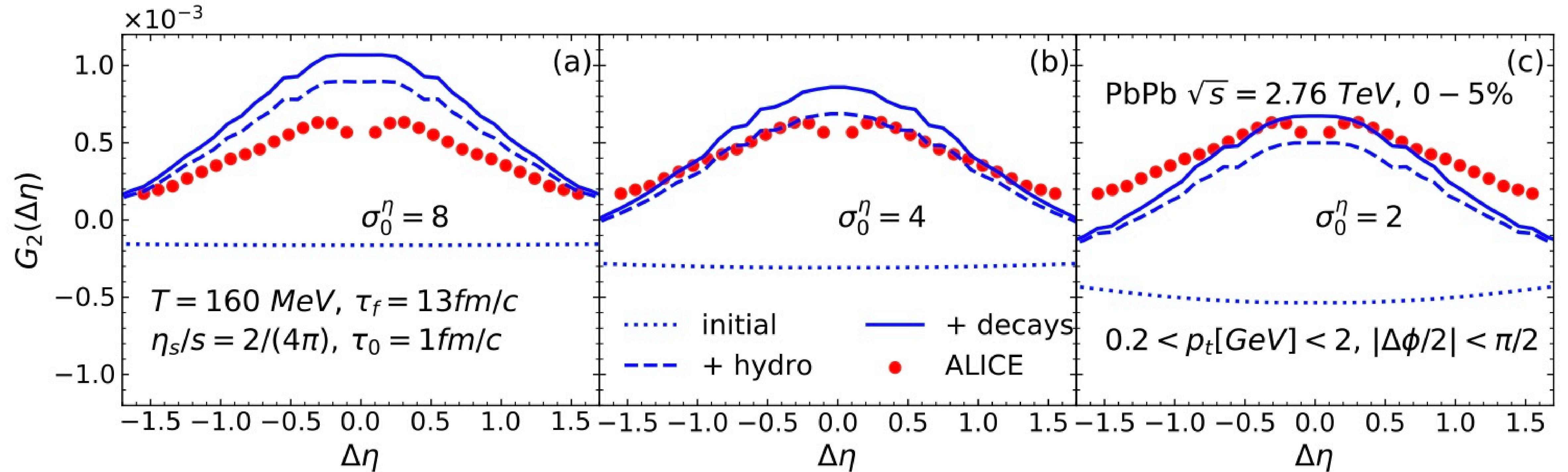


Pair production dynamics of balancing part



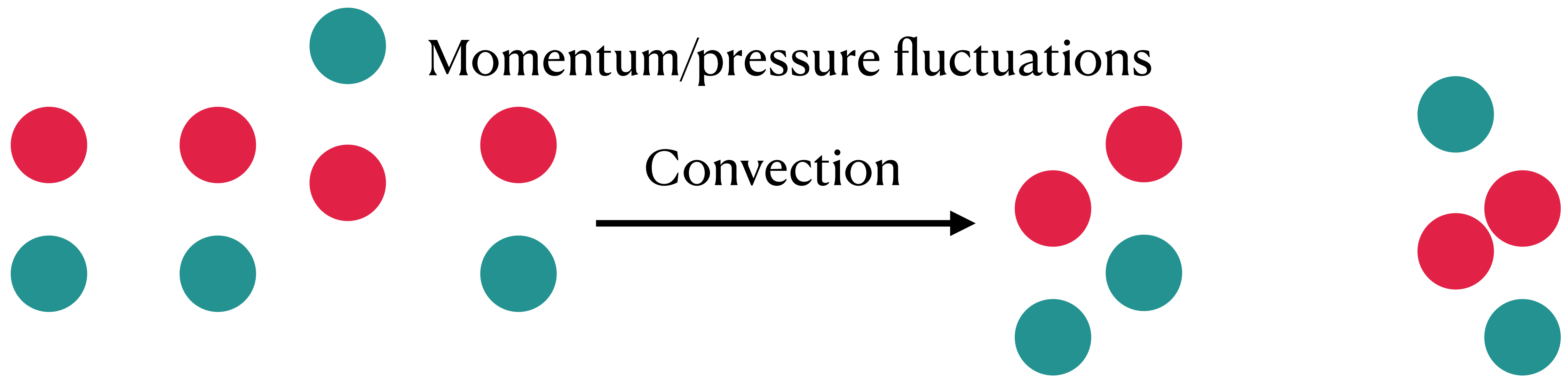
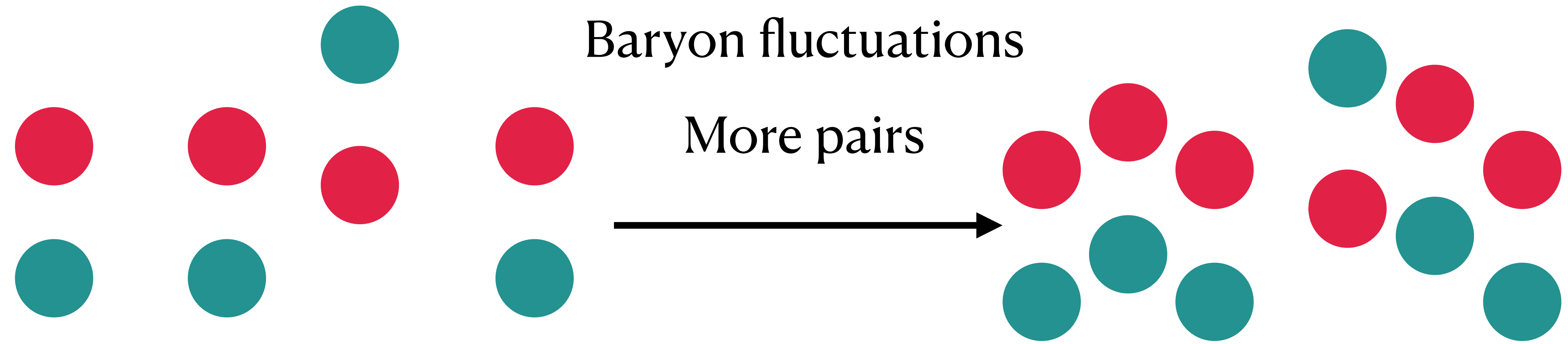
Momentum-momentum correlation

$$G_2(\eta_1, \eta_2, \phi_1, \phi_2) = \frac{1}{\langle p_{t,1} \rangle \langle p_{t,2} \rangle} \left[\frac{\int dp_{t,1} dp_{t,2} p_{t,1} p_{t,2} \rho_2(\vec{p}_1, \vec{p}_2)}{\left(\int dp_{t,1} \rho_1(\vec{p}_1) \right) \left(\int dp_{t,2} \rho_1(\vec{p}_2) \right)} - \langle p_{t,1}(\eta_1, \phi_1) \rangle \langle p_{t,2}(\eta_2, \phi_2) \rangle \right]$$



We include - hydrodynamic spread of initial correlation, “pair production” in equilibrated medium and decays in the hadronic phase

Example of cross-contamination of different charges



Coupled system of hydro equations is required

Hydrokinetics

Expansion around background

7 conserved charges
intertwined

$$\partial_\tau \left(\varepsilon + P - \frac{4\eta_s}{3\tau} \right) \delta u^\eta = -\frac{2}{\tau} \left(\varepsilon + P - \frac{4\eta_s}{3\tau} \right) \delta u^\eta - c_\rho^2 \frac{\partial_\eta \delta \varepsilon}{\tau} - \partial_\rho P \frac{\partial_\eta \delta \rho}{\tau} + \frac{4\eta_s \partial_\eta^2 \delta u^\eta}{3\tau^2}$$

Advection affects baryon charge

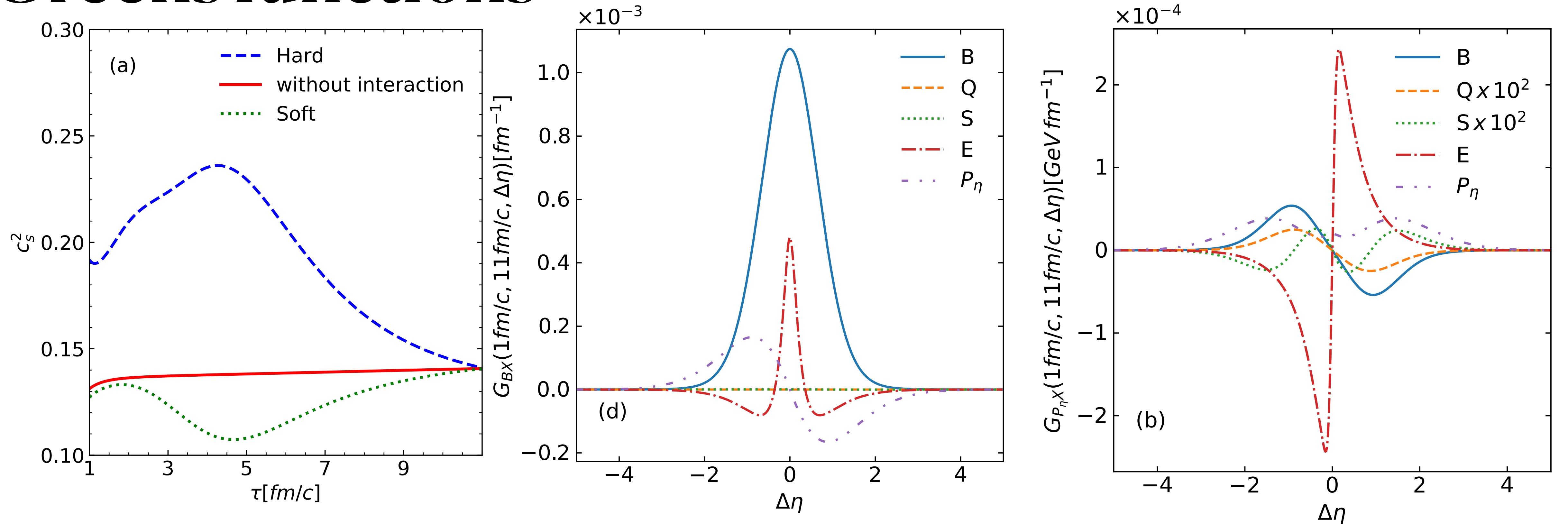
$$\partial_\tau \delta \varepsilon = -\frac{1}{\tau} \delta(\varepsilon + P) - \partial_\eta \frac{\varepsilon + P - \frac{8\eta_s}{3\tau}}{\tau} \frac{\delta u^z}{\cosh \eta} + \text{local fluctuations}$$

$$\partial_\tau \left(\varepsilon + P - \frac{2\eta_s}{3\tau} \right) \delta u^{x,y} = -\frac{1}{\tau} \left(\varepsilon + P - \frac{2\eta_s}{3\tau} \right) \delta u^{x,y} + \frac{\eta_s \partial_\eta^2 \delta u^{x,y}}{\tau^2}$$

Charge diffusion:
$$\partial_\tau \delta \rho = -\frac{\delta \rho}{\tau} + \frac{\rho + D \partial_\tau \rho}{\tau} \partial_\eta \delta u^\eta + \frac{D}{\tau^2} \partial_\eta^2 \delta \rho$$

Sensitive to the EoS

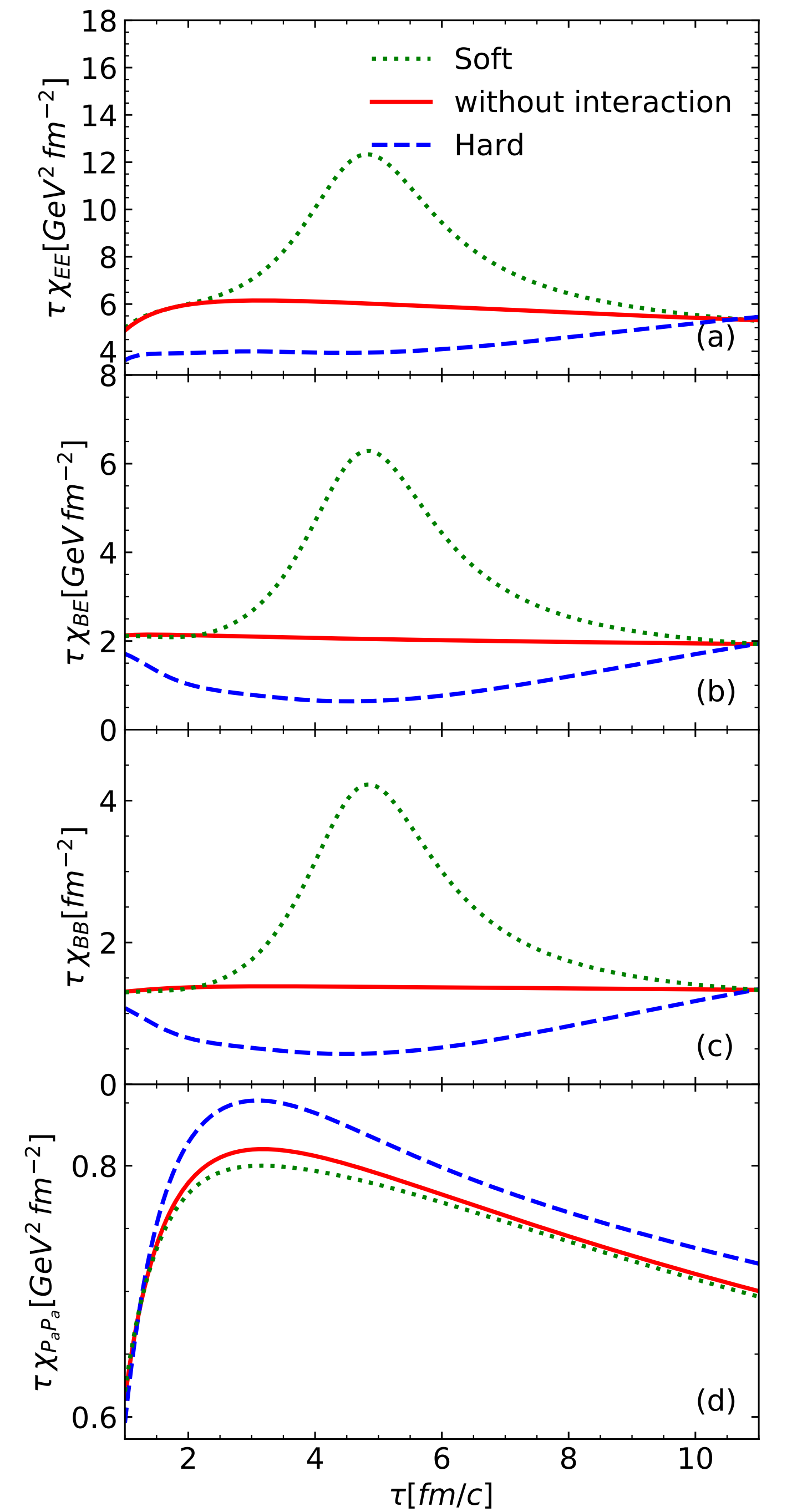
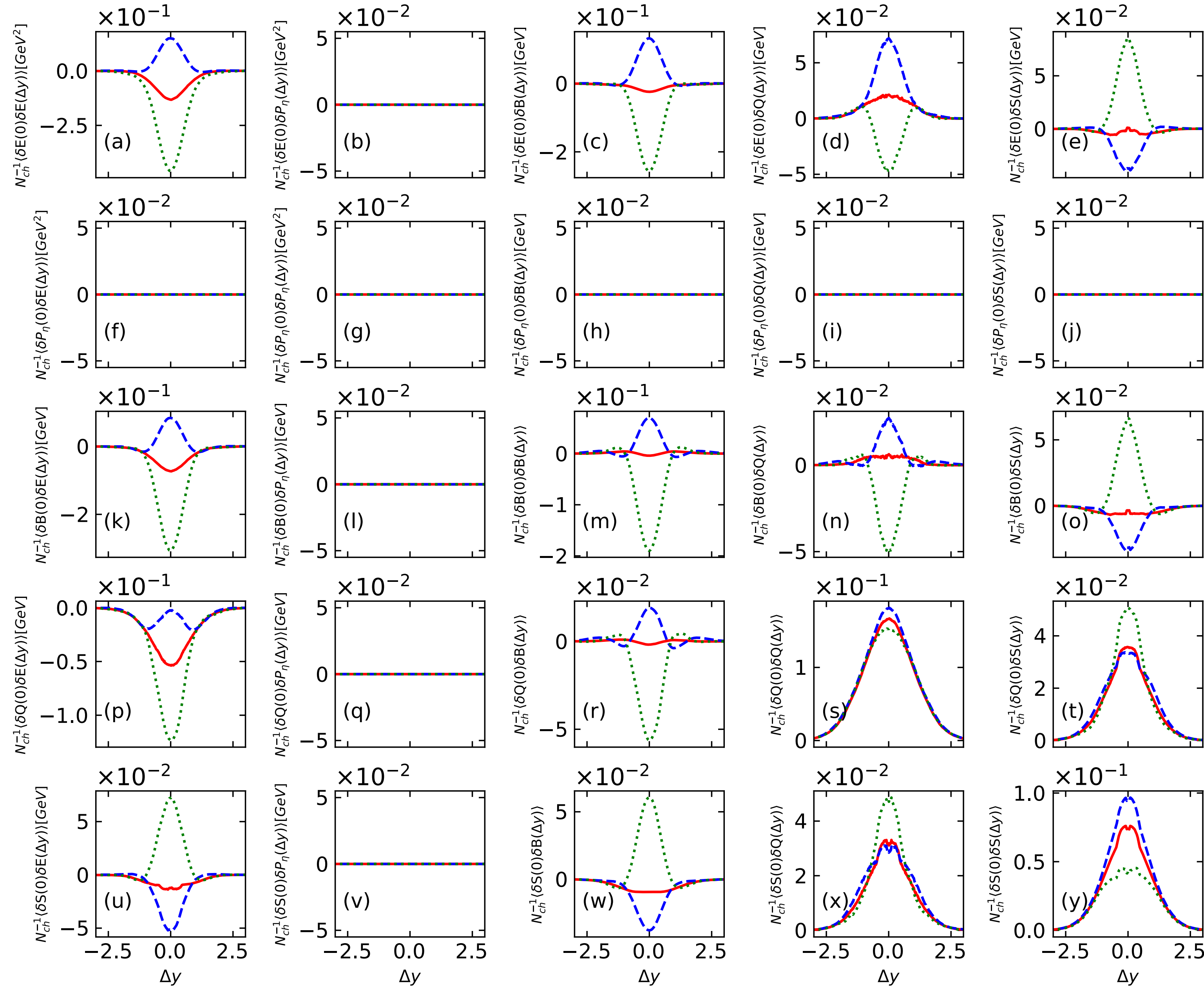
Greens functions



Apart from diffusive mode at finite density charge can relax using Sound wave propagation (spreads further)

Both susceptibilities and speed of sound are functions of EoS

Correlation matrix



Conclusions

- Future and current experimental facilities generate large amount of information about heavy-ion collisions governed by the Quantum Chromodynamics.
- There is a possibility to access dense and hot stages of matter using fluctuations and correlations.
- Baryon stopping and local charge conservation are could be used to effectively describe dynamic development of fluctuations and correlations.
- Onset of deconfinement at the stopping stage can potentially explain deviation from statistical model baselines and show importance of including early stage into the modeling.
- Local charge conservation and stochastic hydrodynamics provide a bridge between theoretically predicted susceptibilities (such as from IQCD) and experimentally measured correlations.
- At finite density different conserved charges merge together giving rise to EoS sensitivity and peculiar behavior.